INDEPENDENT STUDY COMPANION

# BASIC HISTOLOGY

## The Exam & Learning Companion

A structured summary of Junqueira’s Basic Histology, Text & Atlas, 17th edition, written for medical students preparing for examinations.

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**23** chapters condensed

**100** exam questions

**10** master comparison tables

**1** rapid-review chapter

An independent companion to Junqueira's Basic Histology, 17th edition · Not affiliated with or endorsed by the publisher

# BASIC HISTOLOGY

## The Exam & Learning Companion

An independent study companion to Junqueira’s Basic Histology: Text & Atlas, 17th edition

First edition · 2026

**HOW THIS BOOK IS ORGANIZED**

Part I · Cells and techniques: microscopy, the cytoplasm, the nucleus
Part II · The four basic tissues: epithelium, connective tissue, adipose, cartilage, bone, nerve, muscle
Part III · Transport, defense, and blood: circulation, blood, hemopoiesis, immunity
Part IV · Digestion and respiration: gut, associated organs, respiratory system, skin
Part V · Excretion and reproduction: urinary system, endocrine glands, male and female reproductive systems, eye and ear
Part VI · Rapid review, master tables, stains, and the vocabulary worth knowing
Part VII · One hundred examination questions with explained answers

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# Contents

How to Use This Companion 4
The Twelve Laws of Histology 6
Exam Technique 7
About the 17th Edition 8
Chapter and Page Guide 9

**PART I — CELLS AND TECHNIQUES 10**
1 · Histology & Its Methods of Study 11
2 · The Cytoplasm 17
3 · The Nucleus 21

**PART II — THE FOUR BASIC TISSUES 25**
4 · Epithelial Tissue 26
5 · Connective Tissue 31
6 · Adipose Tissue 35
7 · Cartilage 38
8 · Bone 41
9 · Nerve Tissue & the Nervous System 45
10 · Muscle Tissue 49

**PART III — TRANSPORT, DEFENSE, AND BLOOD 53**
11 · The Circulatory System 54
12 · Blood 58
13 · Hemopoiesis 62
14 · The Immune System & Lymphoid Organs 66

**PART IV — DIGESTION AND RESPIRATION 71**
15 · Digestive Tract 72
16 · Organs Associated with the Digestive Tract 77
17 · The Respiratory System 81
18 · Skin 85

**PART V — EXCRETION AND REPRODUCTION 90**
19 · The Urinary System 91
20 · Endocrine Glands 95
21 · The Male Reproductive System 99
22 · The Female Reproductive System 103
23 · The Eye & Ear: Special Sense Organs 109

**PART VI — RAPID REVIEW AND REFERENCE 114**
24 · The Last-24-Hours Rapid Review 115
Appendix A · Stains and What They Prove 122
Appendix B · Vocabulary You Should Know 124

**PART VII — ASSESS YOUR KNOWLEDGE 125**
100 Examination Questions 126
Answer Key with Explanations 137
Scoring and What to Do Next 142

# How to Use This Companion

Histology rewards recognition rather than recall: a small number of structural rules generate most of the detail you will be asked to reproduce. Learn the rules and the detail follows.

This companion condenses **Junqueira's Basic Histology: Text & Atlas, 17th edition (Anthony L. Mescher, McGraw Hill, 2024; ISBN 978-1-26-493039-5)** into a structured revision text. It is built on the book's own contents page: **every chapter follows the same section sequence, in the same order, using the book's own section titles**, with printed page numbers so you can read a section here and then turn straight to the corresponding pages in your copy.

### How each chapter in Parts I–V is built

| Block | What it gives you |
|---|---|
| **Chapter line** | Printed page range and the book's own section titles |
| **Learning objectives** | What you must be able to do, not merely recognize, by the end |
| **The core idea** | The one organizing principle that makes the chapter cohere |
| **Sections** | The book's sections, in the book's order, condensed |
| **Comparison tables** | Side-by-side distinctions between structures that are easily confused |
| **CLINICAL** | The disease link that turns a fact into a mark |
| **MICRO** | How to recognize the tissue on an unknown slide or EM image |
| **TIP / MNEMONIC** | Traps, confusions, and memory hooks |
| **One-line answers** | The chapter compressed to the sentences you would write in an exam |
| **Active recall** | Questions to answer from memory before checking |
| **Exam-style question** | One worked MCQ with an explained answer |
| **Plates to study** | Four to six textbook figures, with the caption subject as a self-test |

At the end of every chapter in the book there is a **Summary of Key Points** and an **Assess Your Knowledge** question set. This companion is designed to be read before those, so that when you reach them you already know the answers.

**Scope and independence.** This is an independent study companion. It is not authorized, sponsored, or endorsed by the author or the publisher, it does not reproduce any plate or figure from the textbook, and it is not a substitute for the textbook or for a histology course. Figure numbers are given so that you can open the corresponding plate in your own copy.

**What the CLINICAL boxes are.** Some name diseases the book itself discusses; others are standard clinical correlates that the book does not cover, included to show why the histology matters. Treat them as orientation, not as claims about the textbook's contents.

**A word about numbers.** Where the book gives a figure — 250 testicular lobules, 2–2.5 million follicles at birth, 70% of semen from the seminal vesicles, a 35-mm cochlea — this companion uses the book's figure. A few standard physiology and clinical numbers (GFR, bladder volumes, sperm-density thresholds) are added as context; they are not claims about the book's text. Nothing here is attributed to the book unless the book states it. Values and clinical terms that the textbook does not contain are marked **supplementary** where they appear.

**Using the practice material.** The 100 questions in Part VII are written to be attempted after the chapter they test; the explanations say why each distractor fails, so a wrong answer is worth reading carefully. A separate flashcard deck and question bank accompany this book for spaced repetition. Suggested spacing: learn a chapter, review its cards on days 1, 3, 7, 16 and 35, and interleave two earlier chapters' cards in every session. Interleaving feels harder and works better.

### Three questions to start any slide question

1. **What tissue is this?** — epithelium, connective tissue, muscle, nervous tissue, or a fluid.
2. **What organ or region?** — the tissue arrangement identifies the organ, not the cell type.
3. **What is it doing here?** — every structure you can see exists because of a function the cell must perform. Naming the function is what makes the structure memorable.

**EXAM TIP**

Never memorize a slide as a picture. Memorize the rule that generates the picture ("stratified squamous with keratin on top = surface exposed to abrasion and drying"). The book's atlas plates teach you the appearance; this companion teaches the rule behind it.

## Three study plans

**If you have six weeks.** Work through Parts I–V at roughly one part per week, reading the corresponding chapter first, then the section here, then covering the tables and redrawing them from memory. Do the book's "Assess Your Knowledge" questions as you go. In the sixth week, revise with Part VI (Rapid Review and the master tables) and work through the 100 questions in Part VII.

**If you have ten days.** Two to three chapters per day, tables only on the first pass. Concentrate on six comparison tables: epithelium types, connective tissue fibers, cartilage, muscle, leukocytes, and the GI tract wall.

**If you have 48 hours.** Read Rapid Review (Chapter 24) end to end, then the 100 MCQs, then the six tables. Do not attempt prose. Histology rewards pattern recognition under time pressure, and patterns live in tables.

# The Twelve Laws of Histology

These twelve rules of thumb hold across most of the book. Each has exceptions, and the chapters point them out. When you are unsure how to approach a structure, reason from them.

1. **Structure follows function, and function follows structure.** Every visible feature is a solution to a physical problem — diffusion distance, mechanical stress, surface area, water loss, conduction speed.
2. **Form is set by the apical surface, the basal surface, and the junctions between cells.** Look at all three before naming a tissue.
3. **Epithelium sits on something and is never crossed by a blood vessel.** Avascularity plus a basement membrane defines it.
4. **Connective tissue is defined by its extracellular matrix, not its cells.** The matrix tells you the mechanical job: tension, elasticity, compression, or filtration.
5. **Basophilia and eosinophilia report what a cell is doing.** Blue-purple staining (hematoxylin) reflects nucleic acid — rough ER and ribosomes in a protein-synthesizing cell. Pink staining (eosin) reflects abundant protein — cytoplasm, secretory granules, collagen. The correlation is strong but not absolute: mucin and lipid droplets stain poorly with either dye.
6. **Cells that make steroid hormone are full of smooth ER and lipid droplets** — hence foamy, pale, or vacuolated cytoplasm (adrenal cortex, Leydig cells, corpus luteum).
7. **Muscle is classified by nucleus position and striation, not by shape alone.** Peripheral multiple nuclei = skeletal; central single nucleus with striation = cardiac; central single nucleus without striation = smooth.
8. **The number of cell layers tells you the mechanical job.** One layer = exchange, absorption, secretion, or a thin barrier. Many layers = protection from abrasion.
9. **Surface area is bought with folds.** Microvilli, villi, plicae circulares, basal infoldings, and cristae all exist for the same arithmetic reason.
10. **Tubes are built in concentric layers, and each layer has one job.** In blood vessels: intima for exchange, media for tone, adventitia for anchorage. In the gut: mucosa for digestion, submucosa for supply, muscularis for movement.
11. **A capsule and a hilum identify a solid organ as lymph node, spleen, thymus, or gland.** The route of fluid flow tells you which one.
12. **Most histological facts have a clinical counterpart.** Linking a structure to a disease makes it easier to recall and easier to apply.

# Exam Technique

## Answering a slide or image question

Never begin with a guess. Use this sequence every single time:

1. **Magnification and staining.** H&E at low power versus PAS versus silver tells you what the preparation was designed to show.
2. **The boundary.** Is there a free surface? A lumen? A capsule? Blood vessels inside the tissue? This immediately separates epithelium-lined organs from solid organs.
3. **The dominant tissue.** Note the arrangement — cords, acini, tubules, follicles, layers, bundles.
4. **The one diagnostic cell.** Parietal cell, Paneth cell, podocyte, Purkinje fiber, Clara cell, Sertoli cell, odontoblast. Examination images often turn on a single diagnostic cell.
5. **Name it, then justify it in the same sentence.** "This is thyroid gland because the parenchyma is arranged in follicles of colloid lined by a single layer of cuboidal cells." A justified answer shows that you understand the image rather than merely recognizing it.

## Answering a "describe and give the function of" essay

Use a fixed skeleton — it works for any organ:

**Capsule → cortex → medulla → parenchyma arrangement → cell types → blood supply → ducts or outflow → function → clinical correlation.**

The headings work for almost any organ. Write them down first, then fill them in.

## Ten common errors

- "It looks like…" without saying why it looks that way.
- Naming a cell without its product or function.
- Confusing "basophilic" (blue, acidic structure) with "eosinophilic" (pink, basic protein).
- Saying "simple squamous" for endothelium without noting it is a type of simple squamous epithelium.
- Using "mucous" for "mucosa" — different words, different things.
- Describing the adrenal medulla and cortex as one tissue; they are embryologically unrelated.
- Calling transitional epithelium "stratified squamous"; the umbrella cells are unique.
- Forgetting that the thymus has **no** follicles and the spleen has **no** afferent lymphatics.
- Mixing up the two adrenal cortex zones that produce cortisol and aldosterone.
- Writing "bone is dead tissue" — osteocytes are metabolically active and connected by canaliculi.

**HIGH-YIELD PEARL**

In a viva or OSCE-style histology station, the task is usually to connect a structure to a function and a disease. Practice every fact in a three-part sentence: "Structurally X, which allows Y, and when it fails you get Z."

# About the 17th Edition

The seventeenth edition (2024) keeps the features that make the book useful for revision: a **concise text**, **clear photomicrographs and labeled illustrations**, and a consistent chapter structure. Three features matter most for examination preparation:

- **Summary of Key Points** at the end of every chapter — the author's own list of what matters.
- **Summary Tables** in each chapter — condensed comparisons of structures that are easily confused.
- **Medical Applications boxes** — the clinical relevance of each structure.

**EXAM TIP**

Where this companion and the book's Summary of Key Points disagree in emphasis, trust the book: it is the primary source for this companion. Use this companion for structure and recall, and the book for the final word.

# Chapter and Page Guide

Each chapter opens with the printed page range it condenses, so the companion and the textbook can be read side by side. The part column refers to the seven parts of this book.

| Ch | Title | Book pages | Part |
|---|---|---|---|
| 1 | Histology & Its Methods of Study | 1–16 | I |
| 2 | The Cytoplasm | 17–52 | I |
| 3 | The Nucleus | 53–70 | I |
| 4 | Epithelial Tissue | 71–95 | II |
| 5 | Connective Tissue | 96–121 | II |
| 6 | Adipose Tissue | 122–128 | II |
| 7 | Cartilage | 129–137 | II |
| 8 | Bone | 138–160 | II |
| 9 | Nerve Tissue & the Nervous System | 161–192 | II |
| 10 | Muscle Tissue | 193–214 | II |
| 11 | The Circulatory System | 215–236 | III |
| 12 | Blood | 237–253 | III |
| 13 | Hemopoiesis | 254–266 | III |
| 14 | The Immune System & Lymphoid Organs | 267–294 | III |
| 15 | Digestive Tract | 295–328 | IV |
| 16 | Organs Associated with the Digestive Tract | 329–348 | IV |
| 17 | The Respiratory System | 349–370 | IV |
| 18 | Skin | 371–392 | IV |
| 19 | The Urinary System | 393–412 | V |
| 20 | Endocrine Glands | 413–438 | V |
| 21 | The Male Reproductive System | 439–459 | V |
| 22 | The Female Reproductive System | 460–489 | V |
| 23 | The Eye & Ear: Special Sense Organs | 490–524 | V |
| App. | Light Microscopy Stains | 525–526 | Appendix A |
| 24 | The Last-24-Hours Rapid Review | synthesis of Chapters 1–23 | VI |

# PART I — CELLS AND TECHNIQUES

Chapters 1–3 · microscopy, the cytoplasm, and the nucleus

# 1 · Histology & Its Methods of Study

**Book pages 1–16**
Preparation of Tissues for Study · Light Microscopy · Electron Microscopy · Autoradiography · Cell & Tissue Culture · Enzyme Histochemistry · Visualizing Specific Molecules · Interpretation of Structures in Tissue Sections

## Learning objectives

By the end of this chapter you should be able to:

1. Trace a tissue from patient to stained slide, and explain what each step preserves.
2. Predict what hematoxylin and eosin will color, and why.
3. Choose the right special technique for a target molecule (carbohydrate, lipid, protein, nucleic acid, antigen).
4. Explain how section thickness, fixation, and artifact can mislead you when you interpret a two-dimensional image.

## The core idea

Histology explores the body's tissues and how they are arranged to produce functional organs. Tissues have two interacting components: **cells and extracellular matrix (ECM)**. The ECM is not packing material — it supports the cells, carries the fluid that brings nutrients and removes waste, and its molecules bind specific cell-surface receptors connected to structural components inside the cell, forming a continuum in which cells and matrix function together.

Organs form by an orderly combination of these tissues. Because cells and matrix components are small, histology depends on microscopes and molecular methods — and every image you will interpret has been fixed, dehydrated, cleared, embedded, sectioned, stained and mounted. Knowing what each step does tells you which appearances are real and which are artefacts.

## Preparation of Tissues for Study

| Step | What the book says it does |
|---|---|
| **Fixation** | Pieces of organ are placed as soon as possible in **stabilizing or cross-linking compounds**. **Formalin** — a buffered solution of formaldehyde, made by diluting the 37–40% stock to about 4% — is widely used for light microscopy; **glutaraldehyde** for electron microscopy. Both react with the **amine groups ($NH_2$) of proteins**, preventing degradation by proteases; glutaraldehyde also **cross-links adjacent proteins**. Fixative must diffuse fully through the tissue, so organs are cut into small fragments, or fixed by **vascular perfusion** |
| **Dehydration** | Water extracted gradually through increasing **ethanol** solutions, ending in 100% ethanol |
| **Clearing** | Ethanol replaced by an organic solvent miscible with both alcohol and the embedding medium; the tissue becomes **translucent** |
| **Infiltration** | Tissue placed in melted **paraffin at 52–60 °C** in an oven, which evaporates the clearing solvent |
| **Embedding** | Paraffin allowed to harden in a small container at room temperature |
| **Sectioning** | Block trimmed and cut on a **microtome**. Paraffin sections are typically **3–10 µm**; EM requires sections **less than 1 µm** |

**A concentration note.** The book describes formalin as "a buffered isotonic solution of 37% formaldehyde." That 37–40% solution is the **stock** from which working fixative is made; tissue is fixed in **10% formalin**, equivalent to about 3.7–4% formaldehyde, buffered to pH 7.2–7.4 (neutral buffered formalin). Keep the two concentrations apart in practical questions.

**The units** the book expects you to handle: 1 µm = 1/1000 mm; 1 nm = 0.001 µm = $10^{-6}$ mm; 1 Å = 0.1 nm = $10^{-4}$ µm.

**Plastic resins** are used for both light and electron microscopy. They are infiltrated after dehydration and harden when cross-linking polymerisers are added; plastic embedding **avoids the higher temperatures of paraffin** and therefore avoids tissue distortion.

**CLINICAL CORRELATION**

**Biopsies** are tissue samples removed during surgery or routine procedures, fixed in vials of formalin. When the result is needed before the procedure ends — typically to know whether a growth is malignant before the patient is closed — the biopsy is **frozen in liquid nitrogen**, which preserves structure and hardens the tissue, and sectioned on a **cryostat** in a subfreezing cabinet. Freezing is also used for histochemistry of **sensitive enzymes or small molecules**, because freezing, unlike fixation, does not inactivate most enzymes; and because clearing solvents dissolve lipids, **frozen sections are used whenever lipid-containing structures must be studied**.

## Staining

Most cells and extracellular material are completely colorless. Dyes behave like acidic or basic compounds and form **electrostatic (salt) linkages** with ionisable radicals of tissue macromolecules.

| Term | Meaning in the book | Examples |
|---|---|---|
| **Basophilic** | Anionic (net negative) components with affinity for **basic dyes** — nucleic acids, glycosaminoglycans | Toluidine blue, alcian blue, methylene blue, **hematoxylin** |
| **Acidophilic** | Cationic components with many ionised amino groups, staining with **acidic dyes** — mitochondria, secretory granules, collagen | **Eosin**, orange G, acid fuchsin |

**H&E** is the most common method. **Hematoxylin** stains DNA in the nucleus, RNA-rich cytoplasm, and cartilage matrix dark blue-purple; **eosin** stains other cytoplasmic structures and collagen pink and acts as the **counterstain**. **Trichrome** stains (eg, Masson) allow greater distinction among extracellular components.

**The PAS reaction** uses the hexose rings of polysaccharides and other carbohydrate-rich structures and stains them distinctly **purple or magenta** — the basis for demonstrating glycogen, mucin, and basement membranes. The **Feulgen reaction** is a modification of PAS that stains **DNA specifically**.

**Enzyme digestion** identifies what a stain is binding: pretreating a section with an enzyme that digests one substrate. Pretreatment with **ribonuclease** greatly reduces cytoplasmic basophilia with little effect on the nucleus, proving that the cytoplasmic staining depended on RNA.

**Lipid-soluble dyes** such as **Sudan black** reveal lipid-rich structures — provided the processing steps that remove lipids (heat and organic solvents) are avoided. **Metal impregnation** with silver salts visualizes certain ECM fibers and specific elements in nervous tissue. The book's Appendix lists the staining procedures used for most of its light micrographs.

**The atlas palette.** The book's Appendix (pp 525–526) lists the stains behind its plates, and the captions name them. Learn what each one makes visible, because a caption that says "Mallory trichrome" is telling you which structures were deliberately contrasted:

| Stain | Nuclei | Cytoplasm | Collagen / other | Where you meet it |
|---|---|---|---|---|
| **H&E** | Blue-purple (hematoxylin) | Pink (eosin) | Collagen pink | Almost every chapter |
| **PT** (pararosaniline–toluidine blue) | Purple chromatin | Lighter violet | Collagen pale violet | Plastic sections needing fine detail |

| Stain | Nuclei | Cytoplasm | Collagen / other | Where you meet it |
|---|---|---|---|---|
| **Mallory trichrome** | Purple | Red-orange (also keratin, erythrocytes) | **Blue** | Connective tissue, small vessels, pituitary, kidney |
| **Picrosirius–hematoxylin (PSH)** | Purple | Pink-violet | **Collagen red**, birefringent under polarized light | Ovary, cartilage, tendon |
| **PAS** | — | Magenta where carbohydrate is abundant | Glycogen, mucin, basement membrane, glycocalyx | Gut, kidney, cartilage |
| **Wright–Giemsa** | Purple | Granules color differentially | Erythrocytes pink | Blood and marrow smears |
| **Silver / gold** | — | — | Nerve filaments and **reticulin (type III)** black-brown | Nervous tissue, lymphoid organs |
| **Elastin stains** (Weigert, aldehyde fuchsin, orcein) | — | — | Elastic fibers brown-purple | Aorta, lung, skin |
| **Lipid stains** (oil red O, Sudan black, osmium tetroxide) | — | — | Lipid droplets and myelin | Adipose tissue, steroid cells, nerve |

Slide preparation, from fixation to observation, takes **12 hours to 2½ days**, depending on tissue size, embedding medium, and staining method.

## Light Microscopy

The **bright-field microscope** has three lens sets:

- **Condenser** — collects and focuses a cone of light onto the slide.
- **Objective** — enlarges and projects the image; routine objectives are **×4** (large field), **×10** (medium), **×40** (detail).
- **Eyepiece (ocular)** — magnifies a further **×10**, giving total magnifications of **×40, ×100, ×400**. Total magnification = objective × ocular.

**Resolving power** — the smallest distance at which two structures are seen as separate — is the critical factor, and its maximum for a light microscope is about **0.2 µm**, permitting clear images at **1000–1500×**. Anything smaller or thinner than 0.2 µm (a single ribosome, a microfilament) cannot be distinguished, and two mitochondria closer than 0.2 µm appear as one. Resolution depends mainly on the **objective lens**; the ocular only enlarges and does **not** improve resolution. Magnification is of value only when accompanied by resolution.

**Virtual microscopy** converts a stained slide into high-resolution digital images captured in a grid at multiple magnifications, stored so a browser can navigate the original slide — now replacing glass slide collections in teaching laboratories.

| Method | Principle | Use |
|---|---|---|
| **Fluorescence** | Fluorophores emit longer-wavelength light when irradiated; bright on a dark background | **DAPI** and **Hoechst** bind DNA (blue); **acridine orange** binds DNA and RNA and emits different colors, localizing them separately; fluorescein-coupled probes and labeled antibodies |
| **Phase-contrast** | Light changes speed through structures of different refractive index, producing contrast in transparent objects | **Living, unstained cultured cells**; **DIC (Nomarski optics)** gives a more three-dimensional image |

| Method | Principle | Use |
|---|---|---|
| **Confocal** | A laser point source and a **pinhole aperture** optically conjugated to the focal plane block out-of-focus light; the beam scans the specimen | Thin **optical sections**, localization with great precision, and 3-D reconstruction |
| **Polarizing** | A polarizing filter passes light vibrating in one direction; oriented macromolecules rotate it and appear bright against a dark background — **birefringence** | Collagen, cellulose, microtubules, actin filaments; only material with repetitive periodic structure is seen |

Digital cameras extend all of these methods: images can be quantified, enhanced, and analyzed by software for features not visible through the eyepieces.

## Electron Microscopy

The wavelength of an electron beam is far shorter than that of light, so the electron microscope can resolve structures far below the light-microscope limit — in principle, a thousandfold improvement.

**Transmission EM (TEM)** resolves to about **3 nm** in routine biological sections — the figure the book gives — enough to view isolated particles magnified as much as **400,000×**. Routine tissue sections of **40–90 nm** embedded in resin are studied at up to about **120,000×**. Electrons pass through the section to produce an image in black, white, and gray: areas that transmitted electrons appear **electron lucent** (bright), while denser areas or those binding heavy metal ions appear **electron dense** (dark). The column is configured like an upside-down light microscope, with electromagnetic lenses and a tungsten filament cathode accelerating electrons at 60–120 kV.

**Scanning EM (SEM)** images the surface in three dimensions.

## Autoradiography

A **radiolabelled precursor** is incorporated into newly synthesized macromolecules and detected by a photographic emulsion applied over the section — so the silver grains mark the **cellular sites where those macromolecules were made**. The classic application is **$^{3}$H-thymidine** labeling of cells in S phase.

## Cell & Tissue Culture

Living cells maintained outside the body, observed with phase-contrast or DIC microscopy, allow behavior, differentiation, and drug responses to be studied directly — without fixation, which is precisely the advantage.

## Enzyme Histochemistry

A substrate is converted by a specific enzyme into an **insoluble, colored product** deposited at the site of the enzyme, localizing enzyme activity within cells. **Acid phosphatase** for lysosomes is the standard example.

## Visualizing Specific Molecules

**Immunohistochemistry (IHC)** depends on the specific reaction between an antigen and an antibody labeled with a visible marker — a fluorescent compound or **peroxidase** for light microscopy, **gold particles** for TEM.

| Method | How it works | Why it is used |
|---|---|---|
| **Direct IHC** | A labeled **primary antibody** binds the antigen directly | Simple, fewer steps |
| **Indirect IHC** | An unlabelled primary antibody is detected by labeled **secondary antibodies** | **More commonly used** — the extra level of binding **amplifies the signal** and gives greater technical flexibility |

**In situ hybridization (ISH)** uses labeled **cDNA probes** to detect specific gene sequences or mRNAs microscopically.

## Interpretation of Structures in Tissue Sections

Two rules from the book's own section:

1. Many processing, preparation, and staining steps introduce **artefacts** — spaces and precipitates not present in living tissue — and these must be recognized for what they are.
2. Sections are essentially **two-dimensional planes through three-dimensional structures**, so what you see is a slice, not the whole object.

**UNDER THE MICROSCOPE**
Before identifying anything, decide whether it could be an artefact. A clear space around every cell is shrinkage; a straight edge through the tissue is a knife mark; granular debris is precipitated stain. True structure has irregular, biological boundaries.

**EXAM TIP**
Three numbers carry the chapter: **0.2 µm** (light microscope resolution), **3–10 µm** (paraffin section thickness), **about 3 nm** (TEM resolution of routine sections). If a question asks why the electron microscope reveals organelles that the light microscope cannot, the answer is **the shorter wavelength of the electron beam**, not the magnification.

**HIGH-YIELD PEARL**
A useful chain to remember from Chapter 1: **dehydration precedes clearing**, clearing makes the tissue translucent, and **freezing avoids both** — which is why frozen sections are used for rapid diagnosis, for enzymes, and for lipids.

## Chapter 1 — one-line answers

- H&E = the routine survey stain; hematoxylin is basic (nuclei, rough ER), eosin is acidic (collagen, cytoplasm).
- Resolution: light microscope about 0.2 µm; TEM about 3 nm in routine sections.
- Fixation cross-links proteins; formalin is the standard fixative, glutaraldehyde for EM.
- Paraffin sections are 3–10 µm; frozen sections are used for urgent diagnosis and for lipid.
- PAS stains carbohydrates magenta; it demonstrates glycogen, mucin, and basement membranes.
- Immunohistochemistry localizes a specific antigen with an antibody; in situ hybridization localizes a specific nucleic acid sequence.
- Artefacts — shrinkage, knife marks, precipitated stain — must be recognized before any structure is identified.
- A 2D section samples one plane, so a tube can look like a ring and a sphere like a circle.

## Active recall

Answer each from memory before checking.

1. **What does formalin do to a tissue?** — Cross-links proteins, preserving structure while making the tissue firm enough to section.
2. **Which dye stains nuclei, and why?** — Hematoxylin, a basic dye that binds acidic nucleic acids (DNA and RNA).
3. **What does PAS demonstrate?** — Carbohydrates, glycogen, and glycoprotein-rich structures such as basement membranes.
4. **Light microscope versus electron microscope resolution?** — About 0.2 µm versus about 3 nm in routine sections, roughly two orders of magnitude.
5. **Why is a 5-µm section dangerous to interpret in one glance?** — It samples a single plane; a tube can look like a ring, a sphere like a circle, and a fold like a septum.

## Exam-style question

**A research group needs to show that a newly described protein is present in the basal layer of the epidermis. Which technique is the most direct choice?**

A. Hematoxylin and eosin
B. Periodic acid–Schiff
C. Immunohistochemistry with a specific antibody
D. Sudan black

**Answer: C** — Immunohistochemistry localizes a specific protein by antibody binding. H&E shows general morphology, PAS shows carbohydrates, and Sudan black shows lipids.

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 1–1** — Sectioning fixed and embedded tissue.
- **Fig. 1–4** — Appearance of cells with fluorescent microscopy.
- **Fig. 1–6** — Principle of confocal microscopy.
- **Fig. 1–9** — Microscopic autoradiography.
- **Fig. 1–11** — Immunocytochemistry techniques.
- **Fig. 1–14** — Interpretation of 3D structures in 2D sections.

# 2 · The Cytoplasm

**Book pages 17–52**
Cell Differentiation · The Plasma Membrane · Cytoplasmic Organelles · The Cytoskeleton · Inclusions

## Learning objectives

By the end of this chapter you should be able to:

1. Describe each membrane-bound organelle and match it to the cell type that needs it most.
2. Explain how proteins are sorted from ribosome to final destination.
3. Distinguish the three cytoskeletal systems by subunit, diameter, and motor protein.
4. Predict the ultrastructure of a cell from its function (steroid synthesis, protein export, phagocytosis, movement).

## The core idea

A cell is a compartmentalization machine. Each organelle is a membrane-bound solution to a specific logistics problem: making protein, making lipid, generating ATP, degrading waste, or moving cargo. Learn the job, and the morphology follows automatically.

## Cell Differentiation

The chapter opens with differentiation, because it explains why one genome produces hundreds of cell shapes. Cells become specialized by **differential gene expression** — the same DNA, different genes switched on. Differentiation is accompanied by changes in the amount and type of organelles, which is exactly what you recognize down a microscope: an erythroblast loses its nucleus, a plasma cell fills with rough ER, a muscle cell fills with contractile filaments.

**Stem cells** retain the capacity to divide and to produce differentiated progeny. They divide **asymmetrically**: one daughter renews the stem-cell pool, the other commits to differentiation. This mechanism explains tissue renewal in the gut, epidermis, and bone marrow (Chapter 3).

## The Plasma Membrane

A **7.5–10 nm phospholipid bilayer** with embedded proteins — the **fluid mosaic model**. Amphipathic phospholipids form the bilayer spontaneously; cholesterol modulates fluidity; glycolipids and glycoproteins project outward as the **glycocalyx**, which mediates cell recognition, adhesion, and immune identity.

| Protein class | Position | Function |
|---|---|---|
| **Integral (transmembrane)** | Span the bilayer | Channels, pumps, receptors, adhesion |
| **Peripheral** | Attached to one face | Signaling, cytoskeletal anchoring, enzymes |

**Lipid rafts** are cholesterol- and sphingolipid-enriched microdomains that concentrate signaling molecules. **Caveolae** are small flask-shaped invaginations involved in uptake and signaling.

### Transport across the membrane

- **Simple diffusion** — small, non-polar molecules down a gradient.
- **Osmosis** — water movement through the bilayer and aquaporins.
- **Facilitated diffusion** — carrier or channel, down the gradient, no energy.
- **Active transport** — against the gradient, ATP-dependent. The **$Na^+/K^+$-ATPase** exchanges 3 $Na^+$ out for 2 $K^+$ in and consumes a large fraction of the cell's resting energy budget.
- **Endocytosis** — **phagocytosis** (large particles, actin-driven), **pinocytosis** (fluid), **receptor-mediated endocytosis** (clathrin-coated pits; the classic example is LDL uptake, and its

failure causes familial hypercholesterolemia).

- **Exocytosis** — vesicle fusion with the plasma membrane; requires SNARE proteins and calcium.
- **Transcytosis** — endocytosis on one face plus exocytosis on the other; used by endothelial cells and antibodies crossing epithelia.

## Cytoplasmic Organelles

**Mitochondria.** Double membrane; the inner membrane folds into **cristae** carrying the electron transport chain and ATP synthase. The matrix contains circular mitochondrial DNA, mitochondrial ribosomes, and calcium granules. Mitochondria are **maternally inherited** and semi-autonomous. Cells with huge energy demand (cardiac muscle, brown fat, proximal tubule, sperm midpiece) are packed with them. **Cells that synthesize steroid hormone have mitochondria with tubular rather than lamellar cristae**, a reliable ultrastructural marker.

**Ribosomes.** 15–20 nm particles of rRNA plus protein, made of a large and small subunit. Free ribosomes make cytosolic protein for internal use; ribosomes bound to the rough ER make protein for secretion, membranes, or lysosomes. **Polyribosomes (polysomes)** are strings of ribosomes on one mRNA, and a **polyribosome** is the functional unit of translation.

**Endoplasmic reticulum (ER).** An anastomosing network of intercommunicating channels and cisternae, continuous with the outer nuclear membrane. **Rough ER (RER)** — flattened cisternae studded with ribosomes. Sites of synthesis of secretory proteins, integral membrane proteins, and lysosomal enzymes. It is the structure responsible for **basophilia** — and in neurons, stacked RER plus polysomes form the **Nissl bodies**.

**Smooth ER (SER)** — tubular and ribosome-free. Three jobs: **lipid and steroid synthesis**, **detoxification** (hepatic cytochrome P450), and **calcium storage** (sarcoplasmic reticulum of muscle). Hepatocytes, adrenal cortical cells, Leydig cells, and the corpus luteum are rich in SER — which is why steroid-secreting cells look pale and foamy rather than blue.

**Golgi apparatus.** Vesicles leave the ER in coats marked by **COPII** and fuse with the cis (forming) face; retrograde transport back to the ER is mediated by **COPI**-coated vesicles. It is a stack of 4–8 flattened cisternae with a **cis (forming) face** receiving vesicles from the ER and a **trans (maturing) face** releasing vesicles. Functions: modification (glycosylation, sulfation, phosphorylation), sorting, and packaging. The **trans-Golgi network** is the dispatch center routing proteins to secretory granules, lysosomes, or the membrane. Cells specialized for secretion (plasma cells, pancreatic acinar cells, goblet cells) have an obvious, often supra-nuclear Golgi.

**Lysosomes.** 0.05–0.5 µm vesicles of acid hydrolases at pH ≈ 5 maintained by a proton pump. **Primary lysosomes** are newly formed; **secondary lysosomes** are the product of fusion with a phagosome or autophagosome; **residual bodies** remain after digestion and accumulate as **lipofuscin**, the "wear-and-tear" pigment of long-lived cells such as cardiac myocytes and neurons.

**Peroxisomes.** Small, membrane-bound oxidative organelles containing **catalase** and oxidases. They perform β-oxidation of very long-chain fatty acids, detoxify ethanol, and destroy $H_2O_2$.

**Proteasomes.** Cytosolic barrel-shaped complexes that degrade ubiquitin-tagged proteins.

## The Cytoskeleton

| Element | Diameter | Protein | Main roles |
|---|---|---|---|
| **Microfilament** | 6–8 nm | **Actin** | Microvilli core, cell cortex, stress fibers, cleavage furrow, muscle contraction |
| **Intermediate filament** | 10 nm | **Keratin, vimentin, desmin, GFAP, neurofilament, lamin** | Mechanical strength; tissue-specific identity |
| **Microtubule** | 25 nm | **Tubulin (α/β dimer)** | Spindle, cilia and flagella, intracellular transport |

**Intermediate filaments are the histologist's best friend**, because each cell type keeps its own: **keratin** in epithelium, **vimentin** in mesenchyme, **desmin** in muscle, **GFAP** in astrocytes, **neurofilament** in neurons, **lamins** in the nucleus. This is why immunohistochemistry can identify the origin of a tumor that looks undifferentiated.

**Microtubules are polar. Kinesin** walks cargo toward the plus end (anterograde, away from the cell body); **dynein** walks toward the minus end (retrograde, back to the cell body). Rabies and herpes viruses exploit dynein to travel to the neuronal soma.

**Cilia and flagella** have a **9 + 2** arrangement with dynein arms; the **basal body** has **9 + 0**. Defective dynein causes **primary ciliary dyskinesia**, of which **Kartagener syndrome** (situs inversus, bronchiectasis, sinusitis, male infertility) is the classic form.

## Inclusions

Non-membranous accumulations of metabolites or pigments: **glycogen** (PAS-positive, liver and muscle), **lipid droplets** (Sudan-positive), **melanin**, **lipofuscin**, **hemosiderin**, and crystals.

**CLINICAL CORRELATION**

Lysosomal storage diseases are the direct consequence of a missing acid hydrolase. In **Tay–Sachs disease** hexosaminidase A is absent, so GM2 ganglioside accumulates in neurons, producing developmental regression, a cherry-red macular spot, and death in early childhood. In **I-cell disease** the enzyme that tags hydrolases with mannose-6-phosphate is defective, so the enzymes are secreted instead of delivered to lysosomes.

**UNDER THE MICROSCOPE**

Learn to read a cell's job from its EM. Abundant RER and a large Golgi = protein secretion. Abundant SER and lipid droplets, mitochondria with tubular cristae = steroid secretion. Many mitochondria and few other organelles = energy transduction. Numerous lysosomes and residual bodies = phagocytosis or long life. Bundles of filaments = mechanical work.

**MEMORY HOOK**

**"Keratins Keep Epithelium; Vimentin is Versatile mesenchyme; Desmin Does muscle; GFAP Glues astrocytes; Neurofilaments are Neuronal."**

**EXAM TIP**

Keep two distinctions straight. The Golgi **modifies and sorts** proteins; the RER makes them. The mitochondrion generates ATP; the lysosome degrades material.

## Chapter 2 — one-line answers

- Rough ER makes membrane and secretory proteins; smooth ER makes lipids and steroids and detoxifies.
- The Golgi is polarized: cis face receives, medial cisternae modify, trans face sorts and packages.
- Mitochondria have outer and inner membranes; cristae shape reflects the cell's energy demand.
- The peroxisome contains catalase and performs β-oxidation of very long-chain fatty acids.
- The lysosome is the acid hydrolase compartment; its failure produces storage diseases.
- Microtubules are 25 nm, actin filaments 7 nm, intermediate filaments 10 nm.
- Kinesin walks toward the microtubule plus end (anterograde); dynein toward the minus end (retrograde).
- Cilia are 9 + 2 and motile; basal bodies are 9 + 0; microvilli have an actin core.
- Apoptosis is programmed, ATP-dependent, and non-inflammatory; necrosis is passive and inflammatory.

## Active recall

1. **Rough versus smooth ER — what distinguishes them functionally?** — Rough ER carries ribosomes and makes membrane and secretory proteins; smooth ER makes lipids and steroids and

detoxifies.

2. **Why is the Golgi described as polarized?** — Cis face receives, medial cisternae modify, trans face packages and sorts.
3. **Which organelle contains catalase, and what does it do?** — The peroxisome; catalase breaks down hydrogen peroxide, and peroxisomes also perform β-oxidation of very long-chain fatty acids.
4. **Which cytoskeletal element forms the mitotic spindle, and which motor walks toward its plus end?** — Microtubules; kinesin moves toward the plus end (anterograde), dynein toward the minus end (retrograde).
5. **What ultrastructural feature identifies a cell that secretes steroid hormone?** — Abundant smooth ER and lipid droplets, with mitochondria that have tubular cristae.

## Exam-style question

**A cell in the adrenal cortex is examined by electron microscopy. Which combination of features is most consistent with its function?**

A. Abundant rough ER and apical zymogen granules
B. Abundant smooth ER, lipid droplets, and tubular mitochondrial cristae
C. Numerous lysosomes and residual bodies
D. Dense arrays of actin filaments and desmosomes

**Answer: B** — Steroid synthesis requires cholesterol (stored as lipid droplets), smooth ER for the enzymatic steps, and mitochondria with tubular cristae.

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 2–1** — Lipids in membrane structure.
- **Fig. 2–6** — Three major forms of endocytosis.
- **Fig. 2–12** — Protein localization and cell morphology.
- **Fig. 2–17** — Lysosomal functions.
- **Fig. 2–23** — Dynamic instability of microtubules.
- **Fig. 2–28** — Cellular inclusions.

# 3 · The Nucleus

**Book pages 53–70**
Components of the Nucleus · The Cell Cycle · Mitosis · Stem Cells & Tissue Renewal · Meiosis · Apoptosis

## Learning objectives

By the end of this chapter you should be able to:

1. Describe the nuclear envelope, lamina, and pore complex as a selective transport system.
2. Explain how DNA is packaged into nucleosomes, chromatin, and chromosomes.
3. Sequence the cell cycle and name the checkpoints that guard it.
4. Contrast mitosis, meiosis, and apoptosis, and explain what goes wrong when each fails.

## The core idea

The nucleus stores the genome, transcribes it, and duplicates it before division. Its morphology is a readout of transcriptional activity: **dark, condensed chromatin means silent genes; pale, dispersed chromatin means active genes.**

## Components of the Nucleus

**Nuclear envelope.** Two membranes separated by a **perinuclear cisterna**, continuous with the rough ER. Beneath the inner membrane lies the **nuclear lamina**, a meshwork of **lamins A, B, and C** that supports the envelope and organizes chromatin. **Mutations in lamin A cause Hutchinson–Gilford progeria.**

**Centrosomes.** Each cell has a pair of **centrioles** — nine microtubular triplets — surrounded by pericentriolar material: the **centrosome**, the microtubule-organizing center that builds the mitotic spindle, and the **basal body** of a cilium.

**Nuclear pores.** 3,000–4,000 per nucleus. Each is a **nuclear pore complex** about 100 nm wide, built from roughly 34 nucleoporins, and can pass up to about 1,000 macromolecules per second. Small molecules diffuse; proteins and RNA require **importins and exportins** and a Ran-GTP gradient.

**Chromatin.** Each human cell holds about **2 meters of DNA — 3.2 billion base pairs** — which must be packaged into a nucleus a few micrometers across. The DNA wraps around a core of **eight histones** (two copies each of H2A, H2B, H3, H4) with **about 150 bp of DNA per nucleosome** (about 1.75 turns around the octamer), plus a larger **H1** histone on the outside: that unit is the **nucleosome**, which in EM looks like **10-nm beads on a string**. Further folding into 30-nm fibers and beyond involves nonhistone scaffold proteins and is less well understood. **Heterochromatin** is condensed and transcriptionally inactive; **euchromatin** is dispersed and active. **Constitutive** heterochromatin is permanently silent; **facultative** heterochromatin can be reactivated. The **Barr body** is a facultatively heterochromatic, inactivated X chromosome — one per somatic cell in a female.

**Nucleolus.** Not membrane-bound; a factory for ribosomal subunits, with a **pars fibrosa** (rRNA transcription) and a **pars granulosa** (ribosome assembly), organized around **nucleolar organizer regions** on the acrocentric chromosomes **13, 14, 15, 21, and 22**. Cells that make large amounts of protein have prominent nucleoli, and malignant cells often do too.

## The Cell Cycle

| Phase | What happens | Duration (typical cycling cell) |
|---|---|---|
| **G1** | Cell grows, checks nutrients and DNA integrity; restriction point | ~10 h |

| Phase | What happens | Duration (typical cycling cell) |
|---|---|---|
| **S** | DNA replication; histones synthesized | ~8 h |
| **G2** | Repair check, prepare for mitosis | ~4 h |
| **M** | Mitosis and cytokinesis | ~1 h |

The durations above are typical of a **rapidly dividing cultured cell**; most adult tissues cycle far more slowly or not at all, so do not transplant these numbers to the body. Progression is driven by **cyclin–CDK complexes**. **p53** halts the cycle at G1 to allow repair and triggers apoptosis if repair fails — hence "guardian of the genome" — and it is mutated in about half of human cancers. **Rb** is the G1/S brake lost in retinoblastoma. **G0** is the quiescent state of cells that have left the cycle.

## Mitosis

- **Prophase** — chromosomes condense, centrioles move to opposite poles, the nuclear envelope begins to break down.
- **Metaphase** — chromosomes align at the equatorial plate; the spindle attaches to kinetochores.
- **Anaphase** — sister chromatids separate and are pulled poleward.
- **Telophase** — nuclear envelopes re-form, chromosomes decondense.
- **Cytokinesis** — an **actin** contractile ring pinches the cell in two.

## Stem Cells & Tissue Renewal

- **Totipotent** — can form the whole organism (zygote, early blastomeres).
- **Pluripotent** — can form all three germ layers (inner cell mass; embryonic stem cells; induced pluripotent stem cells reprogrammed with the Yamanaka factors Oct4, Sox2, Klf4, c-Myc).
- **Multipotent** — form a restricted set of cell types (hematopoietic stem cell).
- **Unipotent** — one cell type.

Tissues are **labile** (continuous turnover — epidermis, gut, marrow), **stable** (low turnover, can be triggered — liver, kidney, fibroblasts), or **permanent** (most neurons, cardiac myocytes, lens fibers), with little or no regenerative capacity under normal conditions.

## Meiosis

Two divisions produce **four haploid cells** from one diploid cell, with **crossing over** between homologous chromosomes during **pachytene of prophase I**, held together by the **synaptonemal complex**. Meiosis generates genetic diversity and halves the chromosome number; **non-disjunction** produces aneuploid gametes, the mechanism behind trisomy 21.

## Apoptosis

Programmed cell death: **caspases** are activated, **cytochrome c** escapes mitochondria, the **Bcl-2 family** decides survival versus death, chromatin condenses, the cell shrinks and buds into **apoptotic bodies** that are phagocytosed **without inflammation**.

| Feature | Apoptosis | Necrosis |
|---|---|---|
| Trigger | Physiological or programmed | Injury, ischemia, toxin |
| Energy | ATP-dependent | Passive |
| Cell size | Shrinks | Swells |
| Nucleus | Condenses, fragments | Pyknosis, karyorrhexis, karyolysis |
| Membrane | Intact, blebbing | Ruptures |
| Inflammation | **Absent** | **Present** |
| Spread | Single cells | Contiguous sheets |

The contrast is a teaching simplification. **Necroptosis** and **pyroptosis** are regulated lytic deaths, and apoptotic cells that are not cleared promptly can provoke inflammation, so the presence or absence of inflammation is a strong rule of thumb rather than an absolute distinction.

**CLINICAL CORRELATION**

**Lamin A mutation** causes progeria with premature aging. Changes in nuclear shape and chromatin texture are also one of the oldest cancer-screening tools: the irregular, hyperchromatic nucleus of a dysplastic cervical cell is what a Pap smear is looking for.

**UNDER THE MICROSCOPE**

An active cell has a pale nucleus with a prominent, reticulated nucleolus and lots of euchromatin. A quiescent cell — a small lymphocyte, a spermatozoon, a superficial keratinocyte — has dense heterochromatin and little visible nucleolus. **Nuclear texture is a functional readout.**

**EXAM TIP**

**One Barr body** in a normal female somatic cell, **zero** in a male, one in Klinefelter syndrome (47, XXY), none in Turner syndrome (45, X). Count Barr bodies and you have counted X chromosomes minus one.

## Chapter 3 — one-line answers

- Nuclear lamina = lamins; lamin A mutation = progeria.
- Nuclear envelope = two membranes + perinuclear cisterna, continuous with rough ER.
- Nuclear pores = 3,000–4,000 per nucleus; importins and exportins, Ran-GTP gradient.
- Nucleosome = eight histones + about 150 bp of DNA; H1 sits outside; 10-nm beads on a string.
- Heterochromatin = condensed and silent; euchromatin = dispersed and active.
- Barr body = facultative heterochromatin of one inactivated X chromosome.
- Nucleolus = rRNA synthesis and ribosome assembly; NORs on 13, 14, 15, 21, 22.
- Cell cycle = G1 → S → G2 → M; DNA replicates in S.
- Cyclin–CDK complexes drive the cycle; p53 arrests at G1; Rb is the G1/S brake.
- p53 = G1 checkpoint and apoptosis; mutated in about half of human cancers.
- Mitosis order = prophase, metaphase, anaphase, telophase; actin ring divides the cell.
- Meiosis = four haploid cells, crossing over in pachytene, held by the synaptonemal complex.
- Non-disjunction = aneuploid gametes, the mechanism of trisomy 21.
- Apoptosis = caspases, cytochrome c, Bcl-2 family, apoptotic bodies, no inflammation.
- Necrosis = swelling, membrane rupture, inflammation.
- Labile tissue turns over constantly; stable tissue can be triggered; permanent tissue has little or no regenerative capacity.

## Active recall

1. **What is a nucleosome?** — About 150 bp of DNA wrapped around an octamer of histones (two each of H2A, H2B, H3, H4), with H1 on the outside.
2. **Heterochromatin versus euchromatin?** — Heterochromatin is condensed and transcriptionally silent; euchromatin is dispersed and active.
3. **What is a Barr body?** — The facultatively heterochromatic, inactivated X chromosome in a female somatic cell.
4. **Which checkpoint does p53 guard, and what happens if repair fails?** — The G1 checkpoint; p53 triggers apoptosis if the DNA cannot be repaired.
5. **Name two ways apoptosis differs from necrosis.** — Apoptosis is ATP-dependent, shrinks the cell, keeps the membrane intact, and causes no inflammation; necrosis is passive, swells and ruptures the cell, and inflames the tissue.

## Exam-style question

**A cell is blocked in metaphase with colchicine. Its chromosomes number 46. Which statement is correct?**

A. It is a haploid gamete
B. It is a diploid somatic or pre-meiotic germ cell
C. It has completed meiosis II
D. It is an anucleate cell

**Answer: B** — Colchicine arrests cells in metaphase, when sister chromatids are still attached; a count of 46 means diploid. Haploid cells would show 23.

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 3–1** — Nuclei of large, active cells.
- **Fig. 3–5** — Nuclear pores.
- **Fig. 3–9** — Chromosome territories of a human fibroblast nucleus.
- **Fig. 3–12** — The cell cycle.
- **Fig. 3–16** — Mitotic cells in adult tissues.
- **Fig. 3–20** — Late apoptosis—apoptotic bodies.

# PART II — THE FOUR BASIC TISSUES

Chapters 4–10 · the four basic tissues and their specializations

# 4 · Epithelial Tissue

**Book pages 71–95**
Characteristic Features of Epithelial Cells · Specializations of the Apical Cell Surface · Types of Epithelia · Transport Across Epithelia · Renewal of Epithelial Cells

## Learning objectives

By the end of this chapter you should be able to:

1. Classify any epithelium by the number of layers and the shape of its surface cells.
2. Match each junctional complex to its molecular components and its function.
3. Explain how microvilli, stereocilia, and cilia differ in structure and purpose.
4. Predict where a given epithelium will be found from the mechanical or transport job it must do.

## The core idea

Epithelial tissues are composed of closely aggregated polyhedral cells adhering strongly to one another and to a thin layer of ECM, forming cellular sheets that line the cavities of organs and cover the body surface. Epithelia (Gr. epi, upon + thele, nipple) line all external and internal surfaces of the body, and **all substances that enter or leave an organ must cross this type of tissue**. Everything else — the number of layers, the shape of the cells, the specializations on the apex — is a solution to one of four problems: **protection, absorption, secretion, or transport**.

**The four basic tissues** (the book's Table 4–1, and the framework for the whole subject):

| Tissue | Cells | Extracellular matrix | Main functions |
|---|---|---|---|
| **Epithelial** | Aggregated polyhedral cells | **Small amount** | Lining of surface or body cavities; glandular secretion |
| **Connective** | Several types of fixed and wandering cells | **Abundant amount** | Support and protection of tissues/organs |
| **Muscle** | Elongated contractile cells | Moderate amount | Strong contraction; body movements |
| **Nervous** | Elongated cells with extremely fine processes | **Very small amount** | Transmission of nerve impulses |

Most organs are divided into the **parenchyma** — the cells responsible for the organ's specialized functions — and the **stroma**, which plays a supporting role. **Except in the brain and spinal cord, the stroma always consists of connective tissue**, which is why the CNS has no connective-tissue framework of its own.

## Characteristic Features of Epithelial Cells

The principal functions of epithelial tissues are **covering, lining, and protecting surfaces** (eg, epidermis), **absorption** (eg, the intestinal lining), and **secretion** (eg, parenchymal cells of glands). Specific cells of certain epithelia may be contractile (**myoepithelial cells**) or specialized sensory cells, such as those of taste buds or the olfactory epithelium.

Cell shape and nuclear shape correspond: **columnar** cells have elongated nuclei, **squamous** cells flattened nuclei, and **cuboidal or pyramidal** cells spherical nuclei. This is the fastest identification rule in the whole subject.

The defining features:

1. Cells are closely apposed with minimal intercellular space.
2. There is a **free (apical) surface** facing air, a lumen, or a cavity.
3. Cells rest on a **basement membrane**.

4. The tissue is **avascular** — nourishment arrives by diffusion from vessels in the underlying connective tissue.
5. Cells are **polarized**: apical, lateral, and basal domains differ in structure and function.

**The basement membrane**, visible with PAS or silver in the light microscope and as two layers in EM:

| Layer | Composition | Made by |
|---|---|---|
| **Lamina lucida** | Laminin, entactin, perlecan | Epithelial cells |
| **Lamina densa** | **Type IV collagen**, heparan sulfate proteoglycan | Epithelial cells |
| **Lamina reticularis** | **Type III (reticular) collagen** | Fibroblasts of underlying connective tissue |

### Cell junctions

| Junction | Transmembrane protein | Cytoskeletal link | Function |
|---|---|---|---|
| **Tight junction** (zonula occludens) | Claudins, occludins | Actin | Seals the intercellular space; creates the **blood–brain barrier** and apical/basal polarity |
| **Adherent junction** (zonula adherens) | **E-cadherin** ($Ca^{2+}$-dependent) | Actin | Mechanical adhesion; the belt that holds sheets together |
| **Desmosome** (macula adherens) | **Desmoglein, desmocollin** | **Keratin** | Spot-welds resisting shear; autoantibodies cause **pemphigus vulgaris** |
| **Gap junction** (nexus) | **Connexins** | None | Direct cytoplasmic communication |
| **Hemidesmosome** | **Integrins** | Keratin → laminin | Anchors basal cells; antibodies cause **bullous pemphigoid** |

The pattern is worth memorizing: **tight seals, adherens belts to actin, desmosomes to keratin, gap junctions for talk, hemidesmosomes for feet.**

## Specializations of the Apical Cell Surface

**Microvilli.** Finger-like extensions with a core of **actin** filaments cross-linked by actin-bundling proteins (villin, fimbrin) and anchored in a **terminal web**. They multiply surface area for absorption. In the proximal tubule and small intestine they form a **brush border** visible as a fuzzy eosinophilic rim. In the epididymis and inner ear, unusually long, stiff microvilli are called **stereocilia** — a misnomer, because they contain no microtubules.

**Cilia.** Motile, **9 + 2** microtubule axoneme with dynein arms, anchored to a **basal body (9 + 0)**. They move mucus, cerebrospinal fluid, and the ovum. A **primary cilium** is a single non-motile 9 + 0 cilium acting as a sensory antenna — its failure underlies polycystic kidney disease.

**Basal infoldings.** Deep folds of the basal membrane packed with mitochondria — the "striated" appearance in the ducts of salivary glands and the distal tubule of the kidney. They are the morphological signature of **active ion transport**.

## Types of Epithelia

| Type | Structure | Location | Function |
|---|---|---|---|
| **Simple squamous** | One flat layer, central nucleus | Alveoli, endothelium, mesothelium, Bowman capsule | Diffusion, filtration, lubrication |
| **Simple cuboidal** | One layer, nuclei round and central | Kidney tubules, small ducts, thyroid follicles, surface of ovary | Secretion, absorption, conduit |

| Type | Structure | Location | Function |
|---|---|---|---|
| **Simple columnar** | One layer, tall cells | Stomach, intestine, gallbladder, uterine tube | Absorption, secretion, protection |
| **Pseudostratified ciliated columnar** | Single layer with nuclei at different heights; cilia and goblet cells | Trachea, bronchi, nasal cavity | Mucus clearance ("respiratory epithelium") |
| **Stratified squamous, keratinized** | Many layers, outermost anucleate keratin | Epidermis | Protection against abrasion and dehydration |
| **Stratified squamous, non-keratinized** | Many layers, outermost live nucleated cells | Oral cavity, esophagus, vagina, anal canal, cornea | Protection in a wet environment |
| **Stratified cuboidal / columnar** | Two or more layers | Ducts of sweat glands, conjunctiva | Conduit with protection |
| **Transitional (urothelium)** | Stratified, with dome-shaped **umbrella cells** whose apical membrane is stiffened by **uroplakin** plaques | Renal pelvis, ureter, bladder, proximal urethra | Stretchable barrier to urine |
| **Epithelioid** | Cells in clusters with no free surface | Adrenal cortex, islets of Langerhans, liver cords | Endocrine secretion |

Two naming traps to avoid. **Endothelium and mesothelium are both simple squamous epithelium** — one lines blood vessels, the other body cavities; they are not separate tissue types. And **pseudostratified** epithelium is truly a **single layer** in which every cell touches the basement membrane, even though the nuclei sit at different levels.

## Transport Across Epithelia

An epithelium that moves material from one side to the other is built like a machine: **entry at the apical face, exit at the basal face**, with the tight junction preventing back-leak and mitochondrial-rich basal infoldings powering the pumps. In the intestinal enterocyte, glucose enters apically via **$Na^+$-glucose cotransport (SGLT1)** and exits basolaterally via **GLUT2**; the $Na^+$ gradient is maintained by basolateral $Na^+/K^+$-ATPase. This is **transcellular transport**. **Paracellular** transport passes between cells and is limited by tight junctions.

## Renewal of Epithelial Cells

Epithelia wear out. They are replaced from **stem cells** whose location varies with the tissue: the **basal layer** of stratified squamous epithelium, the **crypt base** of the intestine, the **isthmus** of the gastric gland, and the basal layer of the epidermis. Renewal rate is a diagnostic index: the intestinal epithelium turns over in days, the epidermis in weeks. Because the stem cells rest on the basement membrane, an epithelium can regenerate completely after superficial injury — but not after the basement membrane itself is destroyed.

## Glands

- **Exocrine** — connected to a surface by a duct. **Endocrine** — ductless, secreting into the blood.
- **Unicellular** — the goblet cell (mucin; empty cup-shaped apex after secretion).
- **Simple** — one unbranched duct (simple tubular = intestinal crypt; simple coiled tubular = eccrine sweat gland; simple branched acinar = sebaceous gland).
- **Compound** — branched duct (salivary glands, pancreas, mammary gland).
- **Mechanism: merocrine** (exocytosis — pancreas, sweat glands), **apocrine** (membrane and cytoplasm bud off — the lipid component of mammary secretion), **holocrine** (whole cell disintegrates — sebaceous gland).

**Myoepithelial cells** sit between secretory cells and the basement membrane and contract to expel product — visible in mammary and salivary glands.

**CLINICAL CORRELATION**
Epithelial junctions are the target of autoimmune blistering. In **pemphigus vulgaris**, antibodies against **desmoglein** break desmosomes, so the epidermis detaches within the epithelium and thin-walled blisters form. In **bullous pemphigoid**, antibodies against **hemidesmosomal** antigens split the epidermis from the basement membrane and the blister is thick-walled and tense. The level of the split is dictated by which junction fails.

**UNDER THE MICROSCOPE**
Count nuclei, not cells. More than one nuclear level with a flat continuous surface = stratified. Then the top layer: **flat and anucleate** = keratinized; **flat with nuclei** = non-keratinized; **dome-shaped and large** = transitional; **tall with cilia** = respiratory.

**MEMORY HOOK**
**"Simple for exchange, stratified for stress, transitional for stretch."** For cilia, remember **9 + 2 moves, 9 + 0 anchors.**

**EXAM TIP**
Where is ciliated simple columnar epithelium found? The **uterine tube** — cilia move the ovum — and parts of the respiratory tract, where the epithelium is pseudostratified. Students often wrongly assign all ciliated epithelium to the lung.

## Chapter 4 — one-line answers

- Epithelium sits on a basal lamina, is avascular, and is always adjacent to connective tissue.
- Simple = one layer (exchange, absorption, secretion); stratified = protection; pseudostratified = all cells touch the basement membrane.
- Squamous = flat; cuboidal = height ≈ width; columnar = height > width.
- Keratinized stratified squamous = dry, abrasive surfaces; nonkeratinized = wet surfaces (mouth, esophagus, vagina).
- Urothelium (transitional) = dome-shaped umbrella cells; stretches.
- Tight junction (zonula occludens) = seal and fence; it creates apical–basal polarity.
- Zonula adherens anchors actin; desmosomes anchor intermediate filaments; gap junctions couple cells.
- Hemidesmosomes attach the basal cell to the basal lamina.
- Basal lamina = lamina lucida + lamina densa (type IV collagen, laminin); the reticular lamina (type III) belongs to connective tissue.
- Microvilli = actin core, increase surface; stereocilia = long, stiff microvilli; cilia = 9 + 2 microtubules, motile.
- Basal bodies anchor cilia and have a 9 + 0 arrangement.
- Apical specializations serve absorption or movement; basal specializations anchor and receive signals.
- Epithelia renew from stem cells: labile (gut, skin) versus stable (liver, kidney).

## Active recall

1. **How do you classify an epithelium in one pass?** — First count the layers (simple, stratified, pseudostratified), then name the surface cell shape (squamous, cuboidal, columnar).
2. **What does a tight junction do besides sealing?** — It fences membrane proteins, creating and maintaining apical–basal polarity.
3. **Which junction attaches a cell to the basal lamina?** — The hemidesmosome.
4. **Microvillus, stereocilium, cilium — core structure?** — Microvillus: actin core. Stereocilium: long, stiff actin bundle. Cilium: 9 + 2 microtubules anchored by a 9 + 0 basal body.
5. **Why is urothelium suited to the bladder?** — Umbrella cells with uroplakin plaques and tight junctions form an osmotic barrier, and the epithelium thins as the bladder fills.

## Exam-style question

**An epithelium lines a dry, abrasion-prone surface. Which description fits it?**

A. Simple columnar with goblet cells
B. Stratified squamous keratinized
C. Simple squamous with fenestrations
D. Pseudostratified ciliated columnar

**Answer: B** — Keratinized stratified squamous epithelium is the body's answer to mechanical wear plus water loss (skin, hard palate, gingiva).

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 4–1** — Epithelia and adjacent connective tissue.
- **Fig. 4–6** — View of a tight junction after cryofracture.
- **Fig. 4–12** — Simple squamous epithelium.
- **Fig. 4–17** — Pseudostratified epithelium.
- **Fig. 4–23** — Apocrine secretion in the mammary gland.
- **Fig. 4–28** — Features of absorptive cells.

# 5 · Connective Tissue

**Book pages 96–121**
Cells of Connective Tissue · Fibers · Ground Substance · Types of Connective Tissue

## Learning objectives

By the end of this chapter you should be able to:

1. Name the cells, fibers, and ground substance of connective tissue and give each one's job.
2. Match collagen types I–IV to their locations and to the diseases caused by their failure.
3. Distinguish loose from dense connective tissue, and regular from irregular, by collagen arrangement.
4. Explain how connective tissue repairs and why scurvy and Ehlers–Danlos syndrome look the way they do.

## The core idea

Connective tissue is the tissue whose defining feature is **abundance of extracellular matrix**. The cells are widely separated because the matrix — not the cells — does the mechanical work. Name the matrix and you have named the tissue's job: collagen for tension, elastin for recoil, ground substance for compression and diffusion.

## Cells of Connective Tissue

| Cell | Origin | Function | Identifying feature |
|---|---|---|---|
| **Fibroblast / fibrocyte** | Mesenchyme | Synthesizes collagen, elastin, GAGs; **fibrocyte** is the quiescent form | Elongated, pale, abundant RER; **myofibroblast** also has actin for wound contraction |
| **Macrophage** | Monocyte | Phagocytosis, antigen presentation, cytokines | Irregular, abundant lysosomes; **Kupffer cell** in liver, **microglia** in CNS, **osteoclast** in bone (a fused macrophage) |
| **Mast cell** | Bone marrow | Releases **histamine** and **heparin**; IgE-mediated allergy | Round, many dark granules; degranulates on IgE cross-linking |
| **Plasma cell** | B lymphocyte | Secretes antibody | **Clock-face nucleus**, eccentric, perinuclear clear zone (Golgi) |
| **Adipocyte** | Mesenchyme | Store fat | Signet-ring when unilocular |
| **Pericyte** | Mesenchyme | Contractile support of capillaries | On the abluminal surface of capillaries |

## Fibers

| Fiber | Protein | Properties | Where |
|---|---|---|---|
| **Collagen** | Tropocollagen (types I–XXVIII) | High tensile strength, inelastic | Everywhere; the most abundant protein in the body |
| **Reticular** | **Type III collagen** | Fine, branched; silver-staining | Lymphoid organs, bone marrow, liver, endocrine glands |
| **Elastic** | **Elastin** core + **fibrillin** microfibrils | Stretch and recoil | Aorta, skin, lung, elastic cartilage, ligaments |

| Type | Form | Main location | Disease if defective |
|---|---|---|---|
| **I** | Thick fibrils | Bone, dermis, tendon, cornea | **Osteogenesis imperfecta** |

| Type | Form | Main location | Disease if defective |
|---|---|---|---|
| **II** | Fibrils | Hyaline and elastic cartilage, vitreous | Chondrodysplasias |
| **III** | Reticular fibers | Lymph node, spleen, bone marrow, skin | **Ehlers–Danlos vascular type** |
| **IV** | Network | **Basement membrane** | Alport syndrome |
| **V** | Fibrils with type I | Cornea, bone, placenta | Ehlers–Danlos |

**Collagen synthesis in five steps** — a reliable structured-question answer:

1. **Translation** on the RER as preprocollagen.
2. **Hydroxylation** of proline and lysine — **requires vitamin C**.
3. **Glycosylation** in the ER and Golgi.
4. **Procollagen** is secreted and the propeptides cleaved → **tropocollagen**.
5. **Fibril assembly** with cross-links formed by **lysyl oxidase**, which requires **copper**.

This pathway explains two diseases. Without vitamin C, hydroxylation fails and the result is **scurvy** — bleeding gums, poor wound healing, perifollicular hemorrhage. Without copper, cross-linking fails, as in **Menkes disease**. In **Ehlers–Danlos syndrome** the defect is in collagen processing or type; in **Marfan syndrome** the defect is in **fibrillin**, so the aorta dilates and the lens dislocates.

## Ground Substance

An amorphous, hydrated gel of **glycosaminoglycans (GAGs)**, **proteoglycans**, and **glycoproteins**, which resists compression, lubricates, and controls diffusion.

- **GAGs** (also called mucopolysaccharides) are long polymers of repeating disaccharide units, usually with many negative charges that attract water: **hyaluronic acid** (the only **unsulfated** one, not attached to protein, and the backbone of the gel) and the **sulfated GAGs** — chondroitin sulfate, keratan sulfate, dermatan sulfate, and heparan sulfate.
- **Proteoglycans** are a core protein with GAG side chains — the classic is **aggrecan**, a very large molecule with a **250-kDa core protein** heavily bound with chondroitin and keratan sulfate and joined to hyaluronan by a **link protein**. Its water-binding is why cartilage resists compression.
- **Perlecan** is the key proteoglycan of **all basal laminae** — the one name that links ground substance to the basement membrane.
- **Glycoproteins** — **fibronectin** (binds cells to matrix via integrins, important in wound healing) and **laminin** (basement membrane).

## Types of Connective Tissue

- **Loose (areolar)** — abundant ground substance, scattered fibers, many cells. Fills spaces, surrounds vessels, underlies epithelia.
- **Dense irregular** — thick collagen bundles in many directions; dermis, organ capsules. Resists stress from all directions.
- **Dense regular** — parallel collagen bundles; **tendon** and ligament. Resists stress in one direction.
- **Elastic** — many elastic fibers; elastic ligament, vocal cord, large arteries.
- **Reticular** — type III collagen network; lymphoid organs and marrow.
- **Mucous (Wharton's jelly)** — gelatinous, in the umbilical cord.
- **Embryonic** — mesenchyme.

Besides fibroblasts, the resident population includes **adipocytes**, **macrophages**, mast cells, and the **leukocytes** that migrate in (especially **lymphocytes** and plasma cells in chronic inflammation) — which is why the book lists several types of fixed and wandering cells.

**Repair.** Inflammation → **granulation tissue** (proliferating capillaries, fibroblasts, macrophages) → collagen deposition → **scar**. **Primary intention** occurs in clean apposed wounds; **secondary intention** in larger, open wounds with more granulation and contraction.

**CLINICAL CORRELATION**

**Scurvy** is the purest histology-to-clinic link in the chapter: inadequate hydroxylation produces defective collagen, which produces fragile capillaries (bleeding gums, petechiae) and failed wound healing. **Scleroderma** is the opposite failure — excess collagen in skin and internal organs. **Keloid** is an overgrowth of dense collagen beyond the original wound margin, common in darker skin.

**UNDER THE MICROSCOPE**

**Collagen is pink, wavy, and in bundles; elastin is sparse, often darker, and best shown by orcein.** A plasma cell has an **eccentric clock-face nucleus with a pale perinuclear halo**; a mast cell's granules stain metachromatically; the eosinophil's bilobed nucleus is unmistakable.

**MEMORY HOOK**

**"1 bone, 2 cartilage, 3 reticulin, 4 basement membrane."**

**EXAM TIP**

The distinction that matters most here is **dense regular versus dense irregular**, and the answer is one word: **direction**. Regular = parallel = tendon = one axis of force. Irregular = random = dermis = many axes.

## Chapter 5 — one-line answers

- Connective tissue = cells + fibers + ground substance; it is vascular except in cartilage.
- Type I collagen = bone, tendon, dermis, scar. Type II = hyaline and elastic cartilage. Type III = reticulin. Type IV = basal lamina.
- Collagen synthesis needs vitamin C for proline and lysine hydroxylation; deficiency = scurvy.
- Elastic fibers = elastin core + fibrillin microfibrils; orcein stains them.
- Ground substance = GAGs and proteoglycans; **hyaluronan is the only unsulfated GAG**.
- Fibroblast makes collagen; myofibroblast contracts (wound closure).
- Plasma cell = eccentric clock-face nucleus, perinuclear Golgi halo, antibody factory.
- Mast cell = heparin, histamine, proteases; metachromatic granules; IgE receptor.
- Macrophage = monocyte-derived phagocyte; foreign-body and Langhans giant cells are fused macrophages.
- Adipocytes, mast cells, and leukocytes are resident connective tissue cells.
- Loose (areolar) tissue = cells and vessels; dense irregular = dermis; dense regular = tendon.
- Reticular tissue = type III collagen framework of lymphoid organs and marrow.
- Edema = excess fluid in ground substance; it follows increased pressure, leaky vessels, or low albumin.

## Active recall

1. **Which collagen type forms the basal lamina?** — Type IV (with laminin), over a reticular lamina of type III.
2. **Which collagen type is defective in osteogenesis imperfecta?** — Type I.
3. **What is the only unsulfated GAG?** — Hyaluronan.
4. **How do you tell dense regular from dense irregular connective tissue?** — Direction: regular has parallel bundles (tendon), irregular has randomly oriented bundles (dermis).
5. **What does a plasma cell look like, and what does it make?** — Eccentric clock-face nucleus with a pale perinuclear Golgi halo; it secretes antibody.

## Exam-style question

**A patient has bleeding gums, poor wound healing, and perifollicular hemorrhages. Which step in collagen synthesis is defective?**

A. Transcription of collagen genes
B. Hydroxylation of proline and lysine
C. Cleavage of propeptides
D. Cross-linking by lysyl oxidase

**Answer: B** — Vitamin C is the cofactor for prolyl and lysyl hydroxylase; without hydroxylation the triple helix is unstable — scurvy.

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 5–1** — Embryonic mesenchyme.
- **Fig. 5–5** — Mast cells.
- **Fig. 5–9** — The collagen subunit.
- **Fig. 5–14** — Formation of elastic fibers.
- **Fig. 5–18** — Movement of fluid in connective tissue.
- **Fig. 5–22** — Mucoid tissue.

# 6 · Adipose Tissue

**Book pages 122–128**
White Adipose Tissue · Brown Adipose Tissue

## Learning objectives

By the end of this chapter you should be able to:

1. Distinguish white and brown adipocytes by light and electron microscopy.
2. Explain how white adipose tissue works as an endocrine organ.
3. Explain the mechanism of non-shivering thermogenesis and where brown fat is found.
4. Relate adipose storage limits to obesity, lipotoxicity, and lipodystrophy.

## The core idea

Adipose tissue is an endocrine and metabolic organ disguised as a filling tissue. White fat stores energy; brown fat burns it for heat. Both are built from the same cell with a completely different internal architecture.

## White Adipose Tissue

**Unilocular.** Each white adipocyte is a very large cell, **50–150 µm** in diameter, containing one enormous lipid droplet that displaces the cytoplasm to a thin rim and flattens the nucleus against the plasmalemma — the **signet-ring** appearance. Adipose tissue is normally **15–20% of body weight in men** and somewhat more in women. The lipid dissolves during routine processing, so on an H&E slide the cell appears as an empty circle with a thin border.

Functions: energy storage, insulation, cushioning of organs (retro-orbital, perirenal, mesenteric), and **endocrine signaling**.

**Where the stored fat comes from — three sources, straight from the book:** dietary fat carried as **chylomicrons**; triglycerides made in the **liver** and carried as **VLDLs**; and fatty acids and glycerol **synthesized by the adipocytes themselves**. Mobilization is by **hormone-sensitive lipase**, activated by **norepinephrine** from the adrenal gland and sympathetic nerves, and by growth hormone; **insulin inhibits** that lipase and stimulates lipid synthesis. The cells are supported by **reticular fibers**, with connective tissue septa dividing the tissue into lobules.

**Adipose-derived factors** — the exam list (note that the last one is an enzyme, not an adipokine):

- **Leptin** — from adipocytes, acts on the hypothalamus to reduce appetite.
- **Adiponectin** — increases insulin sensitivity; low in obesity.
- **Resistin, TNF-α, IL-6** — pro-inflammatory, part of why obesity is a chronic inflammatory state.
- **Aromatase** — an enzyme, not a cytokine: it converts androgens to estrogens, making adipose tissue a peripheral estrogen source (the reason obesity raises estrogen levels).

## Brown Adipose Tissue

**Multilocular.** Many small lipid droplets, a **central round nucleus**, and — the diagnostic feature — **numerous mitochondria with abundant cristae** containing **uncoupling protein 1 (UCP1, thermogenin)**, which short-circuits the proton gradient so energy is released as heat instead of ATP.

Found in the fetus and newborn (interscapular, axillary, perirenal, along the great vessels), much reduced in adults. Sympathetic innervation drives it. **"Beige" adipocytes** appear within white fat after cold exposure.

| Feature | White | Brown |
|---|---|---|
| Lipid droplets | One (unilocular) | Many (multilocular) |

| Feature | White | Brown |
|---|---|---|
| Nucleus | Peripheral | Central |
| Mitochondria | Few | **Many, cristae-rich** |
| UCP1 | Absent | **Present** |
| Main role | Energy storage, endocrine | **Non-shivering thermogenesis** |
| Sympathetic innervation | Sparse | Rich |

**CLINICAL CORRELATION**

In **lipodystrophy** fat is absent or redistributed, and the metabolic consequences are paradoxical: severe insulin resistance with hypertriglyceridemia, because the remaining fat cannot store lipid safely. In **obesity** the same storage limit is approached, so lipid spills into liver (steatosis), muscle, and pancreas — **lipotoxicity**. Brown fat is a therapeutic target: activating it increases energy expenditure, and PET imaging shows active depots in adults.

**UNDER THE MICROSCOPE**

**One big hole per cell with a flattened nucleus at the rim = white. Many small holes with a round central nucleus and granular eosinophilic cytoplasm = brown.**

**EXAM TIP**

Do not confuse adipose tissue with **steatosis**. Adipocytes are a normal tissue with a capsule and vessels; steatosis is lipid accumulating inside hepatocytes or myocytes as vacuoles.

## Chapter 6 — one-line answers

- White adipocyte: **one large lipid droplet**, nucleus flattened at the rim, few mitochondria.
- Brown adipocyte: **many small droplets**, round central nucleus, many cristae-rich mitochondria.
- White fat = energy storage plus an **endocrine organ**: leptin, adiponectin, and other adipokines.
- Brown fat = **non-shivering thermogenesis** through **UCP1**, which uncouples the proton gradient.
- Brown fat is richest in the newborn (neck, back, mediastinum) and persists in adult neck and supraclavicular depots.
- Beige/brite adipocytes are an inducible, brown-like population within white fat.
- Adipose tissue is vascular and innervated; white fat's sympathetic supply is sparse, brown fat's is rich.
- Obesity is expansion by hypertrophy and hyperplasia; the storage limit produces lipotoxicity.
- Lipodystrophy = absent or redistributed fat with severe insulin resistance.
- Leptin signals energy stores to the hypothalamus; adiponectin increases insulin sensitivity.

## Active recall

1. **One histological difference between white and brown fat?** — White: one large droplet, peripheral flattened nucleus. Brown: many small droplets, central round nucleus, eosinophilic granular cytoplasm.
2. **What is UCP1 and where does it act?** — Uncoupling protein 1 in the inner mitochondrial membrane; it lets protons leak back without ATP synthesis, releasing heat.
3. **Name two adipokines.** — Leptin (signals stored energy) and adiponectin (increases insulin sensitivity).
4. **When is brown fat most abundant?** — In the newborn (neck, back, mediastinum); it persists in adult neck and supraclavicular depots.
5. **What is lipotoxicity?** — When adipose storage capacity is exceeded, lipid accumulates in liver, muscle, and pancreas, impairing their function.

## Exam-style question

**A newborn's PET scan shows active supraclavicular fat that burns energy without shivering. Which feature explains this?**

A. Unilocular droplets and few mitochondria
B. Multilocular droplets, abundant mitochondria, and UCP1
C. Dense collagen bundles and few cells
D. Abundant rough ER and zymogen granules

**Answer: B** — Brown adipocytes are built for heat production: many small droplets for rapid lipolysis, cristae-rich mitochondria, and UCP1.

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 6–1** — White adipose tissue.
- **Fig. 6–2** — Lipid storage and mobilization from adipocytes.
- **Fig. 6–3** — Development of white and brown fat cells.
- **Fig. 6–4** — Brown adipose tissue.

# 7 · Cartilage

**Book pages 129–137**
Hyaline Cartilage · Elastic Cartilage · Fibrocartilage · Cartilage Formation, Growth, & Repair

## Learning objectives

By the end of this chapter you should be able to:

1. Distinguish hyaline, elastic, and fibrocartilage by matrix, cells, and perichondrium.
2. Explain why cartilage heals so poorly and what tissue replaces it.
3. Describe appositional and interstitial growth and when each is possible.
4. Identify cartilage types in a micrograph and relate each to its mechanical role.

## The core idea

Cartilage is a **tough, resilient** connective tissue in which the extracellular matrix — collagen, proteoglycans, and water — is the bulk of the tissue; **chondrocytes** occupy only a small fraction of its volume, sitting in lacunae. **Aggrecan** binds a large amount of water, giving the matrix its compressive resilience. Cartilage supports soft tissues, notably in the respiratory tract, and provides **cushioned, low-friction surfaces in joints**. It is **avascular and aneural** — no blood vessels, lymphatics, or nerves — and is usually surrounded by a vascularized **perichondrium** of dense connective tissue whose fibroblasts form its outer fibrous layer. The absence of vessels explains both its durability under load and its poor healing.

## The shared architecture

- Cells: **chondroblasts** (immature, at the surface) and **chondrocytes** (mature, in **lacunae**).
- Matrix: **territorial matrix** immediately around the lacuna (rich in proteoglycan, basophilic) and **interterritorial matrix** between cells (more collagen, eosinophilic).
- **Isogenous groups** — clusters of 2–8 chondrocytes from one parent cell, indicating recent **interstitial growth**.
- **Perichondrium** — fibrous outer layer plus chondrogenic inner layer; present on most cartilage, **absent on fibrocartilage and on articular surfaces**.

## Hyaline Cartilage

The commonest type and the template for most of the fetal skeleton. Glassy, homogeneous matrix; **collagen makes up 40% of hyaline cartilage's dry weight**, all of it embedded in a hydrated gel of proteoglycans and structural glycoproteins, with more proteoglycan and less collagen immediately around the lacunae (the **territorial matrix**). Perichondrium is present except on **articular surfaces** and the epiphyses of growing long bones, where it forms the smooth, low-friction bearing surface of joints. Found in the **trachea, bronchi, nose, ribs, epiphyseal plate, and articular surfaces**.

## Elastic Cartilage

Type II collagen plus a network of **elastic fibers**, giving flexibility with recoil. Found in the **epiglottis, pinna, external auditory canal, and Eustachian tube** — that is, everywhere something must bend and spring back.

## Fibrocartilage

**Type I and type II collagen** in coarse bundles, with chondrocytes arranged in **parallel rows** between them. **No perichondrium.** Found where compressive and tensile forces meet: **intervertebral disc, symphysis pubis, menisci, temporomandibular joint**. The disc is the perfect example, with an inner **nucleus pulposus** (notochord remnant, rich in hyaluronic acid) and an outer **annulus fibrosus** of concentric fibrocartilage lamellae.

| Feature | Hyaline | Elastic | Fibrocartilage |
|---|---|---|---|
| Collagen | **Type II** | Type II + **elastic fibers** | **Type I + II** |
| Cells | Isogenous groups | More numerous | **Rows** between bundles |
| Perichondrium | Present (not on articular surfaces) | Present | **Absent** |
| Locations | Trachea, bronchi, nose, ribs, epiphyseal plate, joints | Epiglottis, pinna, auditory canal | Disc, symphysis pubis, menisci |
| Function | Smooth bearing surface, template for bone | Flexible support | Shock absorption, tensile strength |

## Cartilage Formation, Growth, & Repair

- **Appositional growth** — new cartilage added at the periphery by chondroblasts of the perichondrium. Increases width.
- **Interstitial growth** — chondrocytes divide within lacunae and secrete matrix, pushing cells apart. Increases length; possible only in young cartilage.

Because cartilage is avascular, nutrition depends on diffusion from the perichondrium or synovial fluid. This limits repair: defects fill with **fibrocartilage** rather than hyaline cartilage, and that patch is mechanically inferior.

**CLINICAL CORRELATION**

**Osteoarthritis** is a failure of the chondrocyte to maintain matrix: proteoglycans are depleted, water content rises, collagen fibrils fragment, and the surface fissures. Because hyaline cartilage has no vessels and no nerves, pain comes late and from the underlying bone, capsule, and synovium. **Rheumatoid arthritis** is different: an autoimmune synovitis destroys cartilage from the surface inward, producing a pannus.

**UNDER THE MICROSCOPE**

Identify cartilage by the **lacuna**: a cell alone in a clear space within a solid matrix. Then the matrix: **glassy purple with cells in clusters = hyaline; wavy dark fibers = elastic; thick parallel pink bundles with cells in rows = fibrocartilage.**

**EXAM TIP**

Two commonly confused points: **hyaline cartilage contains type II collagen, not type I**, and **fibrocartilage has no perichondrium**, which is why disc injuries persist.

## Chapter 7 — one-line answers

- Cartilage = chondrocytes in lacunae + solid matrix; no blood vessels, no nerves.
- Hyaline cartilage = **type II collagen** + aggrecan; glassy; perichondrium present except at articular surfaces.
- Elastic cartilage = type II collagen + **elastic fibers**; perichondrium present; epiglottis, auricle, auditory tube.
- Fibrocartilage = **type I + type II collagen**, rows of chondrocytes, **no perichondrium**; discs, menisci, pubic symphysis.
- Territorial matrix surrounds each lacuna; interterritorial matrix lies between groups.
- Isogenous groups = daughter cells from recent division, still in one lacuna.
- Appositional growth = from perichondrium, adds width.
- Interstitial growth = division inside lacunae, adds length, young cartilage only.
- Articular cartilage is hyaline **without perichondrium** and heals poorly.
- Cartilage repair fills with fibrocartilage, not hyaline.
- Osteoarthritis = matrix failure; rheumatoid arthritis = synovial inflammation with pannus.

- Chondroblasts make matrix; chondrocytes maintain it.

## Active recall

1. **Which cartilage has no perichondrium, and why does that matter?** — Fibrocartilage; without a perichondrial source of new chondroblasts, disc and meniscal injuries heal poorly.
2. **Which collagen types are in hyaline and fibrocartilage?** — Hyaline: type II. Fibrocartilage: type I and type II.
3. **What is an isogenous group?** — Daughter chondrocytes from a recent division, still sharing one lacuna.
4. **Appositional versus interstitial growth?** — Appositional adds cartilage at the periphery and increases width; interstitial growth divides chondrocytes inside lacunae, increases length, and occurs only in young cartilage.
5. **Why is articular cartilage a poor repairer?** — It is avascular, aneural, and lacks perichondrium, so nutrition depends on synovial fluid diffusion.

## Exam-style question

**A tissue shows rows of chondrocytes between thick parallel bundles of type I collagen and has no perichondrium. Which is it?**

A. Hyaline cartilage
B. Elastic cartilage
C. Fibrocartilage
D. Decalcified bone

**Answer: C** — Rows of chondrocytes plus type I bundles and no perichondrium identify fibrocartilage (intervertebral disc, meniscus, pubic symphysis).

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 7–1** — Distribution of cartilage in adults.
- **Fig. 7–2** — The structure of cartilage matrix and cells.
- **Fig. 7–3** — Hyaline cartilage.
- **Fig. 7–5** — Fibrocartilage.
- **Fig. 7–6** — Chondrogenesis.
- **Fig. 7–7** — Chondrocytes in growing cartilage.

# 8 · Bone

**Book pages 138–160**
Bone Cells · Bone Matrix · Periosteum & Endosteum · Types of Bone · Osteogenesis · Bone Remodeling & Repair · Metabolic Role of Bone · Joints

## Learning objectives

By the end of this chapter you should be able to:

1. Describe the cells and matrix of bone and how each contributes to strength and turnover.
2. Distinguish woven from lamellar bone, and compact from spongy bone.
3. Sequence endochondral and intramembranous ossification.
4. Explain remodeling, calcium homeostasis, and fracture repair.

## The core idea

Bone is the main constituent of the adult skeleton: it gives **solid support**, protects vital organs of the cranial and thoracic cavities, encloses **medullary cavities** where blood cells form, and acts as a **reservoir of calcium and phosphate**, storing or releasing them to keep body fluid concentrations constant. It also forms a system of **levers** that multiply the force of muscle contraction into movement. Its stiffness comes from mineral, its toughness from collagen, and its capacity to adapt from four cell types that continuously demolish and rebuild it — and most skeletal disease reflects failure of one of those cells, or of the balance between them.

## Bone Cells

| Cell | Origin | Function | Marker / feature |
|---|---|---|---|
| **Osteoprogenitor** | Mesenchyme | Divide and differentiate into osteoblasts | Periosteum (inner layer), endosteum |
| **Osteoblast** | Osteoprogenitor (periosteum, endosteum) | Secrete **osteoid**, rich in **type I collagen** plus proteoglycans; mineralize it; regulate osteoclasts via **RANKL** | Cuboidal, basophilic, cover existing bone surfaces; **alkaline phosphatase** high |
| **Osteocyte** | Osteoblast that becomes embedded | Maintain matrix; sense mechanical load | In **lacunae**, connected by **canaliculi** via gap junctions |
| **Osteoclast** | **Monocyte/macrophage lineage** | **Resorb bone** | Large, multinucleated, **ruffled border**, resorption bay (**Howship lacuna**); **TRAP**, carbonic anhydrase II, $H^+$-ATPase |

As osteoid calcifies and hardens it **entraps some osteoblasts**, which then differentiate into osteocytes sitting in **lacunae** between the lamellae, with cytoplasmic processes in **canaliculi**. Osteocytes maintain the calcified matrix and are fed from microvasculature in the **central canals of osteons** through those canaliculi.

**The osteoclast is the exception.** Every other bone cell is mesenchymal; the osteoclast is **monocyte-derived** and is the cell required for bone remodeling. It dissolves mineral by pumping $H^+$ and digests collagen with cathepsin K.

**The RANK–RANKL–OPG axis:** osteoblasts express **RANKL**, which binds **RANK** on osteoclast precursors and drives differentiation; they also secrete **osteoprotegerin (OPG)**, a decoy receptor that blocks RANKL. **PTH** shifts the balance toward resorption; **estrogen** suppresses it.

## Bone Matrix

- **Inorganic (~65%)** — calcium phosphate as **hydroxyapatite** crystals; compressive strength.
- **Organic (~35%) — type I collagen** (90% of the organic fraction), osteocalcin, osteonectin, osteopontin, proteoglycans, growth factors; tensile strength.

Remove the mineral and bone becomes flexible; remove the collagen and it becomes brittle. **Bone is strong because it is both at once.**

## Periosteum & Endosteum

**Periosteum** — dense connective tissue with a mainly **fibrous layer** over a more **cellular** layer; small vessels penetrate the matrix from it. Absent on articular surfaces and at tendon insertions. **Endosteum covers all trabeculae around the marrow cavities** — not merely the shaft. Both are sources of osteoprogenitor cells for growth and repair.

## Types of Bone

**Compact (cortical)** bone — dense, on the surface; built of **osteons (Haversian systems)**: concentric lamellae around a central **Haversian canal** containing vessels and nerves, with **osteocytes** in lacunae linked by canaliculi, and **Volkmann canals** running transversely.

**Spongy (cancellous)** bone — a network of **trabeculae** with marrow between them, oriented along lines of mechanical stress. No true osteons.

**Woven versus lamellar.** Woven bone is disorganized and produced rapidly in the fetus, fracture callus, and disease (Paget, osteosarcoma). Lamellar bone is ordered and mechanically superior; adult bone is lamellar but always being remodelled through **basic multicellular units (BMUs)**.

## Osteogenesis

**Intramembranous ossification** — mesenchyme condenses and differentiates directly into osteoblasts, depositing bone in a fibrous membrane. Produces the **flat bones of the skull, mandible, and clavicle**. No cartilage template.

**Endochondral ossification** — a hyaline cartilage model is replaced by bone. This is how **long bones** form and grow in length. The **epiphyseal plate** has **five zones**:

| Zone (from epiphysis toward diaphysis) | What happens |
|---|---|
| **1. Reserve (resting)** | Small, inactive chondrocytes |
| **2. Proliferative** | Chondrocytes divide in columns — the actual growth in length |
| **3. Hypertrophic** | Chondrocytes enlarge and the matrix calcifies |
| **4. Calcified cartilage** | Chondrocytes die; matrix calcifies |
| **5. Ossification** | Osteoblasts deposit bone on the calcified spicules |

Growth in **width** is **appositional**, by periosteal osteoblasts. The **primary ossification center** appears in the diaphysis; **secondary centers** in the epiphyses. When the plate ossifies, an **epiphyseal line** remains and length growth stops.

## Bone Remodeling & Repair

Fracture → **hematoma** → inflammation → **soft callus** (fibrocartilage) → **hard callus** (woven bone) → **remodelling** to lamellar bone. Immobilization and blood supply decide the outcome. **Wolff law** — bone is laid down where it is loaded and resorbed where it is not — explains both trabecular architecture and the bone loss of bed rest.

## Metabolic Role of Bone

Bone is the body's calcium and phosphate reservoir.

| Hormone | Source | Effect on bone | Net serum calcium |
|---|---|---|---|
| **PTH** | Parathyroid chief cells | Increases resorption, increases renal reabsorption, activates vitamin D | **Raises** |
| **Calcitriol (1,25-(OH)₂D)** | Kidney | Increases intestinal absorption and bone resorption | **Raises** |
| **Calcitonin** | Thyroid **C cells** | Inhibits osteoclasts | **Lowers** |

## Joints

**Fibrous joints (synarthroses)** — bones united by dense connective tissue, no cavity (sutures of the skull, gomphosis of a tooth).

**Cartilaginous joints (amphiarthroses)** — bones united by cartilage: **primary** by hyaline cartilage (epiphyseal plate, first sternocostal joint); **secondary** by fibrocartilage (intervertebral disc, symphysis pubis).

**Synovial joints (diarthroses)** — a joint cavity between **hyaline articular cartilage** on each bone, enclosed by a **capsule** of dense connective tissue lined by **synovial membrane**, which secretes **synovial fluid** for lubrication and nutrition of the avascular cartilage. Some joints add a **meniscus** (fibrocartilage), a **labrum**, or a **fat pad**. Because articular cartilage has no perichondrium and no vessels, it heals hardly at all — the anatomical basis of degenerative joint disease.

**CLINICAL CORRELATION**

Six to learn as a set. **Osteoporosis** — reduced bone mass with normal mineralization; estrogen deficiency drives it, trabeculae are lost first, fragility fractures follow. **Rickets/osteomalacia** — unmineralized osteoid; in children the growth plate widens and deforms (rachitic rosary, bowed legs). **Paget disease** — chaotic excessive remodelling with mosaic woven bone. **Osteopetrosis** — osteoclast failure, dense but brittle bone with a crowded marrow cavity. **Achondroplasia** — failure of chondrocyte proliferation in the plate, short limbs with normal trunk. **Osteogenesis imperfecta** — defective type I collagen, brittle bones, blue sclerae.

**UNDER THE MICROSCOPE**

On a ground section, **concentric lamellae around a central canal** = an osteon = compact bone. On a decalcified H&E section, osteoblasts are the row of cuboidal cells on the surface and osteocytes sit in lacunae. A **multinucleated cell in a scalloped pit** is an osteoclast resorbing.

**MEMORY HOOK**

**"OsteoBlast Builds, OsteoClast Chews, OsteoCytes Communicate."** For the epiphyseal plate, **"Rest, Proliferate, Hypertrophy, Calcify, Ossify"** — read from the joint outward.

**EXAM TIP**

The commonest error is saying osteoclasts are mesenchymal. They are **hematopoietic**, from the monocyte–macrophage line. The second is reversing the epiphyseal zones: growth happens at the **epiphyseal** side, ossification at the **diaphyseal** side.

## Chapter 8 — one-line answers

- Bone matrix is type I collagen plus hydroxyapatite; osteoid must mineralize before it is rigid.
- Osteoblasts make osteoid; osteocytes maintain matrix; osteoclasts resorb bone with a ruffled border.
- Osteoclasts are multinucleated and derive from the monocyte–macrophage lineage, not from osteoblasts.
- Compact bone is built of osteons; spongy bone is a trabecular lattice without osteons.
- Endochondral ossification replaces a cartilage model; intramembranous ossification forms directly in mesenchyme.
- The growth plate runs reserve → proliferative → hypertrophic → calcified → ossification zones.

- Osteogenesis imperfecta = type I collagen; achondroplasia = FGFR3; Marfan = fibrillin.
- PTH raises serum calcium; calcitonin lowers it; vitamin D increases absorption.
- Fracture repair: hematoma → fibrocartilaginous callus → bony callus → remodeling.

## Active recall

1. **Which cell makes bone matrix, and which resorbs it?** — Osteoblasts make osteoid; osteoclasts resorb bone using a ruffled border and acid.
2. **What does osteoid contain, and what mineralizes it?** — Type I collagen and proteoglycans; hydroxyapatite crystals deposit along the collagen.
3. **How do osteocytes stay alive inside lacunae?** — Their processes run through canaliculi and contact processes of neighboring cells and the central canal, allowing diffusion of nutrients.
4. **Endochondral versus intramembranous ossification?** — Endochondral replaces a cartilage model (long bones, base of skull); intramembranous forms directly in mesenchyme (flat skull bones, mandible).
5. **What happens at a growth plate, zone by zone?** — Reserve → proliferative → hypertrophic → calcified matrix → ossification, with chondrocyte death and vascular invasion.

## Exam-style question

**A child has brittle bones, blue sclerae, and recurrent fractures. Which protein is defective?**

A. Type II collagen
B. Type I collagen
C. Fibrillin
D. Dystrophin

**Answer: B** — Osteogenesis imperfecta is a type I collagen defect. Type II defects cause chondrodysplasias; fibrillin causes Marfan syndrome.

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 8–1** — Components of bone.
- **Fig. 8–5** — Osteocytes in lacunae.
- **Fig. 8–10** — Lamellar bone: Perforating canals and interstitial lamellae.
- **Fig. 8–14** — Osteogenesis of long bones by endochondral ossification.
- **Fig. 8–19** — Main features of bone fracture repair.
- **Fig. 8–23** — Articular cartilage.

# 9 · Nerve Tissue & the Nervous System

**Book pages 161–192**
Development of Nerve Tissue · Neurons · Glial Cells & Neuronal Activity · Central Nervous System · Peripheral Nervous System · Neural Plasticity & Regeneration

## Learning objectives

By the end of this chapter you should be able to:

1. Draw a neuron and label its parts, and classify neurons by shape and function.
2. Name the six glial cell types and give each one's job.
3. Explain how a synapse transmits a signal, and how myelin speeds conduction.
4. Compare regeneration in the peripheral and central nervous systems.

## The core idea

Nervous tissue is built for **rapid signaling over distance**. The neuron is a polarized cell with a receiving end (dendrites and soma), a conducting cable (axon), and a transmitting end (synapse). Everything else — myelin, glia, barriers — exists to make that signaling faster, safer, and better supplied.

## Development of Nerve Tissue

The nervous system arises from the **neural plate**, which folds into the **neural tube** (giving the CNS) while **neural crest** cells migrate away to form the peripheral neurons, Schwann cells, satellite cells, adrenal medulla, melanocytes, and much of the facial skeleton. This embryology explains three exam facts: the CNS is lined by **ependyma** derived from the tube; Schwann cells are neural crest; and the adrenal medulla is neural crest while the cortex is mesoderm.

## Neurons

- **Perikaryon (soma)** — large, pale nucleus with a prominent nucleolus and abundant **Nissl bodies** (stacked RER and polysomes). Nissl substance is absent from the **axon hillock** — a useful landmark.
- **Dendrites** — short, branched, receptive; contain Nissl bodies.
- **Axon** — one per neuron, arises at the axon hillock, contains no Nissl bodies, specialized for conduction.
- **Axonal transport** — **kinesin** anterograde; **dynein** retrograde. Rabies, herpes, and tetanus toxin exploit retrograde transport.

**Classification:** multipolar (most CNS neurons and motor neurons), bipolar (retina, olfactory epithelium), pseudounipolar (sensory ganglia), anaxonic. Also by function — sensory, motor, interneuron.

**The synapse.** A chemical synapse has a presynaptic terminal with **synaptic vesicles** and mitochondria, a **synaptic cleft** (~20–30 nm), and a postsynaptic membrane with receptors. Transmission is unidirectional and modifiable. **Electrical synapses (gap junctions)** are fast and bidirectional.

## Glial Cells & Neuronal Activity

| Cell | Location | Function | Exam hook |
|---|---|---|---|
| **Astrocyte** | CNS | **The most numerous cell of the CNS**; hundreds of processes enclose neuronal perikarya, synapses, and capillaries, providing regulated microenvironments | **GFAP**; foot processes on capillaries; glial scar |

| Cell | Location | Function | Exam hook |
|---|---|---|---|
| **Oligodendrocyte** | CNS | Wraps flattened processes around portions of axons, forming **myelin sheaths** | One cell, many segments |
| **Microglia** | CNS | Resident phagocyte, immune surveillance | **Mesodermal** origin — the odd one out |
| **Ependymal cell** | CNS | Lines ventricles; cilia move CSF; choroid plexus | Produces CSF |
| **Schwann cell** (neurolemmocyte) | PNS | **Myelinates one segment of one axon** | Neural crest origin |
| **Satellite cell** | PNS ganglia | Supports sensory and autonomic neurons | — |

The book counts **six major glial types**: oligodendrocytes, astrocytes, ependymal cells, and microglia in the CNS; Schwann cells and satellite cells in the PNS. **Myelination:** in the PNS a Schwann cell wraps one axon segment; in the CNS an oligodendrocyte wraps several. The gaps are **nodes of Ranvier**, where voltage-gated $Na^+$ channels cluster, so the impulse jumps by **saltatory conduction**. Velocity rises with axon diameter and myelin thickness.

## Central Nervous System

- **Cerebral cortex** — six layers; **pyramidal cells** are the output neurons; gray matter outside.
- **Cerebellar cortex** — three layers: **molecular**, **Purkinje** (large flask-shaped cells, the only output), and **granular**.
- **Spinal cord** — gray matter in an H-shaped core, white matter outside; **anterior horn** motor, posterior horn sensory. This is the reverse of the cerebral cortex.
- **Meninges** — dura, arachnoid, pia; CSF in the subarachnoid space.
- **Choroid plexus** — produces CSF.
- **Blood–brain barrier** — tight junctions between endothelial cells, reinforced by astrocyte end-feet and pericytes. Absent in the circumventricular organs (area postrema, median eminence, pineal, posterior pituitary), which is how circulating hormones reach the brain.

## Peripheral Nervous System

- **Sensory ganglia** — pseudounipolar neurons with satellite cells; **no synapses inside the ganglion**.
- **Autonomic ganglia** — multipolar neurons, synapses present.
- **Peripheral nerve** — axons in fascicles surrounded by **perineurium**, the whole nerve by **epineurium**, individual fibers by **endoneurium**.

## Neural Plasticity & Regeneration

After axotomy the distal stump undergoes **Wallerian degeneration**; the soma shows **central chromatolysis**; Schwann cells form **bands of Büngner** to guide regrowth; regeneration proceeds at about **1 mm/day**. CNS axons regenerate poorly because of inhibitory molecules (Nogo, MAG) and glial scarring. **Plasticity** — the remodelling of synapses with use — is the basis of learning and of rehabilitation after injury.

**CLINICAL CORRELATION**

**Multiple sclerosis** is autoimmune demyelination in the CNS. **Guillain–Barré syndrome** is demyelination in the PNS, ascending from the legs. **Parkinson disease** is loss of dopaminergic neurons in the substantia nigra with Lewy bodies. **Alzheimer disease** shows neurofibrillary tangles and amyloid plaques. **Poliomyelitis** destroys anterior horn motor neurons and produces flaccid paralysis.

**UNDER THE MICROSCOPE**

Identify neurons by the **large pale nucleus, prominent nucleolus, and clumped Nissl bodies**. The axon hillock is the pale cone where Nissl substance stops. In peripheral nerve, look for **wavy bundles of axons with myelin rings** and connective tissue sheaths. In the cerebellar cortex, the **single row of enormous Purkinje cells** is unmistakable.

**EXAM TIP**

**Oligodendrocyte = CNS, several axons. Schwann cell = PNS, one segment of one axon.** And **microglia are the only glia of mesodermal origin.**

## Chapter 9 — one-line answers

- Neurons are classified by shape (multipolar, bipolar, pseudounipolar) and by function (sensory, motor, interneuron).
- Nissl bodies are rough ER plus polysomes; their position marks the soma and proximal dendrites.
- Oligodendrocytes myelinate the CNS (several axons each); Schwann cells myelinate the PNS (one segment each).
- Astrocytes form end-feet, contribute to the blood–brain barrier, and scar the CNS.
- Microglia are the resident macrophages, derived from monocytes.
- Ependymal cells line the ventricles and produce cerebrospinal fluid.
- Myelin allows saltatory conduction between nodes of Ranvier.
- The blood–brain barrier is sealed by tight junctions between capillary endothelial cells.
- PNS axons regenerate through bands of Büngner; CNS axons do not regenerate.

## Active recall

1. **Which glial cell myelinates the CNS, and which myelinates the PNS?** — Oligodendrocyte (CNS, several axons); Schwann cell (PNS, one segment of one axon).
2. **Which glial cell is the macrophage of the CNS?** — Microglia, derived from monocytes.
3. **What forms the blood–brain barrier?** — Tight junctions between capillary endothelial cells, reinforced by astrocyte end-feet and pericytes.
4. **How is a signal carried along an axon?** — An action potential; myelin allows saltatory conduction between nodes of Ranvier.
5. **Why does CNS injury scar rather than regenerate?** — Oligodendrocytes and myelin inhibit axon growth, astrocytes form a glial scar, and there is no permissive sheath like the PNS bands of Büngner.

## Exam-style question

**A peripheral nerve is cut and surgically reapproximated. Which cell guides regrowing axons?**

A. Oligodendrocyte
B. Schwann cell
C. Astrocyte
D. Microglial cell

**Answer: B** — Schwann cells dedifferentiate, form bands of Büngner, and provide a physical and trophic guide for the regenerating axon.

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 9–1** — The general organization of the nervous system.
- **Fig. 9–7** — Types of synapses.
- **Fig. 9–13** — Satellite cells around neurons of ganglia in the PNS.
- **Fig. 9–18** — Spinal cord and meninges.
- **Fig. 9–24** — Node of Ranvier and endoneurium.
- **Fig. 9–30** — Regeneration in peripheral nerves.

# 10 · Muscle Tissue

**Book pages 193–214**
Skeletal Muscle · Cardiac Muscle · Smooth Muscle · Regeneration of Muscle Tissue

## Learning objectives

By the end of this chapter you should be able to:

1. Distinguish skeletal, cardiac, and smooth muscle by light microscopy.
2. Explain the sarcomere and the sliding-filament mechanism.
3. Compare excitation–contraction coupling in the three muscle types.
4. Predict which muscle regenerates and which is replaced by scar.

## The core idea

Muscle tissue consists of cells that optimize the universal cell property of **contractility**. Essentially **all muscle cells originate from embryonic mesoderm** and differentiate by elongation and abundant synthesis of the myofibrillar proteins **actin and myosin**. All three types use the same actin–myosin interaction but arrange it differently: **striated and voluntary** (skeletal), **striated and involuntary** (cardiac), **unstriated and involuntary** (smooth).

## Skeletal Muscle

**Organization, large to small:** muscle → fascicles → fibers → myofibrils → myofilaments. Each fiber is a long, cylindrical, **multinucleated** syncytium with **peripheral nuclei**, produced by fusion of myoblasts. Satellite cells persist beneath the basal lamina and mediate limited regeneration.

**The sarcomere**, bounded by two **Z lines**:

| Band | Contents | Behavior in contraction |
|---|---|---|
| **A band** | Thick filaments (**myosin**) | Constant length |
| **I band** | Thin filaments (**actin**) only | Shortens |
| **H band** | Thick filaments only, no overlap | Shortens |
| **Z line** | Anchor for actin | Moves closer |
| **M line** | Center of the A band | — |

**Sliding filament mechanism:** myosin heads bind actin, hydrolyze ATP, and pivot, pulling thin filaments toward the M line. **The A band never changes length.**

**Control.** The **sarcoplasmic reticulum** stores $Ca^{2+}$; **T-tubules** are invaginations of the sarcolemma that encircle each myofibril **near the aligned A- and I-band boundaries**, where each T-tubule associates with **two terminal cisternae** — a **triad**. Depolarization spreads along the T-tubules to trigger simultaneous $Ca^{2+}$ release throughout the fiber. An action potential releases $Ca^{2+}$, which binds **troponin C**, moves **tropomyosin** off the myosin-binding site, and contraction begins. Relaxation requires $Ca^{2+}$ reuptake.

**Innervation.** The **motor end plate** — acetylcholine binds nicotinic receptors, depolarises the sarcolemma, and triggers contraction. One motor neuron plus all the fibers it supplies is a **motor unit**.

### Fiber types:

| Type | Speed | Metabolism | Fatigue | Color |
|---|---|---|---|---|
| **I (slow, red)** | Slow | **Oxidative** | Resistant | Red |
| **IIa (fast, oxidative-glycolytic)** | Fast | Mixed | Intermediate | Intermediate |

| Type | Speed | Metabolism | Fatigue | Color |
|---|---|---|---|---|
| **IIx/IIb (fast, glycolytic)** | Fast | **Glycolytic** | Fast fatigue | White |

**Sensory receptors. Muscle spindles** (intrafusal fibers, Ia afferents, gamma motor neurons) sense length and stretch; **Golgi tendon organs** sense tension.

## Cardiac Muscle

- **Striated**, branched fibers, **one or two central nuclei**.
- **Intercalated discs** — the diagnostic feature — contain **fascia adherens** (anchors actin), **desmosomes** (mechanical strength), and **gap junctions** (electrical coupling), making the heart a functional syncytium.
- **Diads** at the **Z line** (one T-tubule + one cisterna) rather than **triads** of the **transverse tubule system** at the A–I junction.
- Abundant mitochondria and lipid droplets; highly oxidative; resistant to fatigue.
- **Purkinje fibers** — specialized conducting cells, larger and paler, with fewer myofibrils; they conduct rapidly through the ventricles.
- **Regeneration is essentially absent**; damage is repaired by fibrous scar.

## Smooth Muscle

- **Spindle-shaped** cells with a single **central** nucleus and no striations.
- Contractile apparatus: **dense bodies** (analogous to Z lines) linked by desmin and actin.
- **No troponin.** Calcium binds **calmodulin**, which activates **myosin light-chain kinase (MLCK)**.
- **Caveolae** replace T-tubules; the SR is sparse and $Ca^{2+}$ entry across the membrane matters more.
- **Unitary (single-unit)** smooth muscle — gap junctions, spontaneous rhythmicity, in gut, uterus, bladder. **Multiunit** — each cell separately innervated, in the iris and vas deferens.
- Capable of **hypertrophy and hyperplasia** (uterine myometrium in pregnancy).

## Regeneration of Muscle Tissue

Skeletal muscle regenerates weakly, from **satellite cells**, and only if the basal lamina survives; otherwise the defect fills with fibrous tissue. **Cardiac muscle does not regenerate** — infarcted myocardium is replaced by scar. **Smooth muscle** can both hypertrophy and, in the uterus, proliferate.

| Feature | Skeletal | Cardiac | Smooth |
|---|---|---|---|
| Striation | **Yes** | **Yes** | No |
| Nuclei | **Many, peripheral** | 1–2, **central** | 1, **central** |
| Cell shape | Long cylinder | Branched | Spindle |
| T-tubules | A–I junction, **triads** | Z line, **diads** | None (caveolae) |
| $Ca^{2+}$ sensor | **Troponin C** | **Troponin C** | **Calmodulin** |
| Junctions | None between fibers | **Intercalated discs** | Gap junctions (unitary) |
| Control | Somatic, voluntary | Autonomic, involuntary | Autonomic, involuntary |
| Regeneration | Satellite cells (limited) | None | Hypertrophy/hyperplasia |

**CLINICAL CORRELATION**

**Duchenne muscular dystrophy** is an X-linked defect of **dystrophin**, which links the sarcolemma to the extracellular matrix; without it, contraction tears the membrane, fibers die, and fat and fibrous tissue replace muscle (pseudohypertrophy). **Myasthenia gravis** is autoantibody blockade of the nicotinic acetylcholine receptor, producing fatigable weakness. **Cardiac hypertrophy** is an adaptation to pressure overload; beyond a point the myocytes outgrow their blood supply and fail.

**UNDER THE MICROSCOPE**

Look first for **striations**; if present, look at the **nuclei**. **Many nuclei at the edge = skeletal. One or two central nuclei with dark transverse bands = cardiac (intercalated discs).** No striations and central nuclei in spindle-shaped cells = smooth.

**MEMORY HOOK**

**"Skeletal = Surface nuclei. Cardiac = Central nucleus, Cross-striations, interCalated discs. Smooth = Spindle, Single central nucleus, no Striations."**

**EXAM TIP**

The A band does not shorten. This is asked more often than any other muscle fact: I and H shorten, A stays constant, Z lines approach.

## Chapter 10 — one-line answers

- Skeletal muscle: long cylinders, **many peripheral nuclei**, triads at the **A–I junction**.
- Cardiac muscle: branched cells, **one or two central nuclei**, **intercalated discs**, diads at the **Z line**.
- Smooth muscle: spindle cells, **one central nucleus**, no striations, no T-tubules (caveolae).
- Sarcomere = Z line to Z line; **A band constant**, I band and H zone shorten.
- Thin filament = actin + troponin + tropomyosin; thick filament = myosin.
- $Ca^{2+}$ binds **troponin C** in striated muscle and **calmodulin** in smooth muscle.
- Smooth muscle contraction uses **myosin light-chain kinase**; it has no troponin.
- Intercalated discs contain **fasciae adherentes, desmosomes, and gap junctions** — not tight junctions.
- Skeletal muscle regenerates from **satellite cells**, and only if the basal lamina survives.
- Cardiac muscle does not regenerate; infarction heals by scar.
- Type I fibers = slow, oxidative, fatigue-resistant. Type II = fast, glycolytic.
- Duchenne dystrophy = **dystrophin** defect; myasthenia gravis = antibody to the nicotinic receptor.
- Muscle spindle = stretch receptor; Golgi tendon organ = tension receptor.

## Active recall

1. **Which band does not change length during contraction?** — The A band, because it is the length of the thick filaments.
2. **What hides the myosin-binding site at rest, and what moves it?** — Tropomyosin; calcium binds troponin C and shifts tropomyosin.
3. **Cardiac versus skeletal muscle at the T-tubule level?** — Cardiac diads lie at the Z line; skeletal triads lie at the A–I junction.
4. **What is the calcium sensor in smooth muscle?** — Calmodulin, which activates myosin light-chain kinase; smooth muscle has no troponin.
5. **Which muscle types regenerate?** — Skeletal (limited, from satellite cells), smooth (hyperplasia/hypertrophy); cardiac does not.

## Exam-style question

**A muscle biopsy shows branched fibers with one central nucleus and dark transverse bands. What is the diagnosis?**

A. Skeletal muscle
B. Cardiac muscle
C. Smooth muscle
D. Dense regular connective tissue

**Answer: B** — Branching, central nuclei, and intercalated discs are the triad of cardiac muscle.

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 10–1** — Three types of muscle.
- **Fig. 10–5** — Capillaries of skeletal muscle.
- **Fig. 10–9** — Organization of a skeletal muscle fiber.
- **Fig. 10–12** — Sliding filaments and sarcomere shortening in contraction.
- **Fig. 10–16** — Cardiac muscle: Intercalated discs.
- **Fig. 10–20** — Smooth muscle contraction.

# PART III — TRANSPORT, DEFENSE, AND BLOOD

Chapters 11–14 · circulation, blood, hemopoiesis, and immunity

# 11 · The Circulatory System

**Book pages 215–236**
Heart · Tissues of the Vascular Wall · Vasculature · Lymphatic Vascular System

## Learning objectives

By the end of this chapter you should be able to:

1. Name the three tunics of the heart and of a blood vessel, and describe what each contains.
2. Distinguish elastic artery, muscular artery, arteriole, capillary, venule, and vein in section.
3. Match each capillary type to the organ that needs it.
4. Trace the lymphatic system from blind-ended capillary to thoracic duct.

## The core idea

The circulatory system pumps and directs blood cells and substances to all tissues, and in an adult the total length of its vessels is estimated at **100,000–150,000 km**. Almost every vessel is a variation on one three-layered tube. What changes between an aorta and a capillary is **how much muscle and elastic tissue each layer contains** — decided entirely by the pressure the vessel must withstand and the job it must do.

## Heart

- **Endocardium** — endothelium + subendothelial connective tissue + a layer of Purkinje fibers; continuous with the intima of vessels.
- **Myocardium** — the thickest layer; cardiac muscle in complex spirals, atria thinner than ventricles, left ventricle thickest.
- **Epicardium** — connective tissue containing many **adipocytes**, covered by mesothelium; the visceral layer of the pericardium.
- **Cardiac skeleton** — dense irregular connective tissue rings that anchor the valves and electrically insulate atria from ventricles.
- **Valves** — endocardium folded over a core of dense connective tissue; **avascular**. The mitral and tricuspid have chordae tendineae and papillary muscles.

## Tissues of the Vascular Wall

| Layer | Contents | Function |
|---|---|---|
| **Intima** | Endothelium, basal lamina, subendothelial connective tissue, **internal elastic lamina** (in arteries) | Non-thrombogenic surface, exchange, barrier |
| **Media** | **Smooth muscle** circularly arranged, elastic lamellae (in elastic arteries), **external elastic lamina** | Tone and compliance |
| **Adventitia** | Loose connective tissue, **vasa vasorum**, **nervi vascularis** | Anchorage and supply of the wall |

**The rule of the wall:** the thicker the media, the higher the pressure the vessel handles. Arteries have a thick media; veins a thin media and a thick adventitia. That single ratio lets you identify a vessel on any slide.

## Vasculature

| Vessel | Intima | Media | Adventitia | Distinguishing feature |
|---|---|---|---|---|
| **Elastic artery** (aorta, pulmonary, common carotid) | Endothelium + subendothelial | **Many elastic lamellae** alternating with smooth muscle | Thin, with vasa vasorum | **Elastic lamellae** |
| **Muscular artery** | Prominent **internal elastic lamina** | Thick, 3–40 layers of smooth muscle | Moderate | **Well-defined IEL** |
| **Arteriole** | Endothelium + thin subendothelium | 1–2 layers of smooth muscle | Minimal | **The resistance vessel** |
| **Capillary** | Endothelium + basal lamina only | None | None | One cell thick |
| **Venule** | Endothelium | None or sparse pericytes | Thin | Site of **leukocyte diapedesis** |
| **Small/medium vein** | Endothelium | Thin | **Thickest layer** | Often collapsed on section |
| **Large vein** (vena cava) | Endothelium | Thin | **Thick, with longitudinal smooth muscle** | Valve-like intimal folds |

### Capillaries:

| Type | Structure | Location | Function |
|---|---|---|---|
| **Continuous** | Uninterrupted endothelium, tight junctions | Muscle, skin, lung, CNS | Regulated exchange; the **blood–brain barrier** is this type plus tight junctions and astrocyte feet |
| **Fenestrated** | Pores with diaphragms | Kidney glomerulus, intestine, endocrine glands | Rapid fluid and solute exchange |
| **Sinusoidal** | Discontinuous, large gaps, incomplete basal lamina | **Liver, spleen, bone marrow** | Free passage of cells and large molecules |

**Microcirculation.** Arteriole → metarteriole → capillary bed → venule, with **precapillary sphincters** regulating flow. **Arteriovenous anastomoses** shunt blood directly, important in skin thermoregulation.

## Lymphatic Vascular System

Begins as blind-ended **lymphatic capillaries** with overlapping endothelial flaps and **anchoring filaments**, which open when interstitial pressure rises. They converge into collecting vessels with valves, then nodes, trunks, and the **thoracic duct** and **right lymphatic duct**. Lymphatics have **no continuous basal lamina** and no fenestrations — which is exactly why tumor cells and bacteria enter them so easily.

**CLINICAL CORRELATION**

**Atherosclerosis** begins with endothelial dysfunction: LDL crosses the endothelium, is oxidized, and provokes monocyte recruitment. Macrophages become **foam cells**, a fatty streak forms, smooth muscle migrates into the intima, and a fibrous cap covers a necrotic core. Rupture of a thin cap exposes thrombogenic material and produces the acute coronary syndrome. **Aneurysm** follows destruction of medial elastic fibers. **Varicose veins** follow valve failure. **Lymphedema** follows lymphatic obstruction — classically after axillary node clearance.

**UNDER THE MICROSCOPE**

Arteries: **round lumen, thick wall, corrugated internal elastic lamina**. Veins: **irregular collapsed lumen, thin media**. In elastic arteries the media shows wavy pink bands with no orderly muscle coat; in muscular arteries the media is uniform smooth muscle.

**EXAM TIP**
The three capillaries are distinguished by **continuity, not size**: continuous (tight), fenestrated (pores), sinusoidal (gaps wide enough for a red cell). Liver, spleen, marrow = sinusoidal. Kidney glomerulus and intestine = fenestrated. Brain = continuous with tight junctions.

## Chapter 11 — one-line answers

- Three tunics: intima (endothelium + subendothelial tissue), media (smooth muscle, elastic lamellae), adventitia (collagen).
- Elastic arteries have many concentric elastic lamellae; muscular arteries have a prominent internal elastic lamina.
- Arterioles have one or two smooth-muscle layers and control peripheral resistance.
- Continuous capillaries are the default; fenestrated ones serve endocrine glands, gut, and kidney; sinusoids serve liver, spleen, and marrow.
- Veins have thinner walls, larger lumens, and valves in medium-sized veins.
- The heart has endocardium, myocardium, and epicardium; Purkinje fibers conduct the impulse.
- Lymphatic capillaries are blind-ended and more permeable than blood capillaries; they drain to the thoracic duct.
- The thoracic duct empties into the left subclavian vein.

## Active recall

1. **Three layers of a vessel wall, outside in?** — Tunica adventitia, tunica media, tunica intima.
2. **Which vessel has the thickest media relative to its lumen?** — The muscular artery; an arteriole has only one or two smooth-muscle layers.
3. **Which capillary type is fenestrated, and where is it found?** — Fenestrated capillaries are in endocrine glands, gut mucosa, and kidney; they allow rapid exchange.
4. **What distinguishes a vein from an artery in section?** — Veins have a larger, often collapsed lumen and a thinner wall with less muscle; valves are present in medium veins.
5. **Where does lymph begin and end?** — Blind-ended lymphatic capillaries in tissue; it drains through nodes to the thoracic duct and into the subclavian vein.

## Exam-style question

**A section shows a vessel with a thick media containing many elastic lamellae and a wide lumen. Which vessel is it?**

A. Large vein
B. Elastic artery
C. Arteriole
D. Capillary

**Answer: B** — Numerous concentric elastic lamellae in a thick media define an elastic artery (aorta, pulmonary artery).

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 11–1** — The cardiovascular system.
- **Fig. 11–6** — Walls of arteries and veins.
- **Fig. 11–11** — Muscular artery.
- **Fig. 11–15** — Comparison of the simple microvascular pathway with arteriovenous shunts and portal systems.
- **Fig. 11–20** — Sinusoidal capillary.
- **Fig. 11–25** — Lymphatic vessels and valve.

# 12 · Blood

**Book pages 237–253**
Composition of Plasma · Blood Cells

## Learning objectives

By the end of this chapter you should be able to:

1. Describe plasma and explain how serum differs from it.
2. Relate erythrocyte shape, size, and lifespan to gas transport.
3. Identify each leukocyte by nucleus and granules, and state its function.
4. Explain platelet structure and the steps of hemostasis.

## The core idea

Blood is connective tissue with a liquid matrix. Its cells all derive from one stem cell, and their morphology is a direct readout of their job: **red for oxygen, granular white for killing, agranular white for immunity, and fragments for clotting.**

## Composition of Plasma

About **55% of blood volume**. Water plus plasma proteins — **albumin, α- and β-globulins, complement proteins, and fibrinogen**, all secreted in the **liver**, plus the **immunoglobulins** — and electrolytes, nutrients, hormones, and waste.

The book's Table 12–1 quantifies the mixture:

| Component | Share | Job |
|---|---|---|
| **Water** | ~92% of plasma | Solvent in which formed elements are suspended |
| **Plasma proteins** | ~7% of plasma | Buffer pH; maintain osmotic pressure |
| — **Albumin** | ~58% of plasma protein | Osmotic pull (oncotic pressure); carries fatty acids, hormones, drugs |
| — **Globulins** | ~37% of plasma protein | α- and β-globulins transport lipids, metals, and iron; γ-globulins are antibodies |
| — **Fibrinogen** | ~4% of plasma protein | Precursor of fibrin for clotting |
| — **Regulatory proteins** | **>1%** of plasma protein | Enzymes, proenzymes, hormones, complement |
| **Electrolytes** | **~1%** of plasma (with nutrients) | Membrane potentials, pH, osmotic control |
| **Nutrients** | **~1%** of plasma (with electrolytes) | Glucose, amino acids, cholesterol, vitamins — energy and synthesis |

On clotting, some proteins are consumed and others are released from platelets, leaving **serum**, whose biological properties differ from plasma. **Serum is plasma minus fibrinogen** (i.e., after clotting).

## Blood Cells

### Erythrocytes

- **Biconcave disc**, **7.5 µm** in diameter, **2.6 µm** thick at the rim but only **0.75 µm** at the center — flexible enough to pass small capillaries; **no nucleus, no organelles**, so the cell is packed with hemoglobin.
- Lifespan about **120 days**; removed by splenic macrophages; the globin is recycled and the heme converted to bilirubin.
- Shape maintained by **spectrin and ankyrin**; defects produce hereditary spherocytosis.

- **Hemoglobin** — four globin chains, each with a heme group carrying $Fe^{2+}$. Adult HbA ($\alpha_2\beta_2$) and fetal HbF ($\alpha_2\gamma_2$, higher affinity for $O_2$).
- **Reticulocytes** are young red cells with residual RNA, appearing polychromatic; a raised count signals active marrow output.
- **Hematocrit** — the sedimented erythrocyte column — is about **44–45% of total blood volume** in healthy adults; the thin layer between cells and plasma is the **buffy coat** of leukocytes and platelets. An average adult has **5–6 L** of blood (supplementary standard value, not stated in the chapter).

**The leukocytes** — broadly grouped as **granulocytes** (neutrophils, eosinophils, basophils) and **agranulocytes** (lymphocytes, monocytes)

| Cell | % of WBC | Size | Nucleus | Cytoplasm | Main function |
|---|---|---|---|---|---|
| **Neutrophil** | **60–70%** | 10–14 µm | **3–5 lobes**, dense | Pale; fine granules | **Phagocytosis of bacteria**; the acute responder |
| **Eosinophil** | **2–4%** | 10–14 µm | **Bilobed** | **Bright orange-red** granules | **Parasites and allergy**; major basic protein |
| **Basophil** | **0.5–1%** | 10–14 µm | **S-shaped**, obscured | **Large dark blue** granules | **Histamine and heparin** |
| **Lymphocyte** | **20–30%** | 7–12 µm | Round, dense | Pale rim | **T cells**, **B cells**, **NK cells** |
| **Monocyte** | **3–8%** | **14–20 µm** (largest) | **Kidney-shaped** | Abundant blue-gray | Become **macrophages** |

**Neutrophil granules: azurophilic (primary)** granules contain myeloperoxidase and defensins; **specific (secondary)** granules contain lysozyme and lactoferrin. The neutrophil dies after phagocytosis; the accumulation of dead neutrophils and debris is **pus**. Each **phagocyte** kills ingested bacteria inside a phagolysosome.

**Platelets** are **2–4 µm** anucleate fragments of **megakaryocyte** cytoplasm, with a peripheral clear zone (hyalomere) and a central granular zone (granulomere). They number **150,000–400,000/µL** and circulate for about **10 days**. They adhere, activate, and aggregate to form the primary hemostatic plug; **von Willebrand factor** links them to collagen.

**Hemostasis in four steps** (the standard physiology sequence; this chapter describes only the platelet plug and clotting): (1) **Vascular spasm** — the injured vessel constricts, slowing flow. (2) **Primary hemostasis** — platelets bind exposed collagen (directly and through von Willebrand factor), degranulate, and aggregate into a soft plug. (3) **Coagulation** — exposed tissue factor and platelet phospholipid trigger the clotting cascade, converting prothrombin to **thrombin**, which cleaves fibrinogen into **fibrin**; the fibrin mesh stabilizes the plug as a clot. (4) **Retraction and repair** — the clot contracts, and endothelial cells and fibroblasts rebuild the wall.

**CLINICAL CORRELATION**

The differential count is a diagnostic instrument. **Neutrophilia** suggests bacterial infection or steroid use; **eosinophilia** parasites or allergy; **lymphocytosis** viral infection or chronic lymphocytic leukemia; **pancytopenia** marrow failure. In **sickle cell disease** a single glutamic acid → valine substitution in β-globin allows deoxygenated HbS to polymerize; the cell sickles, occludes small vessels, and is removed by the spleen.

**UNDER THE MICROSCOPE**

Count lobes, not granules. **Neutrophil: 3–5 lobes. Eosinophil: exactly two, orange granules. Basophil: granules so dark the nucleus is hidden. Monocyte: kidney-shaped nucleus, largest cell. Lymphocyte: round dense nucleus, almost no cytoplasm.**

**MEMORY HOOK**
**"Never Let Monkeys Eat Bananas"** = **N**eutrophil, **L**ymphocyte, **M**onocyte, **E**osinophil, **B**asophil — in descending order of abundance.

**EXAM TIP**
Eosinophil granules stain with eosin because they are **basic proteins** — the naming is the reverse of what students expect. The basophil's granules contain **heparin and histamine**, so the "basic" cell is the one full of allergy mediators.

## Chapter 12 — one-line answers

- Plasma is the fluid phase; serum is plasma after clotting and lacks fibrinogen.
- Erythrocytes are biconcave, 7.5 µm, anucleate, and live about 120 days.
- Hemoglobin is the gas carrier; the biconcave shape maximizes surface-to-volume ratio.
- Neutrophils are the most numerous leukocyte and the first responder to acute inflammation.
- Eosinophils have bilobed nuclei and orange-red granules (major basic protein); they attack parasites.
- Basophils carry histamine and heparin and mediate immediate hypersensitivity.
- Lymphocytes are the second most numerous; monocytes are the largest and become macrophages.
- Platelets are anucleate fragments of megakaryocytes and form the primary hemostatic plug.
- Hemostasis = vascular spasm → platelet plug → coagulation → clot retraction and repair.

## Active recall

1. **How does serum differ from plasma?** — Serum is plasma after clotting, so it lacks fibrinogen and consumed clotting factors.
2. **Normal erythrocyte dimensions and lifespan?** — About 7.5 µm diameter, biconcave with a 2.6-µm rim, circulating about 120 days.
3. **Which leukocyte is most numerous, and which is largest?** — Neutrophils (~60–70%); monocytes are the largest.
4. **What do eosinophil granules contain?** — Major basic protein and eosinophil cationic protein, active against parasites and in allergy.
5. **What are the four steps of hemostasis?** — Vascular spasm, primary hemostasis (platelet plug), coagulation (fibrin clot), and clot retraction with repair.

## Exam-style question

**A blood smear shows a cell with a bilobed nucleus and bright orange-red granules. What is it?**

A. Neutrophil
B. Eosinophil
C. Basophil
D. Monocyte

**Answer: B** — Bilobed nucleus plus large eosinophilic granules is the eosinophil.

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 12–1** — Composition of whole blood.
- **Fig. 12–4** — Normal human erythrocytes.
- **Fig. 12–6** — Diagram of events involving leukocytes in a postcapillary venule at sites of inflammation.
- **Fig. 12–9** — Eosinophils.
- **Fig. 12–11** — Lymphocytes.
- **Fig. 12–14** — Platelet aggregation, degranulation, and fibrin clot formation.

# 13 · Hemopoiesis

**Book pages 254–266**
Stem Cells, Growth Factors, & Differentiation · Bone Marrow · Maturation of Erythrocytes · Maturation of Granulocytes · Maturation of Agranulocytes · Origin of Platelets

## Learning objectives

By the end of this chapter you should be able to:

1. Trace the shift of hemopoietic sites from embryo to adult.
2. Describe the marrow niche and the stromal cells that support it.
3. Follow each lineage from stem cell to mature cell, naming the recognizable stages.
4. Match growth factors and colony-stimulating factors to the lineages they drive.

## The core idea

All blood cells descend from one **hematopoietic stem cell**, and the pathway from that cell to a mature erythrocyte, granulocyte, or platelet is a sequence of named stages you can recognize by **decreasing nucleus, increasing cytoplasm, and changing color**. It is the most examinable lineage in histology because every stage has a name.

## Sites of hemopoiesis

| Period | Site |
|---|---|
| **First trimester** | Yolk sac (mesoblastic phase) |
| **Second trimester** | **Liver and spleen** (hepatic phase) |
| **Third trimester** | **Bone marrow** — marrow becomes the major organ in the third trimester as skeletal elements ossify |

In the adult, active **red marrow** persists in the **sternum, ribs, vertebrae, pelvis, skull, and proximal femur**. The rest is **yellow (fatty) marrow**, which can revert to red marrow under demand — why chronic anemia produces marrow expansion and frontal bossing.

## Stem Cells, Growth Factors, & Differentiation

The **hematopoietic stem cell** is CD34⁺, self-renewing, and pluripotent, sitting in a **niche** of stromal cells, other cells, and ECM. Committed **progenitor cells** — also called **CFUs** (colony-forming units) — proliferate and differentiate under growth factors, which are also called **CSFs** (colony-stimulating factors) or cytokines. Differentiation is driven by **growth factors**, examinable by name:

| Factor | Target |
|---|---|
| **Erythropoietin (EPO)** | Erythroid progenitors (kidney, some liver) |
| **Thrombopoietin (TPO)** | Megakaryocytes (liver) |
| **G-CSF** | Granulocyte production |
| **GM-CSF, M-CSF** | Granulocyte–monocyte and monocyte lineages |
| **IL-3** | Early multipotent progenitors |
| **IL-5** | Eosinophils |
| **SCF (stem cell factor)** | Stem cell maintenance |

## Bone Marrow

Between the trabeculae lie **hemopoietic cords** and **sinusoids** with a discontinuous endothelium, so mature cells can enter the circulation. The **myeloid:erythroid (M:E) ratio** is normally **2:1 to 4:1**; it shifts toward erythroid in hemolytic anemia and toward myeloid in infection or leukemia.

## Maturation of Erythrocytes

**Proerythroblast → basophilic → polychromatophilic → orthochromatophilic erythroblast → reticulocyte → erythrocyte**, developing in **erythropoietic islands or cords**. The names track the cytoplasm's transition from **RNA-rich to hemoglobin-filled**; at the last stage the nucleus is **extruded**, producing a **reticulocyte** that still contains some polyribosomes when released into the circulation.

The cell **shrinks**, the nucleus **condenses and is extruded**, hemoglobin accumulates so the cytoplasm shifts from blue to pink, and the nucleolus disappears. The sequence takes about **5–7 days**. EPO drives it; iron, vitamin $B_{12}$, and folate are required.

## Maturation of Granulocytes

**Myeloblast → promyelocyte → myelocyte → metamyelocyte → band cell → segmented neutrophil (or eosinophil/basophil).**

The suffixes carry the meaning: **-blast** = dividing precursor (**myeloblast**, **lymphoblast**, **proerythroblast**), **-cyte** = the later, non-dividing cell (**myelocyte**, **lymphocyte**, **reticulocyte**); **thrombocyte** is another name for the platelet.

**Primary (azurophilic) granules** appear at the promyelocyte stage, **specific granules** at the myelocyte stage, the nucleus indents at the metamyelocyte stage, and becomes a **C-shaped band** before segmenting. A **left shift** — more band forms and younger cells in the peripheral blood — indicates accelerated production: immature neutrophils, chiefly **band (stab) cells**, are released prematurely when the circulating neutrophil pool is depleted by bacterial infection.

## Maturation of Agranulocytes

**Lymphopoiesis** begins in the marrow for B cells and in the thymus for T cells. **Monopoiesis** produces monocytes that become tissue macrophages. Both pass through stages in which the nucleus becomes less lobated, the cytoplasm more abundant, and the chromatin pattern characteristic of the mature cell.

## Origin of Platelets

**Megakaryoblasts → megakaryocytes**, which become polyploid and develop **demarcation membranes** that delineate future platelet territories; platelets bud from the cytoplasm under TPO control.

**CLINICAL CORRELATION**

**Aplastic anemia** is marrow failure with an empty, fatty marrow and pancytopenia. **Leukemia** is the opposite: marrow packed with malignant blasts, crowding out normal lineages and producing anemia, infection, and bleeding. **Megaloblastic anemia** ($B_{12}$ or folate deficiency) shows nuclear–cytoplasmic asynchrony — the nucleus matures more slowly than the cytoplasm, so cells are large and hypersegmented. **Iron deficiency** produces microcytic hypochromic cells because hemoglobin synthesis lags behind cell division.

**UNDER THE MICROSCOPE**

Identify the lineage by **nucleus-to-cytoplasm ratio and granule color**. Early cells have a large nucleus, a thin rim of basophilic cytoplasm, and a nucleolus. As cells mature, the ratio falls, granules appear, and the nucleus condenses. **Maturation moves in one direction: nucleus shrinks, cytoplasm grows, color shifts from blue to pink or from blue to granular.**

**EXAM TIP**
Two facts that are easy to invert: **EPO comes from the kidney** (peritubular interstitial cells), and **TPO comes mostly from the liver**. Students routinely invert these.

## Chapter 13 — one-line answers

- Hemopoiesis shifts from yolk sac → liver and spleen → bone marrow; the adult site is red marrow of the axial skeleton.
- The marrow stroma (reticular cells, macrophages, adipocytes, endothelial cells) forms the hematopoietic niche.
- Erythropoiesis: proerythroblast → basophilic → polychromatophilic → orthochromatophilic normoblast → reticulocyte → erythrocyte.
- The nucleus is extruded at the reticulocyte stage; the reticulocyte still contains residual RNA.
- Granulopoiesis: myeloblast → promyelocyte → myelocyte → metamyelocyte → band → segmented cell.
- Specific granules appear at the myelocyte stage; azurophilic granules at the promyelocyte stage.
- Megakaryocytes are polyploid and shed platelets from their cytoplasm.
- Erythropoietin from the kidney drives erythroid production; G-CSF drives granulocytes.
- The normal myeloid-to-erythroid ratio is about 2:1 to 4:1.

## Active recall

1. **Where does hemopoiesis occur in the adult?** — Red marrow of the axial skeleton (vertebrae, ribs, sternum, pelvis, proximal femur).
2. **What is the first recognizable erythroid precursor, and what is lost at the reticulocyte stage?** — Proerythroblast; the nucleus is extruded to form the reticulocyte.
3. **How do you recognize a megakaryocyte?** — A huge polyploid cell with a lobulated nucleus whose cytoplasm fragments into platelets.
4. **Which growth factor drives erythroid production?** — Erythropoietin, made by peritubular interstitial cells of the kidney.
5. **What does the M:E ratio describe?** — The ratio of myeloid (granulocytic) to erythroid precursors in marrow, normally about 2:1 to 4:1.

## Exam-style question

**A bone marrow smear shows a cell with a large, lobulated nucleus and abundant granular cytoplasm producing platelets. What is it?**

A. Megakaryocyte
B. Osteoclast
C. Macrophage
D. Proerythroblast

**Answer: A** — The megakaryocyte is the polyploid platelet factory; osteoclasts are multinucleated but resorb bone, not make platelets.

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 13–1** — Shifting locations of hemopoiesis during development and aging.
- **Fig. 13–4** — Red bone marrow (active in hemopoiesis).
- **Fig. 13–6** — Summary of erythrocyte maturation.
- **Fig. 13–9** — Developing erythrocytes and granulocytes in marrow.
- **Fig. 13–11** — Neutrophilic myelocyte.
- **Fig. 13–14** — Megakaryocyte ultrastructure.

# 14 · The Immune System & Lymphoid Organs

**Book pages 267–294**
Innate & Adaptive Immunity · Cytokines · Antigens & Antibodies · Antigen Presentation · Cells of Adaptive Immunity · Thymus · Mucosa-Associated Lymphoid Tissue · Lymph Nodes · Spleen

## Learning objectives

By the end of this chapter you should be able to:

1. Contrast innate and adaptive immunity in cells, speed, and memory.
2. Explain antigen presentation by MHC I and MHC II and the T-cell classes they activate.
3. Compare the architecture and function of thymus, lymph node, MALT, and spleen.
4. Explain how T and B cells are selected and activated, and what happens when selection fails.

## The core idea

The immune system is a set of organs arranged to **bring antigen to lymphocytes**. Each organ is a different solution to the same problem: how to filter a fluid (blood, lymph, or air/food), trap antigen, and present it to the right cell.

## Innate & Adaptive Immunity

| Feature | Innate | Adaptive |
|---|---|---|
| Timing | **Present from birth**, acts within minutes | **Develops more slowly**, improves with exposure |
| Specificity | Broad molecular patterns | **Exact antigen** |
| Memory | None | **Yes** |
| Cells | **Granulocytes** (mainly neutrophils), macrophages, dendritic cells, NK cells | **T and B lymphocytes** |
| Mediators | **Defensins, complement, lysozyme, interferons**, cytokines | Antibodies, cytotoxic granules |

The book's one-line definition is worth memorizing: innate immunity is present from birth and uses leukocytes (mainly granulocytes) plus soluble proteins; adaptive immunity develops more slowly and is **based on antigen presentation to lymphocytes**.

## Cytokines

**Polypeptide hormones of the immune system** — the book's phrase — secreted by immune cells to regulate one another. The ones worth naming: **interleukins** (IL-1, IL-2 for T-cell proliferation, IL-4 for IgE and Th2 responses, IL-5 for eosinophils, IL-6 for acute-phase response), **interferons** (antiviral; IFN-γ activates macrophages), **TNF-α** (inflammation, fever, cachexia), **chemokines** (directed leukocyte migration), and **colony-stimulating factors** (hemopoiesis, Chapter 13). Cytokines act locally (paracrine) or on the secreting cell (autocrine), and their excessive release produces a cytokine storm and the systemic inflammatory response.

## Antigens & Antibodies

**Antigens** are molecules recognized by the adaptive immune system; **epitopes** are the specific parts bound by a receptor. **Antibodies** are immunoglobulins:

| Class | Structure | Main role |
|---|---|---|
| **IgG** | Monomer | Most abundant; crosses the placenta; opsonizes |
| **IgM** | Pentamer | First responder; complement activation |

| Class | Structure | Main role |
|---|---|---|
| **IgA** | Dimer with secretory component | Mucosal secretions |
| **IgE** | Monomer | Mast-cell binding; allergy and parasites |
| **IgD** | Monomer | B-cell receptor |

## Antigen Presentation

**T cells cannot see free antigen.** They recognize peptide displayed on **MHC** molecules:

- **MHC class I** — on the surface of **all nucleated cells**, bearing fragments of that cell's own proteins (and viral proteins if infected); recognized by **CD8⁺ cytotoxic T cells**.
- **MHC class II** — only on **antigen-presenting cells (APCs)**, mostly derived from monocytes; presents fragments of **endocytosed foreign** proteins to **CD4⁺ helper T cells**. This is the defining difference: MHC I says "here is what I am making"; MHC II says "here is what I have eaten".

This pairing — **CD4/MHC II, CD8/MHC I** — is the most examined immunology fact in the chapter.

## Cells of Adaptive Immunity

**T lymphocytes** mature in the thymus and are responsible for cell-mediated immunity. Their **T-cell receptor (TCR)** is usually an **αβ heterodimer**; each T cell also carries one CD marker that fixes its class:

| T-cell class | Marker | Job |
|---|---|---|
| **Helper T cell** | **CD4⁺** | Recognizes MHC II; secretes cytokines that activate B cells and macrophages |
| **Cytotoxic T cell** | **CD8⁺** | Recognizes MHC I; kills infected or tumor cells |
| **Regulatory T cell** | **CD4⁺ CD25⁺** | Suppresses immune responses and prevents autoimmunity |
| **γδ T cell** | γδ TCR | Mainly in **epithelia**; early defense at body surfaces |

**B lymphocytes** mature in bone marrow and carry a **B-cell receptor (BCR)** — surface **IgM or IgD** — that binds antigen directly, without MHC. On activation they become **plasma cells** (antibody factories) and **memory cells**. **NK cells** kill without prior sensitization and without MHC restriction.

## How a Lymphoid Nodule Forms

This sequence explains every secondary lymphoid organ in the chapter:

1. Lymphocytes are packed in a meshwork of **reticulin** made by **fibroblastic reticular cells**; most APCs among them are **dendritic cells** with many processes.
2. In a secondary organ, **follicular dendritic cells (FDCs)** display antigen–antibody complexes on their large surface. A B cell whose BCR binds that antigen is held there.
3. With cytokines from helper T cells, that B cell **proliferates clonally**, producing a large transient nodule with a pale **germinal center**.
4. The cells disperse as **plasma cells, various T cells, and B and T memory cells** — which is why the same nodule can look "active" (large germinal center) or "resting" (dense primary follicle).

## Thymus

- **Capsule** with trabeculae dividing it into lobules, each with a **cortex** and a **medulla**.
- **No follicles.** The cortex is densely packed with developing **thymocytes** and epithelial reticular cells; the medulla is paler and contains **Hassall corpuscles** — concentric whorls of epithelial cells, unique to the thymus.
- **Blood–thymus barrier** — formed by **squamous thymic epithelial cells** joined by desmosomes and occluding junctions, together with vascular endothelium and pericytes — prevents

unregulated exposure of thymocytes to antigens.

- **Cortex** is packed with **T lymphoblasts (thymocytes)**, macrophages, and **thymic epithelial cells (TECs)**. Stellate TECs form a **cytoreticulum** to which lymphocytes attach instead of reticulin fibers; these TECs are **antigen-presenting cells** expressing **MHC class II** and secreting cytokines — which is why the thymus is also counted among endocrine glands.
- **Hassall corpuscles** — aggregates of medullary TECs, concentrically arranged, **up to 100 µm**, **unique to the medulla**; their cells secrete cytokines that control local dendritic cells and promote regulatory T cells for peripheral tolerance.
- Function: **T-cell maturation and selection**. **Positive selection occurs in the cortex**: only thymocytes whose TCR recognizes the individual's own MHC molecules survive. **Negative selection occurs in the medulla**: thymocytes that bind self-antigen too tightly are deleted, leaving tolerance. Medullary TECs express **Aire**, which lets them display a wide sample of tissue self-antigens for this test.
- **Involutes** after puberty, replaced by adipose tissue.

## Mucosa-Associated Lymphoid Tissue

**Tonsils** (palatine, lingual, pharyngeal/adenoid, tubal) — a partial capsule (hemicapsule), no afferent lymphatics, epithelium invaginates into crypts. **Peyer's patches** in the ileum; **appendix**; **MALT** throughout the gut, respiratory tract, and genitourinary tract.

## Lymph Nodes

| Region | Contents | Function |
|---|---|---|
| **Cortex** | **Lymphoid follicles** — primary (dense) or secondary (with pale **germinal center** and dark mantle) | **B-cell** zone |
| **Paracortex** | Diffuse lymphoid tissue, **high endothelial venules (HEV)** | **T-cell** zone; lymphocytes enter from blood here |
| **Medulla** | Cords and sinuses, macrophages and plasma cells | Filtration; antibody secretion |

**Flow:** afferent lymphatics → subcapsular sinus → trabecular sinuses → medullary sinuses → **efferent lymphatics at the hilum**. The one-way flow explains why nodal metastases lodge first in the subcapsular sinus.

## Spleen

- **No afferent lymphatics.** It filters **blood**.
- **White pulp** — lymphocytes around a central arteriole: the **periarteriolar lymphoid sheath (PALS)** is the T-cell zone; follicles at its periphery are the B-cell zone.
- **Red pulp** — **cords of Billroth** and **sinusoids** whose elongated endothelial **stave cells** are separated by narrow slits.
- **Two routes through the red pulp.** In the **closed circulation**, capillaries from the penicillar arterioles connect directly to sinusoids. In the **open circulation** (about half of the penicillar arterioles), capillaries are open-ended and dump blood into the splenic cords, so plasma and cells must squeeze between stave cells to re-enter the vasculature. Platelets, motile leukocytes, and thin flexible erythrocytes pass; **stiff or swollen erythrocytes at the end of their 120-day life are blocked and removed by macrophages**.
- **Iron recycling.** Iron released from **hemoglobin** during erythrocyte degradation is stored by macrophages in **ferritin**, released bound to **transferrin**, and reused for erythropoiesis. Heme is carried by **hemopexin** and degraded to **bilirubin**, which is excreted in bile.
- **Functions:** blood filtration, removal of old erythrocytes, immune response to blood-borne antigen, and **extramedullary hemopoiesis** in fetal life and marrow failure.

| Organ | Capsule | Afferent lymphatics | Follicles | Unique feature |
|---|---|---|---|---|
| **Thymus** | Yes | None | **No** | **Hassall corpuscles** |
| **Lymph node** | Yes | **Yes** | Yes | Cortex/paracortex/medulla; HEV |
| **Spleen** | Yes | **None** | Yes | **White and red pulp** |
| **Tonsil** | Partial | None | Yes | Crypts |

**CLINICAL CORRELATION**

**DiGeorge syndrome** is thymic aplasia (third and fourth pharyngeal pouches) with T-cell deficiency and hypocalcemia. **Severe combined immunodeficiency** is failure of both arms. **Hodgkin lymphoma** spreads contiguously through nodes and shows Reed–Sternberg cells. **Splenectomy** increases susceptibility to encapsulated organisms — pneumococcus, meningococcus, Haemophilus influenzae — because the spleen's marginal zone is the first line against polysaccharide capsules.

**UNDER THE MICROSCOPE**

Ask **"what is at the center of the nodule?"** Thymus: lobules with dark cortex and pale medulla plus pink whorls (Hassall corpuscles). Lymph node: capsule, subcapsular sinus, hilum. Spleen: **white pulp islands inside a red sea** of sinusoids and cords.

**MEMORY HOOK**

**"Thymus Teaches T cells; Nodes Filter lymph; Spleen Filters blood."** And **"CD4 sees MHC II (2 × 2); CD8 sees MHC I (8 × 1)."**

**EXAM TIP**

Two negative facts are worth more than most positive ones: the **thymus has no follicles**, and the **spleen has no afferent lymphatics**.

## Chapter 14 — one-line answers

- Innate immunity = present at birth, granulocytes and soluble proteins; adaptive = antigen presentation to lymphocytes.
- MHC I is on **all nucleated cells** and presents endogenous peptide to **$CD8^+$** cells.
- MHC II is on **APCs** (mostly monocyte-derived) and presents endocytosed peptide to **$CD4^+$** cells.
- BCR = surface **IgM or IgD**; TCR is usually **αβ** with a CD co-receptor.
- T cells: CD4 helper, CD8 cytotoxic, $CD4^+CD25^+$ regulatory, γδ in epithelia.
- B cells make antibodies (humoral); T cells mediate cell-mediated immunity.
- Plasma cells secrete antibody; memory B and T cells give the rapid second response.
- Primary lymphoid organs: **bone marrow (B)** and **thymus (T)**.
- Secondary: lymph nodes, MALT, spleen — where lymphocytes are activated.
- FDCs display antigen–antibody complexes; activated B cells form germinal centers.
- Thymus: cortex = positive selection, medulla = negative selection; **Aire** displays self-antigens.
- Thymus has **no follicles** and Hassall corpuscles in the medulla.
- Lymph node: cortex B cells, paracortex T cells and HEVs, medulla cords and sinuses.
- Spleen: white pulp (PALS = T zone, follicles = B zone) and red pulp (cords + sinusoids).
- Spleen filters blood; **open circulation** lets macrophages remove stiff 120-day erythrocytes.
- Iron is stored as **ferritin**, carried by **transferrin**, and heme iron binds **hemopexin**; the rest becomes bilirubin.

## Active recall

1. **Which MHC class is on all nucleated cells, and which T cell reads it?** — MHC I; $CD8^+$ cytotoxic T cells.

2. **Which cells present on MHC II?** — Antigen-presenting cells, mostly monocyte-derived: dendritic cells, macrophages, B cells.
3. **Where do T cells undergo positive and negative selection?** — Positive selection in the cortex; negative selection in the medulla.
4. **What is the difference between the thymus and the spleen in one line?** — The thymus makes and selects T cells and has no follicles; the spleen filters blood and has no afferent lymphatics.
5. **What does an FDC do?** — Follicular dendritic cells display antigen–antibody complexes to B cells and drive germinal-center formation.

## Exam-style question

**A child has thymic aplasia with absent T cells and hypocalcemia. Which structure failed to develop?**

A. First pharyngeal arch
B. Third and fourth pharyngeal pouches
C. Neural crest
D. Spleen

**Answer: B** — DiGeorge syndrome is a third/fourth pharyngeal pouch defect affecting the thymus and parathyroids.

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 14–1** — The lymphoid organs and main paths of lymphatic vessels.
- **Fig. 14–6** — Activation of lymphocytes.
- **Fig. 14–11** — Thymic selection of functional but not self-reactive T cells.
- **Fig. 14–16** — Regions of a lymph node.
- **Fig. 14–21** — Spleen.
- **Fig. 14–26** — Structure and function of splenic sinusoids.

# PART IV — DIGESTION AND RESPIRATION

Chapters 15–18 · gut, liver, pancreas, lung, and skin

# 15 · Digestive Tract

**Book pages 295–328**
General Structure of the Digestive Tract · Oral Cavity · Esophagus · Stomach · Small Intestine · Large Intestine

## Learning objectives

By the end of this chapter you should be able to:

1. Draw and name the four layers of the gut wall and their contents.
2. Identify each segment of the digestive tract from its mucosa and muscularis.
3. Match each gastric and intestinal cell type to its product.
4. Explain how the small intestine amplifies surface area and how each nutrient is absorbed.

## The core idea

The gut is one tube with four concentric layers, and every region is a modification of those four layers for a specific job: **grinding, acidifying, absorbing, or dehydrating**. Learn the four layers and their regional variations and you can describe any segment from esophagus to anus.

## General Structure of the Digestive Tract

| Layer | Contents | Function |
|---|---|---|
| **Mucosa** | Epithelium + **lamina propria** + **muscularis mucosae** | Digestion, absorption, secretion, barrier |
| **Submucosa** | Loose connective tissue, vessels, lymphatics, **Meissner (submucosal) plexus**, glands where present | Supply and control of secretion |
| **Muscularis** | Inner circular + outer longitudinal smooth muscle, **Auerbach (myenteric) plexus** between them | Peristalsis |
| **Serosa or adventitia** | Mesothelium + connective tissue (intraperitoneal) or connective tissue only (retroperitoneal) | Suspension and protection |

This is the skeleton of every answer about the gut. Two rules organize it:

- In the digestive tract the **mucosa has three components** — a lining epithelium, a lamina propria of loose connective tissue, and smooth muscle from the **muscularis mucosae**. The epithelium changes; the other two do not.
- **Serosa = suspended in the abdomen; adventitia = embedded in surrounding tissue.** The esophagus develops within the mediastinum and therefore has an adventitia along almost its whole length.

Both plexuses belong to the **enteric nervous system**: Meissner's controls secretion and blood flow, Auerbach's controls motility. The lymphocytes and nodules scattered through mucosa and submucosa are **MALT** (Chapter 14).

## Oral Cavity

The lining is **stratified squamous** everywhere, but the degree of keratinization tracks wear: **keratinized on the hard palate and gingiva**, partly keratinized on the lips, and **nonkeratinized** over most of the soft oral mucosa.

**Lips** — three zones: cutaneous (skin with hair and sebaceous glands), **vermilion** (thin keratinized epithelium over vascular connective tissue — the red of the lip), and mucosal (nonkeratinized, wet). A core of **striated muscle** makes the lips mobile.

**Tongue** — skeletal muscle in bundles, covered by stratified squamous epithelium with four papilla types:

| Papilla | Shape | Taste buds | Notes |
|---|---|---|---|
| **Filiform** | Cone-shaped, **heavily keratinized** | **No** | Most numerous; mechanical; gives the whitish dorsal appearance |
| **Fungiform** | Mushroom-shaped, lightly keratinized | Yes | Scattered among filiform |
| **Foliate** | Leaf-like folds | Yes | Lateral margins |
| **Vallate (circumvallate)** | Largest, **1–3 mm** wide, in a trench | **Yes, ~250 per papilla** | **8–12** in a V in front of the terminal sulcus |

**Taste buds** are ovoid intraepithelial organs of **50–100 cells**, about half of them elongated **gustatory cells** with a **7–10-day** turnover. Their microvilli project toward a **taste pore** in the epithelial surface; support cells and basal stem cells complete the bud. **Filiform papillae never carry taste buds** — a standard trap.

**Teeth.** An adult has **32 permanent teeth**. Each has a crown, a constricted neck, and roots in dental alveoli.

- **Enamel** — hardest material in the body: **96% calcium hydroxyapatite**, only **2–3% organic**, **no collagen**. It is built of **enamel rods ~5 µm wide** running the full **~2 mm** thickness. **Ameloblasts** secrete a collagen-free matrix rich in **amelogenin (20 kDa)**, which initiates and guides crystal growth. Enamel is **acellular**; ameloblasts disappear at eruption, so enamel cannot regenerate, and it is lost from decalcified sections.
- **Dentin** — **70% hydroxyapatite** in a **type I collagen** matrix, harder than bone. **Odontoblasts** (cranial neural crest) secrete **predentin**, which mineralizes within about a day. Odontoblasts persist for life, and their processes occupy **dentinal tubules** across the full thickness of the dentin, accompanied by unmyelinated nerves — the basis of dentin sensitivity.
- **Cementum** — bone-like covering of the roots. **Pulp** — vascular, innervated loose connective tissue entering through apical foramina.
- **Periodontium = cementum + periodontal ligament + alveolar bone + gingiva**. The ligament's collagen bundles (**Sharpey fibers**) bind cementum to bone; it is **150–350 µm** thick.

## Esophagus

A muscular tube about **25 cm** long. **Nonkeratinized stratified squamous** epithelium, a thick muscularis mucosae, **esophageal glands in the submucosa** (mucus-secreting, lubricating), and small **cardiac glands** in the lamina propria near the junction.

The muscularis is the examinable progression: **skeletal muscle at the superior end → mixed in the middle → exclusively smooth in the lower third**. The outer layer is **adventitia** except for the distal **1–2 cm**, which lies in the peritoneal cavity and has a serosa. At the **esophagogastric junction** the epithelium changes abruptly from stratified squamous to simple columnar, and the adventitia gives way to serosa.

## Stomach

**Four regions:** the superior **cardia** and inferior **pylorus** resemble each other histologically (branching mucous glands, few parietal or chief cells), while the intervening **fundus and body** contain the main gastric glands. The cardia and pylorus are **histologically similar** to each other and **lack parietal and chief cells**, secreting mucus instead. The surface is perforated by **gastric pits**, which are lined by the same **surface mucous cells** as the stomach lumen and lead into branching glands. **No goblet cells, no villi.**

The gland epithelium contains surface mucous cells, mucous neck cells, parietal cells, chief cells, and enteroendocrine cells, plus pluripotent stem cells in the neck:

| Cell | Location | Product | Function |
|---|---|---|---|
| **Parietal (oxyntic)** | Upper half of gland | **HCl + intrinsic factor** | Large cell, many mitochondria, **intracellular canaliculi**; intrinsic factor is required for **vitamin $B_{12}$** uptake; **eosinophilic** |
| **Chief (zymogenic)** | Deeper half of gland | **Pepsinogen** | Activated to **pepsin** by luminal acid; **basophilic** with apical zymogen granules |
| **Mucous neck** | Neck | Mucus (less alkaline than surface mucus) | Secretion; the proliferative stem-cell zone lies among them |
| **Surface mucous** | Surface and pits | **Viscous mucus + bicarbonate** | The barrier that protects the mucosa |
| **Enteroendocrine** | Throughout | **Gastrin, histamine**, others | Paracrine and endocrine control of secretion |
| **Stem cells** | Narrow neck region | — | Pluripotent source of all the above |

Parietal cells are controlled both by **parasympathetic innervation** and by paracrine **histamine** and **gastrin** from enteroendocrine cells — the pharmacological basis of proton-pump inhibitors and $H_2$ blockers. The **muscularis has three layers** (oblique, circular, longitudinal), an exception to the four-layer template that allows churning.

## Small Intestine

The small intestine is about **5 m long**. Its absorptive surface is amplified at three levels: **plicae circulares** (mucosa + submucosa, about a 3-fold increase), **villi** (mucosa, about 10-fold), and **microvilli** (cell surface, a further 20-fold), for a total absorptive area of **over 200 m²** — the clearest "structure follows function" example in the book.

Each **villus** is a finger of lamina propria covered by simple columnar epithelium, containing a **lacteal** and smooth muscle. Between villi, simple tubular **crypts (of Lieberkühn)** are the proliferative compartment. Smooth muscle of the lamina propria and muscularis mucosae, controlled by **Meissner's plexus**, moves the villi and propels lymph; the muscularis, controlled by **Auerbach's plexus**, produces peristalsis.

### Cell types and what each absorbs or secretes:

- **Enterocytes** — brush border; **disaccharidases and peptidases in the glycocalyx** complete digestion, and the resulting monosaccharides and amino acids are absorbed across the brush border and pass to capillaries.
- **Goblet cells** — mucus.
- **M cells** — specialized epithelium over the lymphoid nodules, which takes up antigen from the lumen and hands it to the lymphocytes below.
- **Paneth cells** — base of crypts, **eosinophilic** granules, secrete **defensins** (and lysozyme).
- **Enteroendocrine cells** — CCK, secretin, GIP, motilin.
- **Stem cells** — crypt base, replacing the epithelium every few days.

**Lipid absorption** is different and examinable: lipid products associate with **bile salts**, enter enterocytes, are re-esterified to **triglycerides** and packaged as **chylomicrons**, and leave through the **lacteal** rather than the capillary.

| Region | Defining feature |
|---|---|
| **Duodenum** | **Duodenal (Brunner) glands in the submucosa**; leaf-like villi |
| **Jejunum** | Tallest villi; no Brunner glands; Peyer's patches belong to the ileum |
| **Ileum** | **Peyer's patches** (mucosal and submucosal); shortest villi |

## Large Intestine

Three regions: the short **cecum** with the **appendix**; the long **colon** (ascending, transverse, descending, **sigmoid**); and the **rectum**.

**No villi**, straight tubular crypts lined by **colonocytes** (absorptive, water and electrolyte uptake) and **abundant goblet cells**. The outer longitudinal muscle is gathered into three **teniae coli**, which produce the bulging **haustra**; the serosa carries hanging fat masses, the **omental appendages** (appendices epiploicae). Function: water and electrolyte absorption, mucus secretion, bacterial fermentation. The **appendix** is a blind diverticulum of the cecum with abundant lymphoid tissue in its lamina propria and submucosa — part of MALT.

The **rectum** (**10–12 cm**) continues as the **anal canal** (**3–4 cm**). There the mucosa is highly vascular, folded into **anal columns (of Morgagni)** with intervening anal sinuses, and the columnar lining changes to **stratified squamous** epithelium at the rectoanal junction. Swollen mucosal or submucosal veins here are **hemorrhoids**.

**CLINICAL CORRELATION**

**Peptic ulcer** — H. pylori breaches the mucous barrier; acid and pepsin digest the wall, classically in the duodenal bulb or gastric antrum. **Pernicious anemia** — loss of parietal cells means no intrinsic factor, so $B_{12}$ cannot be absorbed in the terminal ileum. **Celiac disease** — villous atrophy, crypt hyperplasia, and intraepithelial lymphocytes in the proximal small intestine, driven by gluten. **Barrett esophagus** — chronic acid reflux causes metaplasia from stratified squamous to **intestinal-type columnar epithelium with goblet cells**, a premalignant change. **Diverticulosis** — mucosa and submucosa herniate between the teniae coli; trapped feces cause diverticulitis.

**UNDER THE MICROSCOPE**

Identify the segment by **the surface, not the glands**. **Villi + crypts = small intestine. No villi + deep pits + thick gland mass = stomach. No villi + straight crypts + many goblet cells = colon.** Then find the regional marker: Brunner glands = duodenum; Peyer's patches = ileum; three muscle layers = stomach.

**MEMORY HOOK**

**"Duodenum has Brunner Below; Ileum has Peyer's Patches."** And **"Filiform = Feel only; Vallate = Carry most taste."**

**HIGH-YIELD PEARL**

The numbers that appear in questions: enamel **96%** mineral and **2 mm** thick; dentin **70%**; esophageal length **25 cm**; **8–12** vallate papillae with about **250** taste buds each. The stomach has **no goblet cells** and **no villi**; the intestine has **both**.

## Chapter 15 — one-line answers

- Four gut layers, outside in: mucosa, submucosa, muscularis, serosa or adventitia.
- Gut mucosa = epithelium + lamina propria + muscularis mucosae.
- Serosa where the gut is suspended; adventitia where it is embedded.
- Auerbach lies between the muscle layers (motility); Meissner lies in the submucosa (secretion, blood flow).
- Keratinized oral epithelium: hard palate and gingiva. Everywhere else, mostly nonkeratinized.
- Filiform papillae are the only type without taste buds.
- Vallate papillae: 8–12, largest, ~250 taste buds each.
- Enamel is acellular and cannot regenerate; ameloblasts disappear at eruption.
- Dentin is 70% hydroxyapatite in type I collagen; odontoblast processes stay in dentinal tubules.
- Esophageal muscle: skeletal above, mixed in the middle, smooth below.
- Esophagogastric junction: squamous → columnar, and adventitia → serosa.
- Parietal cells: HCl + intrinsic factor, intracellular canaliculi. Chief cells: pepsinogen.

- Stomach has three muscle layers and no goblet cells; the colon has teniae coli and no villi.
- Duodenum = Brunner glands; ileum = Peyer's patches; colon = teniae coli.
- Lipids leave the villus as chylomicrons in the lacteal; sugars and amino acids enter capillaries.

## Active recall

1. **Name the four layers of the gut wall in order.** — Mucosa, submucosa, muscularis, serosa or adventitia.
2. **Which cells secrete HCl and intrinsic factor?** — Parietal cells; they have intracellular canaliculi and many mitochondria.
3. **What are the three surface amplifiers of the small intestine?** — Plicae circulares, villi, and microvilli.
4. **How do lipids leave the villus?** — As chylomicrons through the lacteal, not through capillaries.
5. **Which segment has teniae coli and no villi?** — The colon (large intestine).

## Exam-style question

**A biopsy of the small intestine shows submucosal mucous glands. Which segment is it?**

A. Duodenum
B. Jejunum
C. Ileum
D. Colon

**Answer: A** — Brunner (duodenal) glands lie in the submucosa of the duodenum only.

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 15–1** — The digestive system.
- **Fig. 15–8** — Ultrastructure of dentinal tubule.
- **Fig. 15–15** — Wall of the stomach with rugae.
- **Fig. 15–21** — Pyloric glands.
- **Fig. 15–28** — Microvasculature, lymphatics, and muscle in villi.
- **Fig. 15–34** — Mucosa of the rectoanal junction.

# 16 · Organs Associated with the Digestive Tract

**Book pages 329–348**
Salivary Glands · Pancreas · Liver · Biliary Tract & Gallbladder

## Learning objectives

By the end of this chapter you should be able to:

1. Compare the secretory units and ducts of the three major salivary glands.
2. Explain how the exocrine pancreas makes and delivers digestive enzymes and bicarbonate.
3. Trace blood and bile through a hepatic lobule in opposite directions.
4. Explain how the gallbladder concentrates bile and what controls its release.

## The core idea

Four gland systems share one duct system and one embryological origin: the salivary glands **lubricate and begin digestion**, the pancreas **digests**, the liver **processes**, and the gallbladder **stores**. Each is built around its job, and each has one histologic signature that identifies it.

## Salivary Glands

The three major pairs produce about **90%** of saliva; minor glands scattered through the oral mucosa contribute the rest. Total output is **0.75–1.5 L/day**.

| Gland | Secretory units | Secretion |
|---|---|---|
| **Parotid** | **Serous acini only** | Watery, enzyme-rich |
| **Submandibular** | Mixed, **mainly serous** acini | Mixed |
| **Sublingual** | Mixed, **mainly mucous tubules**, some with **serous demilunes** (partly a fixation artifact) | Viscous |

**Serous cells** are protein-secreting and form round or oval acini; **mucous cells** secrete mucin and form elongated tubules. **Myoepithelial cells** sit between the secretory cells and the basal lamina and squeeze the secretion onward.

The duct system is a three-step sequence, and each step has a different epithelium:

1. **Intercalated ducts** — simple cuboidal; the smallest.
2. **Striated ducts** — simple columnar, with **mitochondria-lined basolateral folds** that reabsorb **$Na^+$ and $Cl^-$** and add **$HCO_3^-$**. This is why saliva is **hypotonic** and why the striated duct is the histologic signature of a salivary gland.
3. **Interlobular (excretory) ducts** — unusually, lined by **stratified cuboidal or columnar** epithelium.

## Pancreas

A compound acinar gland, entirely **exocrine** except for the scattered **islets of Langerhans**. It secretes about **1.5 L/day** of alkaline juice.

- **Acinar cells** — **pyramidal**, with a basal nucleus, abundant **rough ER and Golgi** in the basal cytoplasm, and **eosinophilic zymogen granules** apically. The cell is polarized for export: make protein at the base, package it at the apex.
- **Centroacinar cells** — pale cells inserted into the acinar lumen; they begin the duct system inside the acinus.
- **Ducts** — intercalated → intralobular → interlobular → main pancreatic duct. **Centroacinar and intercalated duct cells secrete $HCO_3^-$** under **secretin**, neutralizing chyme and establishing the pH at which pancreatic enzymes work. The high pH also keeps the enzymes inactive until they

reach the duodenum.

- **Endocrine islets** — pale, round clusters between acini (details in Chapter 20).

The pancreas has only a **very thin capsule**, whose septa carry the larger vessels and ducts and separate the parenchyma into lobules; each acinus is surrounded by a basal lamina and a delicate reticular sheath — a useful contrast with the parotid, which has striated ducts and no islets.

## Liver

The liver is both an **endocrine** gland (plasma proteins into blood) and an **exocrine** gland (bile into ducts). That dual output explains its unusual architecture: two fluid streams run in opposite directions through the same lobule.

**Blood supply.** About **75%** of hepatic blood is nutrient-rich, $O_2$-poor **portal vein** blood; **25%** is oxygenated **hepatic artery** blood. Both enter at the portal areas and flow **toward the central vein**.

- **Classic lobule** — a polygon of **plates of hepatocytes radiating toward a central vein**, with **portal areas** at the corners. Connective tissue between lobules is sparse, becoming more abundant in the portal areas.
- **Portal area (triad + one)** — a **portal venule**, a **hepatic arteriole**, a **bile ductule**, and a small **lymphatic**.
- **Hepatocytes** — large polygonal epithelial cells with large central nuclei; about **50% are polyploid** and many are **binucleated**. They contain abundant **rough and smooth ER**, many small **Golgi complexes**, glycogen, and peroxisomes.
- **Sinusoids** — discontinuous, **fenestrated** endothelium. Between endothelium and hepatocytes is the **perisinusoidal space (of Disse)**, where plasma exchanges with hepatocytes.
- **Kupffer cells** — stellate macrophages in the sinusoidal lining; remove effete erythrocytes and release iron and bilirubin.
- **Hepatic stellate (Ito) cells** — in the perisinusoidal space, storing **vitamin A** and other fat-soluble vitamins in lipid droplets; they become myofibroblasts in fibrosis.
- **Bile canaliculi** — grooves **between adjacent hepatocytes**, sealed by tight junctions. Bile flows **toward** the portal area while blood flows **away** from it.
- **Canals of Hering** — where canaliculi join the bile ductules; all ducts eventually form the left and right hepatic ducts.

**Functions:** carbohydrate, lipid, and protein metabolism; **albumin** and clotting factor synthesis; detoxification by SER and peroxisomes; bile production; bilirubin conjugation; storage of glycogen, iron, and vitamins A, D, and $B_{12}$.

**Zonation. Zone I** (periportal) receives the most oxygen and nutrients and performs oxidative metabolism, such as protein synthesis; **zone III** (pericentral) receives the least, and its cells are the first to accumulate fat and undergo ischemic necrosis. Zone III is also where drug biotransformation predominates — the book's basis for many patterns of hepatic injury.

## Biliary Tract & Gallbladder

Every bile-conducting structure after the canaliculi is lined by **cholangiocytes** — simple cuboidal or columnar epithelium.

**Route:** canaliculi → canals of Hering → bile ductules → interlobular ducts → left and right hepatic ducts → **common hepatic duct** → **cystic duct** → gallbladder → **common bile duct** → **hepatopancreatic ampulla (of Vater)** → duodenum.

The **gallbladder** is a sac with a **highly folded mucosa**, a well-vascularized lamina propria, **no muscularis mucosae or submucosa**, and a **muscularis of interwoven bundles**. Its columnar cholangiocytes **absorb water**, concentrating bile; the muscle contracts under **cholecystokinin (CCK)** released by duodenal enteroendocrine cells when food arrives.

**CLINICAL CORRELATION**

**Cirrhosis** is the end result of hepatocyte injury: fibrosis with regenerative nodules, portal hypertension, and synthetic failure. **Acute pancreatitis** is autodigestion by prematurely activated enzymes; **chronic pancreatitis** destroys acini and islets, producing diabetes and steatorrhea. **Carcinoma of the head of the pancreas** obstructs the common bile duct and produces painless obstructive jaundice. **Gallstones** form when bile is supersaturated with cholesterol or pigment; impaction in the cystic or common bile duct produces biliary colic and obstructive jaundice.

**UNDER THE MICROSCOPE**

The liver is recognized by **cords of eosinophilic cells with central veins and portal areas**. The pancreas by **dark purple acini with pink apical granules and pale islets**. Pancreas versus parotid: the pancreas has **islets** and **no striated ducts**; the parotid has **striated ducts** and **no islets**.

**EXAM TIP**

Bile flows **toward** the portal triad; blood flows **away** from it. Students who memorize "everything flows to the center" lose this mark every year.

## Chapter 16 — one-line answers

- Major salivary glands make ~90% of saliva; minor glands ~10%.
- Parotid = serous only. Submandibular = mixed, mainly serous. Sublingual = mixed, mainly mucous with demilunes.
- Intercalated duct = simple cuboidal; striated duct = simple columnar with basal folds; excretory duct = stratified cuboidal or columnar.
- Striated ducts reabsorb $Na^+$ and $Cl^-$ and add $HCO_3^-$, making saliva hypotonic.
- Myoepithelial cells contract to expel secretion.
- Pancreatic acinar cell: basal RER and Golgi, apical zymogen granules — polarized for export.
- Centroacinar cells sit inside the acinus and begin the duct.
- Secretin drives $HCO_3^-$ secretion from centroacinar and intercalated duct cells.
- Liver blood: 75% portal vein, 25% hepatic artery.
- Portal area = portal venule + hepatic arteriole + bile ductule + lymphatic.
- About half of hepatocytes are polyploid; many are binucleated.
- Kupffer cells = sinusoidal macrophages. Ito cells = vitamin A storage.
- Bile canaliculi lie between hepatocytes, sealed by tight junctions, and drain to the canals of Hering.
- Hepatic zone 1 = periportal, most oxygen; zone 3 = pericentral, most biotransformation and first to die in ischemia.
- Gallbladder: folded mucosa, no muscularis mucosae, no submucosa; cholangiocytes absorb water; CCK contracts the muscle.

## Active recall

1. **Which salivary gland is purely serous?** — The parotid. Submandibular is mixed, mainly serous; sublingual is mixed, mainly mucous.
2. **What do striated ducts do?** — Reabsorb $Na^+$ and $Cl^-$ and add $HCO_3^-$, making saliva hypotonic.
3. **What do centroacinar cells secrete, and under what hormone?** — Bicarbonate-rich fluid, under secretin.
4. **Trace bile from canaliculus to duodenum.** — Canaliculi → canals of Hering → bile ductules → interlobular ducts → hepatic ducts → common hepatic duct → cystic duct → gallbladder → common bile duct → hepatopancreatic ampulla.
5. **Which way does blood flow in a hepatic lobule?** — From portal areas at the periphery toward the central vein.

## Exam-style question

**A liver biopsy shows a portal area containing a venule, an arteriole, a bile ductule, and a lymphatic. Which structure is missing from the center of the lobule?**

A. Central vein
B. Portal vein
C. Hepatic artery
D. Bile duct

**Answer: A** — The portal triad (plus lymphatic) sits at the lobule corner; the central vein sits at the center.

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 16–1** — Major salivary glands.
- **Fig. 16–5** — Submandibular gland and sublingual gland.
- **Fig. 16–9** — Pancreatic acini.
- **Fig. 16–12** — Hepatic lobule.
- **Fig. 16–16** — Hepatocyte ultrastructure and major functions.
- **Fig. 16–20** — Gallbladder.

# 17 · The Respiratory System

**Book pages 349–370**
Nasal Cavities · Pharynx · Larynx · Trachea · Bronchial Tree & Lung · Lung Vasculature & Nerves · Pleural Membranes · Respiratory Movements

## Learning objectives

By the end of this chapter you should be able to:

1. Follow the airway from nose to alveolus and name what changes at each level.
2. Identify respiratory epithelium and its cell types.
3. Explain the blood–air barrier, surfactant, and the mechanics of ventilation.
4. Distinguish conducting from respiratory airways in a micrograph.

## The core idea

The respiratory tract is a **conducting** series (nose to terminal bronchiole) followed by a **respiratory** series (respiratory bronchiole to alveolus). The transition is marked by the disappearance of cartilage and the appearance of alveoli. Walking down the tree, each level loses one structural support and gains a thinner epithelium.

## Nasal Cavities

Each nasal cavity has a **vestibule** where air enters, and **three conchae** projecting from the medial wall that create turbulence and force air over a large mucosal surface.

- **Vestibule** — stratified squamous epithelium with **vibrissae** (stiff hairs) that filter large particles.
- **Respiratory region** — **respiratory epithelium**: pseudostratified ciliated columnar with goblet cells, over a highly vascular lamina propria containing seromucous glands. It warms, humidifies, and cleans air; the cilia sweep mucus toward the pharynx.
- **Olfactory region** — the roof and part of the superior concha: **olfactory epithelium**, about **10 cm²** and up to **100 µm** thick, containing **bipolar olfactory neurons**, supporting cells, and basal stem cells. Each neuron has about **20 long nonmotile cilia** bearing odorant receptors. Olfactory neurons are replaced every **1–2 months** — unusual for neurons. **No goblet cells**, but **Bowman glands** in the lamina propria.

**The respiratory epithelium is a mixed population**, and the proportions are examinable:

| Cell | Share | Function |
|---|---|---|
| **Ciliated columnar cells** | Majority | Move the mucus blanket |
| **Goblet cells** | — | Secrete mucus |
| **Brush cells** | ~3% | Chemosensory; short blunt microvilli |
| **Small granule (Kulchitsky) cells** | ~3% | **DNES**; dense-core granules 100–300 nm |
| **Basal cells** | ~30% | Stem cells |

## Pharynx

**Nasopharynx** — respiratory epithelium with **lymphoid tissue** (pharyngeal tonsil) in the lamina propria. **Oropharynx** and **laryngopharynx** — stratified squamous epithelium, continuous with the oral cavity and esophagus. The change from respiratory to stratified squamous epithelium marks where the airway is exposed to abrasion from food.

## Larynx

A short air passage between pharynx and trachea, with skeletal cartilages and intrinsic muscles.

- **Epiglottis** — elastic cartilage, with stratified squamous epithelium on its lingual surface and respiratory epithelium on its laryngeal surface; the switch is abrupt.
- **Vocal folds (true cords)** — **stratified squamous** epithelium, avascular, with the **vocal ligament** and the **vocalis muscle**. Tension and vibration produce sound.
- **False cords (vestibular folds)** — respiratory epithelium and glands.

## Trachea

- **Epithelium** — respiratory epithelium (pseudostratified ciliated columnar with goblet cells); the trachea is **10–12 cm** long and supported by **C-shaped rings** of hyaline cartilage.
- **Lamina propria** — elastic fibers, lymphocytes, seromucous glands.
- **Cartilage** — **C-shaped hyaline rings**, incomplete posteriorly.
- **Trachealis muscle** — smooth muscle bridging the posterior gap, allowing the esophagus to expand.

## Bronchial Tree & Lung

**Bronchi** — the primary bronchus enters each lung and branches repeatedly. Cartilage breaks into **irregular plates** rather than rings, a complete ring of smooth muscle appears, and elastic fibers and glands persist. Each **tertiary (segmental) bronchus** with its branches forms a **bronchopulmonary segment** — a pyramidal portion of lung with its own blood supply, separated from its neighbors by connective tissue septa.

**Bronchioles** — airways of **1 mm or less**, formed after about the **tenth generation** of branching. They have **no cartilage and no glands**, a simple ciliated columnar or cuboidal epithelium, circular smooth muscle, and elastic fibers.

- **Club cells** (formerly Clara cells) replace goblet cells as the main secretory cell; they secrete **surfactant-like material**, help innate immunity, and their **smooth ER P450 system detoxifies inhaled compounds**. They are the progenitor cells of the bronchiolar epithelium.
- **Terminal bronchiole** — the last conducting airway, lined mainly by club cells, with **no alveoli**.

**Respiratory region:** terminal bronchiole → **respiratory bronchiole** (simple cuboidal, interrupted by alveoli) → **alveolar duct** (a linear series of alveoli) → **alveolar sac** (a cluster). Gas exchange begins at the respiratory bronchiole.

| Cell | Shape | Function | Share of alveolar lining |
|---|---|---|---|
| **Type I pneumocyte** | Squamous, extremely thin | **Gas exchange**; forms the blood–air barrier | Lines almost the **entire** alveolar surface |
| **Type II pneumocyte** | Cuboidal, **lamellar bodies** | **Surfactant**; innate immunity; progenitor for type I | A **small part** of each alveolus |
| **Alveolar macrophage** | Large, free ("dust cell") | Phagocytosis | — |

### Numbers to know:

- Each alveolus is about **200 µm** across. Each adult lung has about **200 million alveoli**, with a total internal surface area of about **75 m²**.
- The **blood–air barrier** = surfactant film + **type I pneumocyte** + fused basal laminae + capillary endothelium. Its total thickness is **0.1–1.5 µm**, and the attenuated type I cytoplasm can be as thin as **25 nm**.
- **Surfactant** is an oily mixture of **cholesterol, phospholipids, and surfactant proteins** secreted by **type II cells and club cells**. It lowers surface tension and prevents alveolar collapse at end-expiration.
- **Pores of Kohn** — **10–15 µm** openings that connect adjacent alveoli and equalize air pressure.

## Lung Vasculature & Nerves

The lung has a **double blood supply**: the **pulmonary artery** carries deoxygenated blood for gas exchange and forms a dense capillary network around alveoli, while the **bronchial arteries** (from the aorta) supply the conducting airways and pleura. Pulmonary arteries are unusual in carrying **deoxygenated** blood and in having thin walls suited to low pressure. Autonomic nerves control bronchial smooth muscle and gland secretion — parasympathetic (vagal) constriction, sympathetic relaxation.

## Pleural Membranes

A **serous membrane** of simple squamous mesothelium over connective tissue: **visceral pleura** covers the lung and is continuous with **parietal pleura** lining the thoracic wall. A thin fluid film lets the layers slide while surface tension holds them together, so the lung follows the chest wall.

## Respiratory Movements

Ventilation is pressure-driven: the **diaphragm** and **external intercostal muscles** enlarge the thoracic cavity, pleural pressure falls, the lung expands passively, and air enters. **Elastic recoil** of the lung and surface tension at the alveolar air–liquid interface drive expiration — precisely why surfactant deficiency causes collapse.

**CLINICAL CORRELATION**

**Respiratory distress syndrome of the newborn** — immature type II pneumocytes produce insufficient surfactant, so alveoli collapse; treatment is exogenous surfactant and antenatal steroids. **Emphysema** — destruction of alveolar walls enlarges air spaces and reduces surface area. **Asthma** — bronchoconstriction, mucosal edema, eosinophilic inflammation, thickened basement membrane, and smooth muscle hypertrophy. **Chronic bronchitis** — mucus hypersecretion and gland hypertrophy. **Cystic fibrosis** — defective CFTR thickens secretions, causing bronchiectasis and recurrent infection. **Lung cancer** — the book names three types: **squamous cell carcinoma** (smoking-related, arises in segmental bronchi), **adenocarcinoma** (commonest in non-smokers, arises peripherally in bronchioles and alveoli), and **small cell carcinoma** (less common, highly malignant, arising from small granule Kulchitsky cells of the bronchial epithelium).

**UNDER THE MICROSCOPE**

Walk down the tree and count what disappears. **Cartilage rings + glands = trachea. Cartilage plates + no glands = bronchus. No cartilage, no glands, muscular wall = bronchiole.** Then look for alveoli in the wall: if present, it is a respiratory bronchiole.

**MEMORY HOOK**

**"Rings, Plates, None"** for cartilage from trachea → bronchus → bronchiole. And **"Type I for gas, Type II for surfactant, macrophage for garbage."**

**EXAM TIP**

Club cells are found in **bronchioles**, not alveoli, and are the progenitor cells of the bronchiolar epithelium. A question that places them in the alveolus is testing whether you know the level of the airway.

## Chapter 17 — one-line answers

- The conducting airway is nose → pharynx → larynx → trachea → bronchi → bronchioles → terminal bronchioles.
- Respiratory epithelium is pseudostratified ciliated columnar with goblet cells.
- Cartilage changes down the tree: C-shaped rings → irregular plates → none.
- The respiratory bronchiole is the first airway with alveoli in its wall.
- Type I pneumocytes line almost the entire alveolar surface and form the thin gas-exchange barrier.
- Type II pneumocytes secrete surfactant and divide to replace type I cells.

- The blood–air barrier is surfactant, type I cell, fused basal laminae, and capillary endothelium (0.1–1.5 µm).
- Alveolar macrophages (dust cells) clear particles and debris.
- The pleura is a serous membrane: visceral and parietal layers with a fluid-filled cavity.

## Active recall

1. **What marks the transition from conducting to respiratory airway?** — The appearance of alveoli in the wall — the respiratory bronchiole.
2. **Which cells line almost the entire alveolar surface?** — Type I pneumocytes.
3. **What do type II pneumocytes secrete, and why does it matter?** — Surfactant, which lowers surface tension and prevents alveolar collapse.
4. **Name the components of the blood-air barrier.** — Surfactant film, type I pneumocyte, fused basal laminae, capillary endothelium.
5. **How does cartilage change down the tree?** — C-shaped rings (trachea) → irregular plates (bronchi) → absent (bronchioles).

## Exam-style question

**A premature infant develops respiratory distress because alveoli collapse at end-expiration. Which cell is functionally immature?**

A. Type I pneumocyte
B. Type II pneumocyte
C. Alveolar macrophage
D. Club cell

**Answer: B** — Type II cells produce surfactant; their immaturity causes neonatal respiratory distress syndrome.

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 17–1** — Anatomy of the respiratory system.
- **Fig. 17–4** — Larynx.
- **Fig. 17–8** — Bronchial wall.
- **Fig. 17–11** — Pulmonary circulation, terminal and respiratory bronchioles, and alveoli.
- **Fig. 17–15** — Ultrastructure of the blood-air barrier.
- **Fig. 17–18** — Pleura.

# 18 · Skin

**Book pages 371–392**
Epidermis · Dermis · Subcutaneous Tissue · Sensory Receptors · Hair · Nails · Skin Glands · Skin Repair

## Learning objectives

By the end of this chapter you should be able to:

1. Name the epidermal layers in order and the defining feature of each.
2. Match each epidermal cell type to its origin and function.
3. Identify the dermis, hypodermis, and each skin appendage.
4. Explain the stages of skin repair, and recognize healing by first and second intention.

## The core idea

Skin is a **stratified, keratinizing, self-renewing barrier** with appendages embedded in it. The epidermis is a factory that moves cells outward and turns them into keratin; the dermis provides mechanical and nutritional support; the appendages provide temperature control, sensation, and protection. Every fact in the chapter is a consequence of one of those three jobs.

## Epidermis

**Thickness: 75–150 µm** in thin skin, **400–600 µm** in thick skin. Full skin (epidermis + dermis) is about **4 mm** on the back and **1.5 mm** on the scalp. The epidermis is **avascular** — nutrients and $O_2$ arrive by diffusion from the dermal papillae. It turns over about every **15–30 days**.

| Layer | Cells | Key feature |
|---|---|---|
| **Stratum basale** | Mitotic stem cells, melanocytes, Merkel cells | Single layer on the basement membrane, attached by **hemidesmosomes and integrins**; **keratin filaments ~10 nm** first appear here |
| **Stratum spinosum** | Polyhedral cells with **desmosomes** | "Spiny" because shrinkage pulls apart desmosome-bound bundles (**tonofibrils**); **Langerhans cells** live here |
| **Stratum granulosum** | Flattened cells with **keratohyalin granules** and **lamellar granules** | Keratohyalin contains **filaggrin**, which aggregates keratin; lamellar granules discharge lipid into the intercellular space |
| **Stratum lucidum** | Clear, anucleate, refractile | **Only in thick skin** of palms and soles |
| **Stratum corneum** | Anucleate **squames** | The barrier; keratin packed in an electron-dense matrix with a lipid envelope |

Keratin filaments contain at least **six polypeptides of 40–70 kDa**, synthesized as the cell differentiates. The **lipid envelope** between squames is what makes epidermis impermeable to water.

| Cell | Origin | Function | Identifying feature |
|---|---|---|---|
| **Keratinocyte** | Ectoderm | Barrier, keratin production | 90% of epidermal cells; desmosomes |
| **Melanocyte** | **Neural crest** | **Melanin** via tyrosinase; transferred in melanosomes | Dark, dendritic; basal layer |
| **Langerhans cell** | **Monocyte/bone marrow** | Antigen presentation | Dendritic, stratum spinosum; **2–8%** of epidermal cells |
| **Merkel cell** | Neural crest | **Light touch** (mechanoreception) | Basal, with a sensory nerve ending |

**Melanocytes** — about **one per five or six keratinocytes** (**600–1200/mm²** of thin skin), with processes reaching **10–30 keratinocytes**. **Tyrosinase** converts tyrosine to DOPA and then to **eumelanin** (brown-black) or **pheomelanin** (red-yellow). Mature melanosomes are ellipsoid, **0.5 × 1 µm**, and are transferred to keratinocytes, which hold more pigment than the cells that make it. The pigment sits over the nucleus and **protects nuclear DNA from UV damage** — the reason tanning exists and the reason melanoma risk tracks sun exposure. One melanocyte plus its keratinocytes is an **epidermal-melanin unit**. Melanocytes are about **four times more numerous and active in thin skin** than in thick palmar and plantar skin.

**The melanin rule:** densities of epidermal-melanin units are **similar across populations**; what differs is how rapidly melanin is made and how abundantly it accumulates in keratinocytes. Dark skin is not skin with more melanocytes.

## Dermis

- **Papillary dermis** — loose connective tissue forming **dermal papillae** that interdigitate with the epidermis; contains the **subpapillary microvascular plexus**, which sends capillary loops into the papillae.
- **Reticular dermis** — dense irregular connective tissue with thick collagen bundles and elastic fibers; up to **4 mm** thick on the back.

Dermal vasculature has two jobs: **nutrition** (two plexuses, subpapillary and deep) and **thermoregulation** through **arteriovenous anastomoses** between them.

**Thick versus thin skin.** Thick skin (palms, soles) has a thick stratum corneum, a **stratum lucidum**, and **no hair follicles**; it has many sweat glands. Thin skin covers most of the body, has hair follicles and sebaceous glands, and a thinner epidermis.

## Subcutaneous Tissue

The **hypodermis** — loose connective tissue and adipose tissue (**panniculus adiposus**) binding skin loosely to underlying organs, so the skin can slide. It insulates, cushions, stores energy, and determines how a wound behaves and how an injection is absorbed.

## Cutaneous Sensory Receptors

| Receptor | Location | Structure | Modality |
|---|---|---|---|
| **Free nerve endings** | Epidermis and dermis | Unencapsulated | Pain, temperature extremes, mechanical movement |
| **Merkel cell** | Basal epidermis | Cell + disc-like nerve ending | **Light touch**, sustained pressure |
| **Meissner corpuscle** | Dermal papilla | Elliptical, **~150 µm**, stacked Schwann cells | **Light touch**, fine discrimination |
| **Pacinian corpuscle** | Deep dermis and hypodermis | Oval, **~1 mm**, **15–50 concentric lamellae** | **Pressure and high-frequency vibration** |
| **Krause end bulb** | Penis and clitoris mainly | Thin collagenous capsule | Low-frequency vibration |
| **Ruffini corpuscle** | Dermis | Fusiform capsule anchored to collagen | **Tissue distortion and sustained stretch** |

## Hair

Hairs are keratinized structures formed in **hair follicles**, epidermal invaginations extending into the dermis.

- The **hair bulb** at the base contains the **matrix** (rapidly proliferating keratinocytes) and is invaginated by a vascular **dermal papilla** that supplies it.

- The hair root has three concentric layers — **medulla, cortex, cuticle** — surrounded by **internal and external root sheaths** continuous with the epidermis, a **glassy membrane** (partly basal lamina), and a connective tissue sheath.
- The **arrector pili** smooth muscle raises the hair; a **root plexus** provides sensation.
- Density: about **600 hairs/cm²** on the face and **60/cm²** elsewhere.
- Growth cycles: **anagen** (growth), **catagen** (involution), **telogen** (rest).

## Nails

Hard keratin plates on the dorsal surface of the distal phalanx. The **nail root** lies under a skin fold whose cornified layer forms the **cuticle (eponychium)**; keratinocytes proliferate in the **matrix** and differentiate into the **nail plate**, which slides forward over the **nail bed** (which does not contribute). The **lunula** is the visible part of the matrix.

## Skin Glands

**Sebaceous glands — holocrine**: sebocytes accumulate lipid, die, and discharge **sebum** into the hair follicle (pilosebaceous unit). Absent from palms and soles.

**Eccrine sweat glands — merocrine**, coiled tubular, with a secretory portion of cuboidal cells and a duct of **stratified cuboidal** cells; abundant on palms and soles; watery sweat for thermoregulation.

**Apocrine sweat glands — axillae and perineum**, much **wider lumens**, develop after **puberty**, and secrete protein-rich sweat onto hair follicles; odor follows bacterial action.

## Skin Repair

Repair runs through overlapping stages (the book's Figure 18–20): a **clot** forms and platelets release growth factors; **neutrophils and macrophages** remove bacteria and debris (inflammation lasts **2–3 days**); fibroblasts proliferate under growth-factor and macrophage signals and produce new collagen while capillaries grow into the site, forming **granulation tissue**; **keratinocytes** from the epidermal basal layer migrate beneath and through the clot and restore epithelial continuity; and excess collagen is left as **scar**. Epithelialization is complete before the collagen of the scar remodels.

**Supplementary (standard clinical terminology, not used in this chapter):** healing by **first intention** — clean, apposed edges needing minimal granulation tissue; **second intention** — an open wound with abundant granulation tissue, wound contraction by myofibroblasts, and a larger scar. **Burns** are graded by depth: first-degree (epidermis), second-degree (into dermis, blistering), third-degree (full thickness, including appendages, so it cannot re-epithelialize).

**CLINICAL CORRELATION**

**Psoriasis** — T-cell-driven accelerated production and differentiation of keratinocytes, with thickening, scaling, and a defective barrier. **Skin cancer** — about **one-third of all cancers** begin in skin; most arise from basal or spinous cells (basal cell and squamous cell carcinoma), are diagnosed early, and are rarely lethal; incidence rises with fair skin and solar exposure. **Melanoma** — moles can show dysplasia and progress to malignant melanoma, whose cells penetrate the basal lamina and metastasize through blood and lymph. **Merkel cell carcinoma** is **40 times less common** than melanoma but has **twice its mortality**. **Albinism** — defective tyrosinase or another melanin-pathway component; **vitiligo** — loss or reduced activity of melanocytes. **Pemphigus** (intercellular junctions) and **bullous pemphigoid** (dermal–epidermal junction) are the blistering diseases. **Cystic fibrosis** — salty sweat, because CFTR fails to reabsorb salt in the sweat duct.

**UNDER THE MICROSCOPE**

Count layers and look for appendages. **Thick skin: five layers, thick keratin, no hair. Thin skin: four layers, hair follicles and sebaceous glands.** A dark condensed nucleus at the basal layer with a clear halo suggests a melanocyte.

**EXAM TIP**

"Melanocytes are more numerous in dark skin" is false. **Unit density is similar in all people; melanin production and transfer differ.** Also remember that the **nail bed makes no contribution** to the nail plate — the matrix does.

**MEMORY HOOK**

**"Come Let's Get Some Beer"** — **C**orneum, **L**ucidum, **G**ranulosum, **S**pinosum, **B**asale (superficial to deep) — and remember **lucidum exists only in thick skin**.

## Chapter 18 — one-line answers

- Epidermal layers, deep to superficial: basale, spinosum, granulosum, lucidum (thick skin only), corneum.
- Keratinocytes make keratin; they are the most numerous epidermal cells.
- Melanocytes are neural crest cells that transfer melanin to keratinocytes.
- Langerhans cells are antigen-presenting cells of monocyte origin.
- Merkel cells are mechanoreceptors at the basal layer.
- Dermis = papillary (loose) + reticular (dense irregular); hypodermis is adipose and loose connective tissue.
- Eccrine sweat glands are merocrine; apocrine glands open into hair follicles; sebaceous glands are holocrine.
- Meissner corpuscles sense light touch; Pacinian corpuscles sense pressure and vibration.
- Thick skin has a stratum lucidum and no hair follicles; thin skin has hair and sebaceous glands.

## Active recall

1. **Name the epidermal layers from deep to superficial.** — Basale, spinosum, granulosum, lucidum (thick skin only), corneum.
2. **Which epidermal cells are neural crest derived?** — Melanocytes and Merkel cells; Langerhans cells come from monocytes.
3. **What is the function of lamellar bodies?** — They discharge lipid into the intercellular space, forming the water barrier.
4. **Where are Meissner and Pacinian corpuscles?** — Meissner in the dermal papillae (light touch); Pacinian deep in the dermis/hypodermis (pressure, vibration).
5. **How do sebaceous and eccrine glands differ in secretion?** — Sebaceous: holocrine. Eccrine: merocrine.

## Exam-style question

**A skin section shows a thick stratum corneum, a clear layer, and no hair follicles. Which site is it from?**

A. Scalp
B. Palm
C. Back
D. Axilla

**Answer: B** — Thick skin with a stratum lucidum and no hair follicles is palmar or plantar skin.

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 18–1** — Layers and appendages of skin.
- **Fig. 18–5** — Granular and clear layers: thick skin.
- **Fig. 18–9** — Merkel or tactile cell.
- **Fig. 18–12** — Meissner and lamellated (Pacinian) corpuscles.
- **Fig. 18–16** — Glands of skin.
- **Fig. 18–20** — Major stages of cutaneous wound healing.

# PART V — EXCRETION AND REPRODUCTION

Chapters 19–23 · kidney, endocrine glands, and the reproductive systems

# 19 · The Urinary System

**Book pages 393–412**
Kidneys · Blood Circulation · Renal Function: Filtration, Secretion, & Reabsorption · Ureters, Bladder, & Urethra

## Learning objectives

By the end of this chapter you should be able to:

1. Trace filtrate from glomerulus to ureter, naming each segment and its epithelium.
2. Explain the three layers of the filtration barrier and how each contributes to selectivity.
3. Describe the juxtaglomerular apparatus and its role in blood-pressure control.
4. Identify urothelium and explain how it protects against hypertonic urine.

## The core idea

The kidney is a **countercurrent filtration and reabsorption machine** built from millions of identical tubes. Each segment of the tubule has a distinct epithelium matched to a distinct transport job, so identifying the segment from its histology is the same as naming its function.

## Kidneys

Each kidney is bean-shaped, about **12 cm long, 6 cm wide, and 2.5 cm thick** in adults, with a convex lateral border and a concave medial **hilum**, which admits nerves and vessels and lets the ureter and lymphatics exit. Inside the hilum the ureter expands as the **renal pelvis**, which divides into **two or three major calyces**, then **minor calyces**.

- **Capsule → cortex** (renal corpuscles + convoluted tubules) → **medulla** (loops of Henle, collecting ducts, vasa recta) organized into **8–15 pyramids** (the chapter summary says 8–12).
- **Renal columns (of Bertin)** — cortical tissue between pyramids.
- **Lobe** = pyramid + overlying cortex; **lobule** = a medullary ray + the cortical tissue around it.
- **Papilla** → minor calyx → major calyx → renal pelvis → ureter.
- The cortex receives **more than 10 times** the blood flow of the medulla.

**The nephron** — about **1.25 million per kidney** — begins with the **renal corpuscle**: the **glomerulus** (a tuft of fenestrated capillaries) inside the **glomerular (Bowman) capsule**.

- **Parietal layer** — simple squamous epithelium, continuous with the proximal tubule.
- **Visceral layer** — **podocytes** with primary processes and interdigitating **pedicels** that embrace the capillaries. Between pedicels are **filtration slit pores**, bridged by a zipper-like **slit diaphragm** of P-cadherin, **nephrins**, and the negatively charged glycoprotein **podocalyxin**.
- **Mesangial cells** — between capillaries, covering surfaces not covered by podocytes; **contractile** (like pericytes) and produce external lamina.

**The filtration barrier has three layers: fenestrated endothelium → glomerular basement membrane (GBM)**, about **330 nm** thick, of **type IV collagen** and other proteins made by endothelial cells and podocytes → **podocyte slit diaphragms**. The barrier is both a size and a charge filter.

### Functional numbers to know:

- Total filtration area ≈ **500 cm²**.
- **GFR ≈ 125 mL/min**, or about **180 L/day**.
- Plasma volume ≈ **3 L**, so the kidneys filter a volume equal to the whole plasma volume about **60 times a day**.

**Juxtaglomerular apparatus (JGA)** — at the vascular pole: **juxtaglomerular (JG) cells** in the afferent arteriole (modified smooth muscle, contain **renin**), the **macula densa** (a plaque of tall, closely packed cells in the distal straight tubule sensing NaCl), and **extraglomerular mesangial (lacis) cells**.

## Blood Circulation

Renal artery → segmental → interlobar (between pyramids) → arcuate (at the corticomedullary junction) → interlobular (radiating into the cortex) → **afferent arteriole** → glomerulus → **efferent arteriole** → peritubular capillaries in the cortex and **vasa recta** in the medulla. The **efferent arteriole**, not a venule, drains the glomerulus — an arrangement that keeps glomerular hydrostatic pressure high. Efferent arterioles of **cortical** glomeruli branch into a diffuse peritubular network; those of **juxtamedullary** glomeruli form the long hairpin **vasa recta** that run alongside the loops of Henle and are the anatomical basis of the **countercurrent multiplier**.

## Renal Function: Filtration, Secretion, & Reabsorption

| Segment | Epithelium | Key features | Function |
|---|---|---|---|
| **Proximal convoluted tubule (PCT)** | Simple cuboidal, **eosinophilic** | **Long microvilli (brush border)**, abundant mitochondria, large interdigitating **basolateral folds** | **Major reabsorption**: all glucose and organic nutrients, all small proteins and peptides (degraded to amino acids), much water and electrolyte |
| **Thin descending limb** | Simple squamous | Thin | Water permeability |
| **Thin ascending limb** | Simple squamous | Thin | Passive NaCl reabsorption; much less water-permeable than the descending limb |
| **Thick ascending limb (TAL = distal straight tubule)** | Simple cuboidal, low brush border | Ends as the **macula densa** at the vascular pole | **NaCl reabsorption, impermeable to water** — the diluting segment and loop-diuretic site |
| **Distal convoluted tubule (DCT)** | Simple cuboidal, **pale**, no brush border | Many mitochondria, basal infoldings | Further adjustment of electrolytes; thiazide site; leads to **connecting tubules** |
| **Collecting duct** | Simple cuboidal → columnar in medulla | **Principal cells** (aquaporins, ADH) and **intercalated cells** (acid–base transport) | Final water and acid–base regulation |

**Size contrast:** the straight proximal tubule is about **60 µm** in outer diameter and narrows abruptly to about **30 µm** in the thin limbs of the loop.

The visual contrast that answers most questions: the **proximal tubule is eosinophilic with a ragged brush border and a star-shaped lumen; the distal tubule is pale with a clean lumen and no brush border.**

**Collecting ducts.** Connecting tubules from several nephrons join to form cortical collecting ducts, which enter the medulla parallel to the loops and vasa recta and become larger with more columnar cells. The largest (papillary) ducts — **ducts of Bellini** — open into a minor calyx, where filtrate undergoes no further modification and is called **urine**.

## Ureters, Bladder, & Urethra

**Ureter** — **urothelium (transitional epithelium)**, a folded mucosa, and a **thick muscularis** that moves urine by peristalsis; inner longitudinal and outer circular layers, with a third layer added near the bladder. **Bladder** — urothelium, a highly folded mucosa, and the **detrusor** muscle in three poorly delineated interlacing layers. It holds **400–600 mL**, with the urge to void at about **150–200 mL**.

**Urothelium is a specialized barrier, not just a stratified epithelium:**

- **Umbrella cells** — large superficial cells up to **100 µm** in diameter, with **uroplakin plaques ~16 nm** across, hinged to allow unfolding as the bladder fills.
- The epithelium is organized in **three layers**: superficial **umbrella cells**, an intermediate layer of cuboidal or low columnar cells (one to several deep), and a basal layer. Emptying folds the mucosa and lets umbrella cells retract folded plaques; filling flattens the cells and thins the epithelium.
- Tight junctions plus plaques make it an osmotic barrier against hypertonic urine.

**Urethra** — in both sexes lined initially by urothelium. In men there are three parts: **prostatic** (3–4 cm, urothelium), **membranous** (short, through the urogenital diaphragm; stratified or pseudostratified columnar), and **penile** (long; stratified columnar/pseudostratified proximally, stratified squamous distally). In women the urethra is **3–5 cm**, urothelium proximally, then nonkeratinized stratified squamous; the middle part is surrounded by the **external striated sphincter**. The short female urethra explains the frequency of cystitis.

**CLINICAL CORRELATION**

**Polycystic kidney disease** — inherited cysts arising from nephron epithelium destroy cortical organization. **Glomerulonephritis** — immune-complex or anti-glomerular antibody injury; **proteinuria** means the filter has become permeable to protein. **Diabetic glomerulosclerosis** — thickening and loss of function of the GBM; the leading cause of end-stage kidney disease in the United States. **Sickle cell nephropathy** — sickling in the vasa recta causes papillary infarcts. **Kidney stones** — calculi form in the pelvis or calyces, usually from calcium oxalate/phosphate or uric acid. **Pyelonephritis** — bacteria reach the papilla from a calyx; **cystitis** — the commonest bladder problem, favored in women by the short urethra.

**UNDER THE MICROSCOPE**

Start at the renal corpuscle: the **glomerulus** is a tuft of capillaries inside a capsule. Then compare tubules: **eosinophilic + brush border = proximal; pale + clean = distal; large pale cells in a row = collecting duct.** The **macula densa** is the patch of tall cells on the distal tubule facing the vascular pole.

**EXAM TIP**

The proximal tubule is the **most eosinophilic** structure in the renal cortex, packed with mitochondria for active transport. Students often call it "basophilic" because the brush border looks fuzzy — the cytoplasm is pink.

## Chapter 19 — one-line answers

- Kidney: **8–15 pyramids**; about **1.25 million nephrons**.
- Lobe = pyramid + cortex; lobule = medullary ray + surrounding cortex.
- Cortex receives **more than 10×** the blood flow of the medulla.
- Afferent arteriole → glomerulus → **efferent arteriole** (not a venule), keeping glomerular pressure high.
- Filtration area ≈ **500 cm²**; GFR ≈ **125 mL/min ≈ 180 L/day**.
- Filtration barrier = fenestrated endothelium + **GBM (type IV collagen, ~330 nm)** + slit diaphragms (nephrin, podocalyxin).
- Mesangial cells are contractile and support the capillary tuft.
- **PCT**: brush border, eosinophilic, reabsorbs all glucose and small proteins.
- **TAL**: water-impermeable, reabsorbs NaCl, ends as the **macula densa**.
- **JG cells** in the afferent arteriole secrete **renin**; macula densa senses NaCl.
- Collecting duct: **principal cells** (aquaporins, ADH) and **intercalated cells** (acid–base transport).
- Urothelium: three layers — umbrella cells with uroplakin plaques above intermediate and basal cells; it thins as the bladder fills.
- Bladder holds **400–600 mL**; urge at **150–200 mL**.
- Male urethra: prostatic, membranous, penile; female urethra 3–5 cm — short, hence cystitis.

## Active recall

1. **Which nephron segment is most eosinophilic, and why?** — The proximal convoluted tubule: abundant mitochondria and a brush border for active reabsorption.
2. **What are the three layers of the filtration barrier?** — Fenestrated endothelium, glomerular basement membrane, and podocyte slit diaphragms.
3. **What does the macula densa sense, and what does the JGA release?** — NaCl concentration; juxtaglomerular cells release renin.
4. **Which cells of the collecting duct respond to ADH?** — Principal cells, by inserting aquaporins into the apical membrane.
5. **What makes urothelium a barrier?** — Umbrella cells with uroplakin plaques and tight junctions.

## Exam-style question

**A patient has proteinuria from podocyte injury. Which layer of the filtration barrier is primarily affected?**

A. Fenestrated endothelium
B. Glomerular basement membrane
C. Podocyte slit diaphragms
D. Mesangial matrix

**Answer: C** — Podocyte foot processes and their slit diaphragms form the final size filter; their loss produces proteinuria.

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 19–1** — Kidney.
- **Fig. 19–4** — Microvasculature of the renal cortex.
- **Fig. 19–8** — Renal cortex: proximal and distal convoluted tubules.
- **Fig. 19–11** — Renal medulla: nephron loops and collecting ducts.
- **Fig. 19–15** — Renal papilla, collecting ducts, and minor calyx.
- **Fig. 19–18** — Urethra.

# 20 · Endocrine Glands

**Book pages 413–438**
Pituitary Gland (Hypophysis) · Adrenal Glands · Pancreatic Islets · Diffuse Neuroendocrine System · Thyroid Gland · Parathyroid Glands · Pineal Gland

## Learning objectives

By the end of this chapter you should be able to:

1. Classify endocrine organs by their architectural pattern (cords, follicles, clusters).
2. Map each pituitary hormone to its cell type and hypothalamic control.
3. Match each adrenal zone to its hormone and trophic control.
4. Distinguish thyroid, parathyroid, and pineal tissue in a micrograph.

## The core idea

Endocrine glands release hormone into the blood, so their histology is organized around **storage and release** rather than ducts. The three patterns to recognize are **cords** (pituitary, adrenal, parathyroid), **follicles** (thyroid), and **clusters** (pancreatic islets). Each pattern is dictated by how the hormone is stored.

## Pituitary Gland (Hypophysis)

Two embryologically distinct parts in one sella turcica.

**Adenohypophysis** (oral ectoderm — **Rathke's pouch**). The **pars distalis accounts for 75%** of the adenohypophysis and is covered by a thin fibrous capsule:

| Region | Cells | Hormones |
|---|---|---|
| **Pars distalis** | **Acidophils** | **GH, prolactin** |
| | **Basophils** | **ACTH, TSH, FSH, LH** |
| | **Chromophobes** | Inactive or degranulated |
| **Pars intermedia** | Basophils, colloid cysts | MSH (rudimentary in humans) |
| **Pars tuberalis** | — | Wraps the infundibulum |

**Reading the stain.** Routine H&E divides the pars distalis into **acidophils** (pink), **basophils** (blue), and **chromophobes** (poorly stained, the least numerous). The modern names describe the hormone, not the color: **somatotrophs** (GH), **lactotrophs** (prolactin), **corticotrophs** (ACTH), **thyrotrophs** (TSH), and **gonadotrophs** (FSH, LH). Acidophils are the somatotrophs and lactotrophs; basophils are the corticotrophs, thyrotrophs, and gonadotrophs.

**Neurohypophysis** (neuroectoderm): the **pars nervosa** stores **oxytocin and ADH (arginine vasopressin)** — both **9-amino-acid peptides** — synthesized in the **supraoptic and paraventricular nuclei** of the hypothalamus and carried down axons. In the axon terminals they are bound to **10-kDa carrier proteins, neurophysin I and II**, and accumulate in dilatations called **neurosecretory (Herring) bodies**, visible as faintly eosinophilic structures. **Pituicytes** are the glial cells of the lobe. The neurohypophysis **makes no hormone of its own** — it stores and releases hypothalamic hormones.

**Control:** the hypothalamus releases **releasing and inhibiting hormones** into the **hypophyseal portal system**, carrying them directly to the pars distalis — which is why a tiny hypothalamic signal controls a large pituitary output.

## Adrenal Glands

Two glands in one organ, with entirely different embryology and function:

| Region | Cells | Hormone | Control |
|---|---|---|---|
| **Zona glomerulosa** | Rounded or arched cords under the capsule, **~15% of the cortex** | **Aldosterone** | **Angiotensin II, $K^+$** |
| **Zona fasciculata** | Long cords of pale, foamy cells | **Cortisol** | **ACTH** |
| **Zona reticularis** | Irregular cords, pigmented cells | **Androgens (DHEA)** | ACTH |
| **Medulla** | **Chromaffin cells** (neural crest) | **Epinephrine, norepinephrine** | Sympathetic splanchnic nerve |

Each adrenal gland is a flattened half-moon about **4–6 cm long, 1–2 cm wide, and 4–6 mm thick**, weighing about **8 g** in total. The **fasciculata** is the **thickest** zone, and its cells are pale and vacuolated because they are full of **lipid droplets (cholesterol esters)**, **tubular mitochondrial cristae**, and **smooth ER** for steroid synthesis. The **medulla** is functionally a sympathetic ganglion without axons: **chromaffin cells** contain electron-dense granules **150–350 nm** in diameter, storing catecholamines with $Ca^{2+}$, ATP, and **49-kDa chromogranins**. About **80%** of the catecholamine released is **epinephrine**. Mnemonic for the zones: **"GFR — Salt, Sugar, Sex"** from outside in.

## Pancreatic Islets

Pale, round clusters between the acini of the exocrine pancreas (Chapter 16). The pancreas has **more than 1 million islets**, mostly in the tail, but they occupy only **1–2%** of the organ's volume. Most are **100–200 µm** in diameter and contain several hundred cells. The book recognizes **four major cell types**:

| Cell | Product | Notes |
|---|---|---|
| **β (beta)** | **Insulin** | About **twice as abundant** as α cells; scattered in human islets |
| **α (alpha)** | **Glucagon** | Abundant and scattered through most islets |
| **δ (delta)** | **Somatostatin** | Paracrine inhibition |
| **PP** | Pancreatic polypeptide | — |

Islets are richly vascularized, and their hormones pass directly into the bloodstream rather than into ducts.

## Diffuse Neuroendocrine System

Scattered hormone-secreting cells throughout the body — gut enteroendocrine cells, respiratory neuroendocrine cells, and similar — sharing the ability to take up amine precursors (**APUD** cells). They are the reason a "pure" endocrine organ is not the only source of hormones: the gut alone produces more regulatory peptides than any single gland.

## Thyroid Gland

- **Follicles** of varying size filled with **colloid** (thyroglobulin), lined by a simple epithelium whose height ranges from **squamous to low columnar**.
- **Thyrocytes (follicular cells)** — synthesize **T4 and T3**. Activity is read from the epithelium: **active gland = more follicles of tall columnar cells with scalloped colloid; hypoactive gland = squamous cells with dense colloid.**
- **Parafollicular (C) cells** — large, pale cells in the follicular epithelium or between follicles, of **neural crest** origin, secreting **calcitonin**.
- **Folliculogenesis:** thyroglobulin is synthesized, iodinated at the apical surface, stored as colloid, taken back by endocytosis, and digested to release T3/T4.

## Parathyroid Glands

Four small ovoid glands, each about **3 × 6 mm**, with a total weight of about **0.4 g**, usually embedded in the thyroid capsule.

- **Principal (chief) cells** — small, pale cells with round central nuclei, arranged in dense cords around capillaries; they secrete **PTH**.
- **Oxyphil cells** — much larger, intensely eosinophilic cells packed with abnormally shaped mitochondria. They are **degenerated derivatives of principal cells**, some still showing low-level PTH synthesis, and they accumulate with age.
- With age, adipocytes replace secretory cells and may constitute **more than 50%** of the gland.
- Cords and capillaries only — **no follicles**.

## Pineal Gland

A pine-cone-shaped organ about **5–8 × 3–5 mm**, attached to the third ventricle and covered by pia mater.

- **Pinealocytes** — modified neurons in cords and lobules, with pale nuclei and long processes ending near capillaries.
- **Corpora arenacea (brain sand)** — concretions of calcium and magnesium salts formed on extracellular protein deposits; they appear in childhood and increase with age without affecting function. Their density is why the gland is radio-opaque and easy to identify on imaging.
- **Melatonin** — a tryptophan derivative released in **darkness** under **sympathetic** control; it sets diurnal rhythms.
- Interstitial (glial) cells lie between pinealocytes; unmyelinated sympathetic fibers run in the septa.

**CLINICAL CORRELATION**

**Pituitary adenoma** — benign tumors of acidophils or basophils; a somatotroph adenoma causes **gigantism** before epiphyseal closure and **acromegaly** afterwards. **Diabetes insipidus** — deficient ADH (mutation, tumor compression, or head trauma) produces copious dilute urine. **Addison disease** — autoimmune destruction of the adrenal cortex with loss of glucocorticoid, mineralocorticoid, and androgen output. **Pheochromocytoma** — a benign chromaffin-cell tumor that discharges catecholamines episodically, causing swings between hypertension and hypotension. **Diabetes mellitus** — type 1 from autoimmune β-cell loss; type 2 from insulin resistance with relative insulin deficiency. **Goiter** — dietary iodine deficiency lowers thyroid hormone output, so TSH rises and the gland enlarges. **Graves disease** — autoantibodies chronically stimulate follicular cells, producing hyperthyroidism. **Hypoparathyroidism** — low PTH leaves bone over-mineralized and causes tetanic muscle contraction; **hyperparathyroidism** — excess PTH drives osteoclasts and hypercalcemia. **Pineal tumors** are rare but may be benign or highly malignant.

**UNDER THE MICROSCOPE**

Three patterns, three organs. **Follicles of colloid = thyroid. Cords of pale foamy cells under a capsule = adrenal cortex; the medulla is paler and central. Small pale chief cells in dense cords = parathyroid.** Pituitary: **acidophils and basophils in sheets (anterior); pale fibrillary tissue without epithelial cells (posterior).**

**EXAM TIP**

The adrenal cortex has **three** zones and the medulla has **none** of the cortex's function. Catecholamines are **medullary**.

## Chapter 20 — one-line answers

- Pars distalis = **75%** of the adenohypophysis; acidophils, basophils, chromophobes.
- Acidophils = somatotrophs (GH) and lactotrophs (prolactin); basophils = corticotrophs, thyrotrophs, gonadotrophs.

- Hypothalamic releasing and inhibiting hormones reach the pars distalis through the **hypophyseal portal system**.
- Neurohypophysis stores **oxytocin and ADH** made in the supraoptic and paraventricular nuclei.
- ADH and oxytocin are **9-amino-acid peptides** carried by **neurophysin I and II**; Herring bodies are their axonal dilatations.
- Pituicytes are the glial cells of the pars nervosa.
- Adrenal cortex: glomerulosa (**aldosterone**) → fasciculata (**cortisol**) → reticularis (**androgens**).
- Zona fasciculata is the **thickest** zone; glomerulosa is ~15% of the cortex.
- Adrenal medulla = **chromaffin cells**, neural crest, **80% epinephrine**.
- Pancreatic islets: **β insulin, α glucagon, δ somatostatin, PP pancreatic polypeptide**; β about twice as abundant as α.
- Thyroid: follicles of **colloid (thyroglobulin)**; active epithelium is columnar with scalloped colloid, inactive is squamous.
- **Parafollicular C cells** secrete **calcitonin**.
- Parathyroid: **chief cells** (PTH) and **oxyphil cells** (function uncertain).
- Pineal: **pinealocytes** and **corpora arenacea**; melatonin under sympathetic control.

## Active recall

1. **Which pituitary cells are acidophils?** — Somatotrophs (GH) and lactotrophs (prolactin).
2. **Where are oxytocin and ADH made and stored?** — Made in the supraoptic and paraventricular nuclei; stored and released in the pars nervosa.
3. **Name the adrenal zones and their main hormones.** — Glomerulosa: aldosterone; fasciculata: cortisol; reticularis: androgens.
4. **Which islet cells secrete insulin, and how abundant are they?** — β cells; about twice as abundant as α cells.
5. **What are the two cell types of the parathyroid, and which secretes PTH?** — Principal (chief) cells secrete PTH; oxyphil cells are larger, acidophilic, and increase with age.

## Exam-style question

**A patient has episodic hypertension and palpitations. A tumor is found in the adrenal medulla. Which cells are proliferating?**

A. Zona glomerulosa cells
B. Chromaffin cells
C. Zona fasciculata cells
D. Pituitary corticotrophs

**Answer: B** — Pheochromocytoma is a chromaffin-cell tumor releasing catecholamines.

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 20–1** — Locations of the major endocrine glands and the hormones they release.
- **Fig. 20–6** — Pars distalis: Acidophils, basophils, and chromophobes.
- **Fig. 20–10** — Negative feedback loops affecting anterior pituitary secretion.
- **Fig. 20–15** — Adrenal cortex.
- **Fig. 20–19** — Thyroid follicular cells and parafollicular cells.
- **Fig. 20–24** — Pineal gland.

# 21 · The Male Reproductive System

**Book pages 439–459**
Testes · Intratesticular Ducts · Excretory Genital Ducts · Accessory Glands · Penis

## Learning objectives

By the end of this chapter you should be able to:

1. Describe the compartments of the seminiferous tubule and the blood–testis barrier.
2. Sequence spermatogenesis and spermiogenesis and identify each stage.
3. Trace sperm from tubule to urethra, naming the epithelium at each level.
4. Explain the vascular mechanism of erection and ejaculation.

## The core idea

The testis does two jobs that require opposite conditions: **producing sperm**, which needs a temperature below core (hence the scrotum), and **producing testosterone**, which needs a rich blood supply. The architecture keeps them apart by separating the **seminiferous tubules** from the **interstitium**.

## Testes

- **Tunica albuginea** — thick capsule; the **mediastinum testis** sends septa dividing the gland into about **250 lobules**, each holding **1–4** convoluted seminiferous tubules. Each testis has **250–1000** tubules; each tubule is **150–250 µm** wide and **30–70 cm** long, for a total of about **250 m** per testis.
- **Tunica vaginalis** — the serous sac the testis drags with it during descent: parietal layer lines the scrotum, visceral layer covers the albuginea.
- **Seminiferous tubules** — lined by a **stratified epithelium of germ cells** plus **Sertoli cells**, on a basal lamina, surrounded by **peritubular myoid cells**.
- **Interstitium** — **Leydig cells**, blood vessels, connective tissue.

**Temperature is part of the anatomy.** Sperm formation cannot proceed at 37 °C, so the scrotum holds the testis at about **34 °C** by three mechanisms: evaporation of sweat from scrotal skin; a **countercurrent heat exchanger** in the spermatic cord, where the **pampiniform plexus** of veins cools the incoming testicular artery; and the **dartos** and **cremaster** muscles, which move the testis toward or away from the body.

**Sertoli cells** extend from base to lumen; their nuclei are pale, oval, and have a prominent nucleolus. They are **not** germ-line cells, and each one supports **30–50** developing germ cells. Functions:

- **Physical support and nutrition** of developing germ cells.
- **Blood–testis barrier** — tight junctions between adjacent Sertoli cells near the base, creating basal and adluminal compartments. This is why the immune system does not reject sperm.
- **Phagocytosis** of residual bodies.
- **Secretion** of **androgen-binding protein (ABP)**, **inhibin** (a 39-kDa glycoprotein that suppresses FSH), **Müllerian-inhibiting substance** (MIS, 140 kDa, which regresses the paramesonephric ducts in the fetus), **transferrin**, and fluid that carries sperm out of the testis.
- **Response to FSH** — which acts on Sertoli cells, not on germ cells.

**Spermatogenesis: spermatogonia → primary spermatocyte → secondary spermatocyte → spermatid → spermatozoon. Meiosis I** halves the chromosome number and **meiosis II** separates chromatids, giving four spermatids per primary spermatocyte. The germ cells do not divide in isolation: cytokinesis is incomplete, so clones stay joined by **cytoplasmic bridges** — a syncytium that lets a haploid cell use products of the full diploid genome, including genes on X or Y. The whole

sequence from the last spermatogonial mitosis to spermatid formation takes **about 2 months**, and the testis produces about **200 million sperm per day**.

**Spermiogenesis** is the metamorphosis of a round spermatid into a sperm, and the book divides it into **four phases**:

1. **Golgi phase** — proacrosomal vesicles from the Golgi coalesce into a single **acrosomal cap**; centrioles migrate to the opposite pole and one becomes the basal body of the flagellum.
2. **Cap phase** — the acrosome spreads over about half the nucleus. It is a specialized **lysosome** containing **hyaluronidase** and the trypsin-like protease **acrosin**.
3. **Acrosome phase** — the nucleus condenses and elongates, its histones replaced by **protamines**; a microtubule **manchette** shapes the head; mitochondria gather around the proximal flagellum to form the **middle piece**.
4. **Maturation phase** — excess cytoplasm is shed as a **residual body**, intercellular bridges are lost, and the sperm is released into the lumen.

Because the barrier makes the adluminal compartment immunologically privileged, sperm antigens are not tolerized — which is why vasectomy can produce anti-sperm antibodies.

**Leydig (interstitial) cells** — large, polygonal, **eosinophilic** cells with central nuclei, abundant SER, mitochondria, and lipid droplets, lying singly or in small groups near capillaries. They secrete **testosterone** under **LH**, which the book also calls **interstitial cell-stimulating hormone (ICSH)** — the same hormone named LH in females. Fetal Leydig cells are active in the third and fourth months of pregnancy, then become quiescent until puberty.

## Intratesticular Ducts

Sperm leave each tubule through a short **straight tubule**, lined at first only by Sertoli cells and then by simple cuboidal epithelium. It enters the **rete testis**, a labyrinth of cuboidal-lined channels in the mediastinum testis. From there **10–20 efferent ductules** carry sperm to the head of the epididymis. Their epithelium has a characteristic **scalloped** profile in section — patches of absorptive, nonciliated cuboidal cells with microvilli alternating with patches of taller ciliated cells. Absorption of Sertoli-cell fluid plus ciliary beating creates the flow that moves sperm out of the testis.

## Excretory Genital Ducts

| Segment | Epithelium | Function |
|---|---|---|
| **Epididymis** | **Pseudostratified columnar with stereocilia** | **Maturation and storage of sperm**; about **6 m** long, and transit takes **2–4 weeks** |
| **Ductus (vas) deferens** | Pseudostratified columnar | Transport; **very thick muscular wall** (inner longitudinal, middle circular, outer longitudinal) |
| **Ejaculatory duct** | Columnar | Delivers into the prostatic urethra |

## Accessory Glands

- **Seminal vesicles** — paired, highly tortuous tubes about **15 cm** long, with thin complex mucosal folds and inner circular plus outer longitudinal smooth muscle. Their viscid yellowish secretion makes up about **70%** of the ejaculate and supplies **fructose** (sperm energy), inositol, citrate, **prostaglandins** (which stimulate the female tract), and **fibrinogen** (which lets semen coagulate).
- **Prostate** — a **20-g** organ of **30–50 branched tubuloacinar glands** in a dense **fibromuscular stroma**; **corpora amylacea** are small spherical concretions (up to **2 mm**, partly calcified, made of glycoproteins and keratan sulfate) that become more numerous with age but have no clinical significance. The gland has three zones around the urethra:
- **Peripheral zone** — about **70%** of the tissue, with the longest ducts; the site of most **carcinomas**.
- **Central zone** — about **25%**, with submucosal glands.

- **Transition zone** — about **5%**, with mucosal glands around the upper urethra; the site of **benign prostatic hyperplasia**.
- **Prostate-specific antigen (PSA)** — a **34-kDa serine protease** that liquefies the coagulated ejaculate; it leaks into blood, so a raised serum PSA points to carcinoma or inflammation.
- **Bulbourethral (Cowper) glands** — paired, **3–5 mm** mucous glands in the urogenital diaphragm; their clear secretion lubricates the urethra before ejaculation.

### Penis

Three erectile bodies: two dorsal **corpora cavernosa** and one ventral **corpus spongiosum** containing the urethra and expanding distally as the **glans**. Each is a sponge of **endothelium-lined cavernous spaces** separated by trabeculae of smooth muscle and connective tissue, with a dense **tunica albuginea** around the corpora cavernosa. Most of the penile urethra is **pseudostratified columnar**, becoming **stratified squamous** in the glans.

Erection is a vascular event with a molecular explanation worth memorizing: parasympathetic acetylcholine makes the endothelium release **nitric oxide**, which activates **guanylate cyclase** and raises **cGMP** in smooth muscle; the muscle relaxes, **helicine arteries** dilate, cavernous spaces fill, and the swollen corpora compress the dorsal veins against the tunica albuginea. **Sildenafil** blocks the phosphodiesterase that degrades cGMP and therefore sustains an erection. Sympathetic discharge at ejaculation constricts the helicine arteries and trabecular muscle, emptying the spaces.

**CLINICAL CORRELATION**

**Cryptorchidism** — failure of descent, present in about **4%** of male neonates; most descend in the first year, and bilateral cases cause infertility if not corrected by **2–3 years**. An undescended testis is exposed to core temperature, so spermatogenesis fails and malignancy risk rises. **Hydrocele** — serous fluid in the tunica vaginalis, the commonest cause of scrotal swelling. **Varicocele** — dilation of the pampiniform plexus raises testicular temperature and impairs spermatogenesis. **Prostatic carcinoma** — arises in the peripheral zone, is palpable on rectal examination, and metastasizes to bone. **BPH** — nodular hyperplasia of the transition zone, compressing the urethra. **Testicular tumors** — **95%** are germ-cell tumors, appearing after puberty and more often in untreated cryptorchidism; a painless mass is malignant until proven otherwise.

**UNDER THE MICROSCOPE**

Identify the two compartments instantly. **Inside the tubule: Sertoli cells (pale, tall) and germ cells at various stages. Outside the tubule: Leydig cells (eosinophilic, polygonal, in clusters).** If asked for the cell that forms the blood–testis barrier, the answer is Sertoli.

**EXAM TIP**

FSH acts on **Sertoli** cells; LH acts on **Leydig** cells. One sentence, and the two are hard to confuse.

## Chapter 21 — one-line answers

- About 250 lobules per testis, each with 1–4 seminiferous tubules, 150–250 µm wide and 30–70 cm long.
- Spermatogenesis takes about 2 months and yields about 200 million sperm per day.
- Sertoli tight junctions form the blood–testis barrier and divide basal from adluminal compartments.
- Sertoli cells secrete ABP (FSH-dependent), inhibin, MIS, and transferrin, and phagocytose residual bodies.
- Leydig cells secrete testosterone under LH (ICSH).
- Spermiogenesis has four phases: Golgi, cap, acrosome, maturation.
- The acrosome is a lysosome containing hyaluronidase and acrosin; protamines replace histones.
- Straight tubules → rete testis → efferent ductules → epididymis → ductus deferens → ejaculatory duct.

- Seminal vesicles contribute about 70% of semen; the prostate contributes PSA and corpora amylacea.
- Prostate zones: peripheral 70% (cancer), central 25%, transition 5% (BPH).
- Erection is parasympathetic, NO–cGMP mediated; ejaculation is sympathetic.

## Active recall

1. **What forms the blood–testis barrier, and what does it separate?** — Tight junctions between Sertoli cells; basal and adluminal compartments.
2. **Which hormone acts on Sertoli cells, and which on Leydig cells?** — FSH on Sertoli; LH on Leydig.
3. **Name the four phases of spermiogenesis.** — Golgi, cap, acrosome, maturation.
4. **What epithelium lines the epididymis, and what is its hallmark?** — Pseudostratified columnar with stereocilia.
5. **What are corpora amylacea, and where are they found?** — Calcified glycoprotein concretions in prostatic glands.
6. **Which prostatic zone gives rise to cancer, and which to benign hyperplasia?** — Cancer: peripheral (70% of the gland). Benign hyperplasia: transition (5%).
7. **What cools the testicular arterial blood, and how?** — The pampiniform plexus, by countercurrent heat exchange in the spermatic cord.

## Exam-style question

**A 25-year-old man has a painless testicular mass. Which cell type is most likely the origin of a germ-cell tumor?**

A. Leydig cell
B. Sertoli cell
C. Spermatogenic cell
D. Peritubular myoid cell

**Answer: C** — Germ-cell tumors arise from the spermatogenic lineage; Leydig and Sertoli tumors are sex-cord stromal tumors and less common.

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 21–1** — The male reproductive system.
- **Fig. 21–5** — Spermatogenesis and spermiogenesis.
- **Fig. 21–8** — Differentiating spermatid.
- **Fig. 21–12** — Ductus deferens.
- **Fig. 21–15** — Organization of the prostate gland.
- **Fig. 21–19** — Penile urethra and erectile tissue.

# 22 · The Female Reproductive System

**Book pages 460–489**
Ovaries · Uterine Tubes · Major Events of Fertilization · Uterus · Embryonic Implantation, Decidua, & the Placenta · Cervix · Vagina · External Genitalia · Mammary Glands

## Learning objectives

By the end of this chapter you should be able to:

1. Sequence follicular development from primordial follicle to corpus albicans.
2. Explain the two-cell, two-gonadotropin mechanism of estrogen production.
3. Correlate ovarian and endometrial cycles and identify each phase histologically.
4. Describe implantation, placental structure, and the hormonal control of pregnancy.

## The core idea

The ovary and the endometrium are **rebuilt through a monthly cycle**. Every histological feature of the ovary, tube, uterus, and vagina is a stage in one of two cycles — **ovarian** (follicle to corpus luteum) and **endometrial** (proliferative to secretory) — locked together by the same four hormones.

## Ovaries

Each ovary is an almond-shaped body about **3 cm long, 1.5 cm wide, and 1 cm thick**.

- **Germinal (surface) epithelium** — simple cuboidal mesothelium (a misnomer; it does not produce germ cells).
- **Tunica albuginea** — dense connective tissue.
- **Cortex** — follicles at all stages, plus corpus luteum and corpus albicans.
- **Medulla** — loose connective tissue with the vessels entering at the hilum. Cortex and medulla have no sharp border.

**The oocyte supply is fixed before birth and only falls.** Primordial germ cells reach the gonads in the first month; oogonia peak at about **7 million** in the fifth month of gestation. About **2–2.5 million** primordial follicles are present at birth and about **400,000** at puberty; only about **450** are ovulated in a lifetime, and the rest undergo atresia. Menopause arrives when the reserve falls below about **1000** follicles.

| Stage | Structure | Hormonal influence |
|---|---|---|
| **Primordial follicle** | Oocyte + single layer of flat granulosa cells | None; present from birth |
| **Primary follicle** | Oocyte enlarges (to ~120 µm); granulosa cells become **cuboidal** (unilaminar), then stratified (multilaminar); **zona pellucida** appears | FSH-dependent from this stage |
| **Secondary (antral) follicle** | Antrum forms among the granulosa cells; **theca interna** (steroidogenic, vascular) and **theca externa** (fibrous, contractile) differentiate | FSH and LH |
| **Graafian (mature) follicle** | Single large antrum up to **2 cm**; oocyte on the **cumulus oophorus**, wrapped by the **corona radiata** | LH surge triggers ovulation |
| **Atretic follicle** | Degenerating; **glassy membrane** remains | — |
| **Corpus luteum** | **Granulosa-lutein cells (progesterone, ~80% of the gland)** and **theca-lutein cells (estrogens)** | LH maintains it |
| **Corpus albicans** | White fibrous scar | — |

### The corpus luteum is a temporary gland with two cell populations:

- **Granulosa-lutein cells** — the old granulosa cells; they enlarge to **20–35 µm** without dividing, make up about **80%** of the gland, and secrete **progesterone** (their SER expands, and they retain some aromatase).
- **Theca-lutein cells** — from the theca interna; about **half the size**, darker staining, lying in the peripheral folds, and secreting **androstenedione** and **estrogens**.

The LH surge sustains progesterone secretion for **10–12 days**. (The chapter summary reverses the two cell types, giving estrogen to the granulosa-lutein cells; the body text and Figure 22–12, followed here, give progesterone to the granulosa-lutein cells and estrogens to the theca-lutein cells.) Without pregnancy, both populations undergo apoptosis, macrophages clear the debris, and fibroblasts lay down a collagen scar — the **corpus albicans**. Note that **involution is not atresia**: atresia is the death of an unovulated follicle.

**The zona pellucida** is a **5–10 µm** glycoprotein coat between the oocyte and the granulosa cells, secreted by the oocyte. Its glycoproteins **ZP1, ZP2, and ZP3**, all made by the oocyte, form the sperm-binding receptor; binding of a capacitated sperm to this coat begins the acrosomal reaction. The book's Summary of Key Points calls them "ZO proteins" — the body text and Figure 22–6 give ZP, and the figure notes four related glycoproteins.

**Ovulation, step by step.** The LH surge triggers: completion of **meiosis I** with extrusion of the **first polar body**; granulosa production of fluid rich in prostaglandins, proteoglycans, and proteases; **hyaluronan** release that loosens the cumulus; activation of **plasmin**, which digests collagen at the **stigma** (the ischemic bulge on the ovarian surface); and contraction of **theca externa** smooth muscle. The follicle ruptures, releasing the **secondary oocyte, arrested in metaphase II**, wrapped in the **corona radiata** — viable for about **24 hours**. If a sperm arrives, the oocyte completes meiosis II and emits the **second polar body**; the two pronuclei fuse to form the **zygote**.

A mature follicle takes about **90 days** to develop from a primordial follicle, and only one per cycle reaches that stage while its neighbors become atretic.

**Two-cell, two-gonadotropin theory: LH** stimulates **theca interna** cells to make **androstenedione**; it crosses the basement membrane (the granulosa layer is avascular) and **FSH** stimulates **granulosa** cells to **aromatize** it to **estradiol**. The estrogen returns to the theca capillaries and reaches the whole body. This division of labor explains why both gonadotropins are needed for normal estrogen production — and why the granulosa layer must stay avascular.

## Uterine Tubes

About **10–12 cm** long, in four parts — **infundibulum with fimbriae → ampulla (longest; fertilization happens here) → isthmus → intramural portion**. The mucosa has complex branching folds, tallest in the ampulla, and is lined by **ciliated columnar cells** that sweep fluid toward the uterus and **peg (secretory) cells** that make the nutritive fluid. The muscularis has interwoven circular and longitudinal layers and provides most of the transport — women with immotile cilia still carry the embryo to the uterus.

## Major Events of Fertilization

Fertilization normally occurs in the **ampulla**, and only a few hundred of the millions of ejaculated sperm reach it. The sequence:

1. **Capacitation** in the female tract makes sperm able to fertilize.
2. The sperm releases **hyaluronidase** to pass between the corona radiata cells.
3. At the zona, binding to the **ZP1–ZP3** complex triggers the **acrosomal reaction**: **acrosin** and other proteases digest a path through the zona.
4. The first sperm to cross fuses with the oolemma. Membrane depolarization and a **$Ca^{2+}$ wave** trigger the **cortical reaction** — proteases from cortical granules modify the zona glycoproteins and convert the zona into the impenetrable **perivitelline barrier**, a permanent block to polyspermy.

5. The oocyte completes **meiosis II**, emits the **second polar body**, and forms the female pronucleus; the sperm head decondenses into the male pronucleus. Fusion gives the diploid **zygote**.

Cleavage follows as the embryo travels to the uterus — about **5 days** from fertilization to the blastocyst in the uterine cavity.

## Uterus

The uterus is pear-shaped: **body**, **fundus**, **isthmus**, and **cervix**, with the cervical canal opening at the internal and external os.

- **Perimetrium (serosa)**, **myometrium** (thick smooth muscle in interlacing bundles), **endometrium**.
- **Endometrium** — **stratum functionalis** (shed each month) and **stratum basalis** (retained, regenerates it). The basal layer has a distinctive stroma: fibroblasts, ground substance, **fine type III collagen**, and **no adipocytes**. **Spiral arteries** supply the functionalis and constrict to cause shedding.

| Phase | Days | Hormone | Histology |
|---|---|---|---|
| **Menstrual** | 1–4 | Falling progesterone | Shedding, hemorrhage, neutrophilic infiltrate |
| **Proliferative** | 5–14 | **Estrogen** | Glands straight and narrow, mitoses, dense stroma |
| **Secretory** | 15–28 | **Progesterone** | Glands **tortuous and dilated with secretion**, edematous stroma, coiled spiral arteries |

The single best discriminator: **straight, narrow glands = proliferative; saw-toothed, dilated, secreting glands = secretory.**

**Thickness is a second discriminator.** At the end of the proliferative phase the endometrium is **2–3 mm**; in the secretory phase it reaches **5–6 mm**, and the functional layer becomes about **four times** thicker than the basal layer.

**How menstruation happens.** Progesterone and estrogen fall **8–10 days after ovulation**. The spiral arteries contract and relax intermittently, then constrict under local **prostaglandins**, producing ischemia. Hypoxic cells release cytokines that attract leukocytes, whose **collagenases and matrix metalloproteinases** digest the matrix. The functional layer — epithelium, most of each gland, stroma, and blood-filled lacunae — separates from the basal layer and is shed. The basal layer, which does not depend on the spiral arteries, survives and regenerates the next functionalis under estrogen.

## Embryonic Implantation, Decidua, & the Placenta

The blastocyst reaches the uterine cavity about **5 days** after fertilization and floats in the glandular secretion for about **2 days**. Implantation (nidation) takes about **3 days** and is complete by about **day 9**: trophoblast receptors bind glycoproteins on the endometrial epithelium, the **syncytiotrophoblast** digests the stroma, and the embryo embeds. The trophoblast has two layers — **cytotrophoblast** (mitotic, inner) and **syncytiotrophoblast** (invasive, multinucleated, outer, and the source of **hCG**).

**Chorionic villi** project into maternal blood in the **intervillous space** and develop in three stages: **primary** villi (cytotrophoblast covered by syncytiotrophoblast) about 2 days after implantation; **secondary** villi (mesenchymal cores) about day 15; **tertiary** villi (with capillary loops) a few days later. By the end of the first month there are thousands of tertiary villi.

**Decidua** — the modified endometrium of pregnancy: **decidua basalis** (under the embryo, forms the maternal placenta), **decidua capsularis** (over it), **decidua parietalis** (the rest).

**Placental hormones: hCG** (maintains the corpus luteum early; the basis of pregnancy tests), a **lactogen**, **relaxin**, growth factors, progesterone, and estrogen. hCG keeps the corpus luteum of pregnancy alive for **4–5 months**, by which time the placenta itself makes enough progesterone.

## Cervix

The cervix is the lower cylindrical part of the uterus and differs from the body: its mucosa is **2–3 mm** thick, has **no spiral arteries**, and **does not shed** with the cycle.

- **Endocervix** — simple columnar mucus-secreting cells with large branched **cervical glands**.
- **Ectocervix** — **stratified squamous** nonkeratinized epithelium, continuous with the vagina.
- **Transformation zone** — the area just outside the external os where columnar epithelium is replaced by squamous; the site of HPV-related dysplasia and the target of the **Pap smear**.

Cervical mucus changes with hormones: watery and abundant at ovulation (helps sperm), viscous in the luteal phase (hinders them), and a thick **mucus plug** in pregnancy. Before birth, **effacement** dissolves collagen so the cervix softens and dilates.

## Vagina

**Stratified squamous nonkeratinized** epithelium **150–200 µm** thick and rich in **glycogen**, with **no glands** (lubrication comes from cervical and Bartholin glands) and a lamina propria with elastic fibers and venous plexuses. **Lactobacilli** metabolize glycogen to lactic acid, keeping the pH low and protecting against pathogens.

## External Genitalia

The **vulva**: the **labia majora** (skin with hair and sebaceous glands, containing the homologue of the scrotum), **labia minora** (thin skin without hair, with sebaceous glands), the **clitoris** (erectile tissue homologous to the penis), and the **vestibule** containing the urethral and vaginal openings and the openings of the **greater vestibular (Bartholin) glands**, which provide lubrication.

## Mammary Glands

A modified **apocrine sweat gland** whose functional unit is the **terminal duct lobular unit (TDLU)**.

- **Non-lactating** — mostly dense connective tissue and fat with small ducts and lobules.
- **Pregnancy** — lobules proliferate under estrogen and progesterone.
- **Lactation** — alveoli distend with secretion; **protein is released by exocytosis** and **lipid droplets by apocrine secretion**, so milk fat is wrapped in apical plasma membrane. Prolactin drives synthesis, oxytocin the ejection reflex.
- **Duct system** — intralobular → interlobular → **lactiferous ducts** → **lactiferous sinuses** beneath the areola → nipple.

Hormonal control: **estrogen** for ductal growth, **progesterone** for lobuloalveolar development, **prolactin** for milk synthesis, **oxytocin** for milk ejection.

**CLINICAL CORRELATION**

**Polycystic ovary syndrome** — multiple arrested antral follicles, anovulation, hyperandrogenism. **Endometriosis** — functional endometrial tissue outside the uterus, bleeding cyclically. **Leiomyoma (fibroid)** — a benign smooth-muscle tumor of the myometrium. **Endometrial carcinoma** — unopposed estrogen, with atypical hyperplasia. **Cervical carcinoma** — HPV in the transformation zone. **Ectopic pregnancy** — implantation outside the uterine cavity, usually in the ampulla. **Breast carcinoma** — arises from ductal or lobular epithelium; the **terminal duct lobular unit** is the functional unit to remember.

**UNDER THE MICROSCOPE**

Place the ovary in the cycle by the **dominant structure**: many small primordial follicles = inactive/early; a large antral follicle = late follicular; a corpus luteum with folded, pale, luteinized cells = secretory phase; a hyaline scar = post-ovulatory. In the uterus, look at the **glands**, not the stroma.

**MEMORY HOOK**

**"Proliferative = Pencil-thin glands; Secretory = Saw-tooth and swollen."**

**SOURCE NOTE** — The chapter's Summary of Key Points says granulosa-lutein cells produce estrogen and theca-lutein cells produce progesterone. That reverses the book's own body text and figure legend (granulosa-lutein → progesterone; theca-lutein → estrogens), and it reverses the physiology. This guide follows the body text. If you meet the summary line in a test, answer **progesterone from granulosa-lutein cells**.

**EXAM TIP**

The corpus luteum secretes **progesterone**, and it is maintained by **LH** (and, in pregnancy, by **hCG**). Do not say estrogen maintains the corpus luteum.

## Chapter 22 — one-line answers

- Ovary has three regions: cortex with follicles, medulla with vessels, and a surface of **germinal epithelium** over a tunica albuginea.
- Follicle sequence: primordial → primary → secondary (antral) → mature (Graafian) → ovulation.
- **Zona pellucida** = glycoprotein coat around the oocyte; **corona radiata** = granulosa cells carried with it at ovulation.
- **Theca interna** makes androgens; **granulosa cells** aromatize them to estrogen under FSH.
- **Cumulus oophorus** = granulosa mound carrying the oocyte into the antrum.
- After ovulation: **corpus luteum** (progesterone) → **corpus albicans** (hyaline scar).
- Uterine tube: **infundibulum with fimbriae → ampulla → isthmus**; fertilization in the ampulla.
- Endometrium has a **functional layer** that sheds and a **basal layer** that regenerates it.
- Proliferative phase = estrogen, straight narrow glands, mitoses; secretory phase = progesterone, saw-toothed glands, edema.
- Menstrual phase = spiral artery constriction and functionalis shedding.
- Cervix: columnar endocervix, stratified squamous ectocervix, squamocolumnar junction = transformation zone.
- Vagina = stratified squamous nonkeratinized epithelium with glycogen, no glands; acidity comes from lactobacilli.
- Mammary gland: **TDLU** is the functional unit; estrogen drives ducts, progesterone drives lobuloalveolar growth, prolactin drives milk synthesis.

## Active recall

1. **Which cells produce estrogen, and from what precursor?** — Granulosa cells aromatize androgens from theca interna cells.
2. **What is released at ovulation, and at what meiotic stage?** — The secondary oocyte, arrested in metaphase II, with the corona radiata.
3. **What prevents polyspermy?** — Cortical granule exocytosis modifies the zona pellucida into the perivitelline barrier.
4. **Which hormone maintains the corpus luteum early in pregnancy?** — hCG.
5. **How do proliferative and secretory endometrium differ?** — Proliferative: straight narrow glands, mitoses. Secretory: tortuous dilated glands with secretion and edematous stroma.

## Exam-style question

**A curettage specimen shows tortuous, saw-toothed glands with secretion and a coiled spiral artery. Which phase is this?**

A. Menstrual
B. Proliferative
C. Secretory
D. Postmenopausal

**Answer: C** — The secretory phase is defined by tortuous secreting glands under progesterone.

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 22–1** — The female reproductive system and overview of ovary.
- **Fig. 22–6** — Ultrastructure of primary follicle and zona pellucida.
- **Fig. 22–12** — Corpus luteum.
- **Fig. 22–17** — Correlation of ovarian and menstrual cycles with levels of their controlling hormones.
- **Fig. 22–23** — Cervix.
- **Fig. 22–28** — Secretion in the mammary gland.

# 23 · The Eye & Ear: Special Sense Organs

**Book pages 490–524**
Eyes: The Photoreceptor System · Ears: The Vestibuloauditory System

## Learning objectives

By the end of this chapter you should be able to:

1. Name the three tunics of the eye and the five layers of the cornea.
2. Trace aqueous humor and explain glaucoma.
3. Describe the layers of the retina and the path of a light signal.
4. Compare the vestibular and cochlear sensory epithelia and their transduction mechanisms.

## The core idea

Both organs convert a physical signal into a nerve impulse through a **specialized epithelium** — the retina for light, the organ of Corti for sound. Their histology is therefore about **transduction**, not merely structure.

**Numbers to anchor the chapter:** the adult eyeball is about **22 mm** in diameter; the **sclera** is 0.5 mm thick (1 mm posteriorly); the **cornea** is the anterior one-sixth and is **avascular**; the retina has **nine neural layers plus a pigmented layer**; the cochlea is about **35 mm** long with **2¾ turns**.

## Eyes: The Photoreceptor System

### Three tunics:

| Tunic | Components | Function |
|---|---|---|
| **Fibrous** | **Cornea** and **sclera** | Protection and support; the cornea also refracts |
| **Vascular (uvea)** | **Choroid, ciliary body, iris** | Nutrition, accommodation, pupil control |
| **Nervous** | **Retina** | Phototransduction |

**The cornea has five layers**, and the order is a standard question:

1. **Corneal epithelium** — stratified squamous, nonkeratinized, **5–6 cells thick**, about **10%** of corneal thickness; its stem cells sit in the **limbus**, and it has one of the richest sensory innervations in the body.
2. **Bowman's membrane** — a thick (**8–10 µm**) basement membrane.
3. **Stroma** — about **90%** of the thickness, some **60 layers** of type I collagen at right angles to each other, with flattened **keratocytes** between them; **avascular**.
4. **Descemet's membrane** — the thick basement membrane of the endothelium.
5. **Corneal endothelium** — simple squamous, the most metabolically active cells of the cornea; their **$Na^+/K^+$ ATPase** pumps control stromal hydration and therefore transparency.

The cornea is **transparent because it is avascular and its collagen is regularly arranged**. When it becomes edematous or scarred, it loses transparency — the anatomical explanation for corneal blindness.

- **Choroid** — pigmented, vascular; two layers: the inner **choroidocapillary lamina** (nourishes the outer retina) and **Bruch's membrane**, an elastin sheet between two collagen layers, next to the retinal pigment epithelium.
- **Ciliary body** — **ciliary muscle** (fibers in three complementary orientations; accommodation) and numerous **ciliary processes** covered by a **bilaminar epithelium**: a pigmented layer next to the stroma and a nonpigmented superficial layer. The nonpigmented cells pump aqueous humor and also secrete the **fibrillin** microfibrils of the **ciliary zonule** that suspends the lens.

- **Iris** — a stroma with melanocytes and, posteriorly, two pigmented epithelial layers. The anterior surface has **no epithelium**, only fibroblasts and melanocytes. The **dilator pupillae** is a layer of **myoepithelial cells** (sympathetic); the **sphincter pupillae** is a circular smooth-muscle bundle near the pupil (parasympathetic).

**Aqueous humor** is secreted by the ciliary processes into the **posterior chamber**, flows through the **pupil** into the **anterior chamber**, and drains at the **iridocorneal angle** through the **trabecular meshwork** into the **scleral venous sinus (canal of Schlemm)**, then into scleral veins. It has the ions of plasma but almost no protein. Blockage raises intraocular pressure — **glaucoma**; a narrow angle that closes when the pupil dilates gives **angle-closure glaucoma**.

**The lens: capsule** (10–20 µm, type IV collagen and proteoglycans) → **subcapsular epithelium** (simple cuboidal, anterior surface only) → **lens fibers** (7–10 mm long, 2 × 8 µm in cross-section, packed with **crystallins**, their organelles and nuclei removed by autophagy). New fibers are added at the equator throughout life, so the lens grows and its nucleus becomes denser. It is **avascular** and suspended by the zonule.

**Accommodation:** at rest the ciliary muscle is relaxed, the zonule is taut, and the lens is **flat** for distant vision. To focus close, the ciliary muscle **contracts**, the ciliary body moves forward, tension on the zonule falls, and the lens becomes **rounder**. Loss of lens elasticity in the fourth decade is **presbyopia**; loss of transparency is **cataract**.

**The retina** — the neural retina has **nine layers**, and with the pigment epithelium there are ten. From the vitreous inward-outward (following the path of light): **inner limiting membrane → nerve fiber layer → ganglion cell layer → inner plexiform layer → inner nuclear layer → outer plexiform layer → outer nuclear layer → outer limiting layer → rod and cone layer → pigmented layer.**

| Layer | Contents |
|---|---|
| **Retinal pigment epithelium (RPE)** | Cuboidal, pigmented; absorbs stray light, forms the **blood–retina barrier** with tight junctions, **phagocytoses shed rod discs**, regenerates **11-cis-retinal**, and secretes trophic factors |
| **Photoreceptor layer** | **Rods** (rhodopsin, dim light, ~92 million) and **cones** (iodopsin, color, ~4.6 million). Rod discs are **membrane stacks**; cone outer segments are **conical**. |
| **Outer nuclear layer** | Photoreceptor nuclei |
| **Inner nuclear layer** | Bipolar, horizontal, amacrine, and Müller cell nuclei |
| **Ganglion cell layer** | Ganglion cells whose axons form the optic nerve |

**Rods versus cones at the level of the disc** — a classic exam distinction: in **cones** the membranous discs of the outer segment are **continuous with the plasma membrane**; in **rods** they are **separate**, pinched-off discs. A single rod contains about **one billion rhodopsin** molecules, and each day it loses and replaces about **90 discs** in a cycle that takes **10 days**.

**Phototransduction in one paragraph.** In darkness, cGMP keeps **cGMP-gated $Na^+$ channels** open, so the photoreceptor is depolarized and steadily releases **glutamate**. A photon isomerizes **11-cis-retinal** to **all-trans-retinal** within a picosecond; the opsin changes shape and activates **transducin**, a heterotrimeric G protein whose α subunit stimulates **phosphodiesterase** to hydrolyze cGMP. The channels close, the cell **hyperpolarizes**, and transmitter release falls — which **depolarizes** the bipolar cells and sends signals to the ganglion cells. The free all-trans-retinal is carried to the RPE, re-isomerized to 11-cis, and returned; this **bleaching** cycle takes a minute or more and underlies slow dark adaptation.

The retina is **inverted**: light passes through the ganglion cell and inner layers before reaching the photoreceptors, which sit against the pigment epithelium. The **fovea centralis**, at the center of the **macula lutea**, contains **no rods** and a very high density of **cones**; the overlying layers are displaced sideways so light reaches the cones with little scattering. That is why acuity is highest there, but only in good light. The **optic disc** is the **blind spot**: ganglion-cell axons converge there to form the optic

nerve, and photoreceptors are absent.

**Müller cells** are the principal glia — modified astrocytes whose cell bodies lie in the inner nuclear layer and whose processes span the whole retina. They provide trophic support, buffer ions and water, regulate blood flow, form the **inner limiting membrane**, and join photoreceptors by zonulae adherentes to form the **outer limiting layer**.

## Ears: The Vestibuloauditory System

**External ear** — the **auricle** is a plate of **elastic cartilage** covered by tightly adherent skin; the **external acoustic meatus** is lined by skin with hairs, sebaceous glands, and modified apocrine **ceruminous glands**. Their product, **cerumen**, contains proteins, saturated fatty acids, and shed keratinocytes and is antimicrobial. Elastic cartilage supports the outer third of the meatus; the temporal bone encloses the inner part. The **tympanic membrane** is fibroelastic connective tissue covered by epidermis outside and simple cuboidal mucosa inside.

**Middle ear** — an air-filled **tympanic cavity** lined by simple cuboidal epithelium, containing the **ossicles** (malleus, incus, stapes) linked by synovial joints, with two small muscles (**tensor tympani** on the malleus, **stapedius** on the stapes) that damp loud sounds. The **auditory (eustachian) tube** connects to the pharynx and equalizes pressure; the **oval window** receives the stapes and the **round window** relieves the pressure wave.

**Inner ear — bony labyrinth** containing the membranous labyrinth:

- **Cochlea** — a **35-mm** canal with **2¾ turns** around a bony core, the **modiolus**. It has three compartments: **scala vestibuli** (perilymph) above, **scala media / cochlear duct** (endolymph) in the middle, and **scala tympani** (perilymph) below. Vestibuli and tympani communicate at the apex through the **helicotrema**. The **vestibular (Reissner) membrane** separates vestibuli from media with tight junctions that keep the two fluids apart; the **basilar membrane** separates media from tympani.
- The **organ of Corti** sits on the basilar membrane: **about 12,000 outer hair cells in three rows (five near the apex)** with V-shaped stereocilia bundles, and a **single row of inner hair cells** with a linear bundle. The inner row is far more heavily innervated. **Phalangeal cells** support the hair cells and, with them, form the **apical plate**; **pillar cells** enclose the stiff **inner tunnel**. The tallest stereocilia of the outer cells are embedded in the acellular **tectorial membrane** (collagens II, V, and IX, proteoglycans, and tectorin).
- The **stria vascularis** in the lateral wall secretes endolymph, pumping $K^+$ into the lumen.
- **Vestibule — utricle** and **saccule**, each with a **macula** containing hair cells and otoliths (gravity and linear acceleration).
- **Semicircular canals — cristae ampullares** with hair cells in a cupula (rotational acceleration).

**Endolymph** fills the membranous labyrinth: **$K^+$ about 150 mM, $Na^+$ about 16 mM** — like intracellular fluid, and the source of the driving force for transduction. **Perilymph** fills the bony labyrinth and resembles cerebrospinal fluid. Perilymph drains through the perilymphatic duct to the subarachnoid space; endolymph drains through the endolymphatic duct to dural venous sinuses.

**Transduction** is the same in the macula, crista, and organ of Corti: bending the stereocilia toward the tallest opens **mechanically gated $K^+$ channels**, $K^+$ enters, the cell depolarizes, voltage-gated $Ca^{2+}$ channels open at the base, and transmitter release rises. Bending the other way closes the channels and silences the cell.

**CLINICAL CORRELATION**

**Glaucoma** — raised intraocular pressure from impaired drainage through the trabecular meshwork, damaging the optic nerve. **Cataract** — lens opacity from crystallin denaturation. **Age-related macular degeneration** — degeneration of the RPE and photoreceptors at the macula. **Retinal detachment** — separation of the neural retina from the RPE, which starves the photoreceptors. **Diabetic retinopathy** — microvascular damage with hemorrhage and neovascularization. **Otosclerosis** — ossification of the stapes footplate, producing conductive deafness. **Presbycusis** — age-related loss of hair cells and neurons. **Ménière disease** — endolymphatic hydrops with vertigo, tinnitus, and fluctuating deafness.

**Eyelids and tear film** — the inner surface of the eyelid and the anterior sclera are covered by **conjunctiva**, a **stratified columnar epithelium with goblet cells** that is continuous with the corneal epithelium. **Lacrimal glands** produce the tear film, which drains through the lacrimal apparatus into the nasal cavity.

**UNDER THE MICROSCOPE**

The **cornea** is identified by five layers and a regular, avascular stroma. The **retina** by its ten layers with a pigmented layer at the outer border and ganglion cells at the inner border. The **organ of Corti** by the tunnel of Corti between inner and outer hair cells on the basilar membrane, with the tectorial membrane above.

**EXAM TIP**

Rods detect **dim light**, cones detect **color and detail**; the **fovea has only cones**, which is why acuity is highest there and color vision is lost first in the periphery.

## Chapter 23 — one-line answers

- The eyeball is about 22 mm; the sclera is 0.5 mm thick (1 mm posteriorly).
- Cornea: epithelium, Bowman membrane, stroma, Descemet membrane, endothelium — avascular and transparent.
- Aqueous humor: ciliary processes → posterior chamber → pupil → anterior chamber → trabecular meshwork → canal of Schlemm.
- The lens is avascular, suspended by the fibrillin zonule; it becomes rounder when the ciliary muscle contracts.
- The retina has nine neural layers plus the pigmented epithelium.
- Rods (rhodopsin, dim light, ~92 million) and cones (iodopsin, color, ~4.6 million); the fovea contains cones only.
- Phototransduction: retinal isomerization → transducin → phosphodiesterase → cGMP falls → channels close → hyperpolarization.
- The RPE absorbs light, forms the blood–retina barrier, phagocytoses discs, and regenerates 11-cis-retinal.
- Endolymph is high-$K^+$; perilymph resembles CSF.
- Organ of Corti: about 12,000 outer hair cells in 3–5 rows and a single row of inner hair cells.
- Maculae sense gravity and linear acceleration; cristae sense rotation; the cochlea hears.

## Active recall

1. **Name the five layers of the cornea, outside in.** — Epithelium, Bowman membrane, stroma, Descemet membrane, endothelium.
2. **Where is aqueous humor produced and where does it drain?** — Ciliary processes; drains through the trabecular meshwork into the canal of Schlemm.
3. **Which retinal cells form the optic nerve?** — Ganglion cells.
4. **What is the fovea, and why is acuity highest there?** — A cone-only pit at the macula; it has the highest photoreceptor density and minimal overlying tissue.

5. **Endolymph versus perilymph?** — Endolymph (high $K^+$) fills the membranous labyrinth; perilymph (high $Na^+$) fills the bony labyrinth.

## Exam-style question

**A patient loses vision from increased intraocular pressure. Where is the obstruction?**

A. Ciliary processes
B. Pupil
C. Trabecular meshwork and canal of Schlemm
D. Central retinal artery

**Answer: C** — Glaucoma results when aqueous humor cannot drain through the trabecular meshwork into the canal of Schlemm.

## Plates to study

Each figure is listed with the subject its caption names. Cover the caption, say what the figure shows and the one structural feature that gives it away, then check. Figure numbers are those printed in the textbook.

- **Fig. 23–1** — Internal anatomy of the eye.
- **Fig. 23–7** — Epithelium of ciliary processes.
- **Fig. 23–14** — General organization and specialized areas of the retina.
- **Fig. 23–20** — Lacrimal gland.
- **Fig. 23–27** — Mechanotransduction in hair cells.
- **Fig. 23–33** — Interpretation of sound waves in the cochlea.

# PART VI — RAPID REVIEW AND REFERENCE

Chapter 24, the master tables, and the reference appendices

# 24 · The Last-24-Hours Rapid Review

**HIGH-YIELD PEARL**
This chapter is designed to be read in one sitting, twice. Read it once slowly, then again out loud. Everything in it is either a rule that generates other facts or a fact worth being able to state from memory.

## The book as a concept map

If you remember nothing else, remember the shape of the subject. Each level is generated by the level above it, and each arrow is a question the subject naturally poses.

- **Molecules** — proteins, lipids, carbohydrates, nucleic acids
- **Organelles** — membrane, cytosol, nucleus (Chapter 2–3)
- **Cells** — each specialized by which organelle it expands
- **Epithelium** — sits on a basal lamina, avascular, joined by junctions (Chapter 4)
- **Surface epithelium** — number of layers and shape of the top cell
- **Glandular epithelium** — merocrine, apocrine, holocrine
- **Connective tissue** — defined by its matrix (Chapter 5)
- **Loose / dense / reticular / elastic / adipose** — amount and arrangement of fibers
- **Cartilage and bone** — matrix stiffened by proteoglycan or mineral (Chapters 7–8)
- **Muscle** — converts ATP into movement (Chapter 10)
- **Skeletal** — voluntary, peripheral nuclei, triads at the A–I junction
- **Cardiac** — branched, central nucleus, intercalated discs, diads at the Z line
- **Smooth** — no striations, calmodulin, dense bodies
- **Nervous tissue** — converts stimulus into impulse (Chapter 9)
- **Neuron** — soma, dendrites, axon, synapse
- **Glia** — oligodendrocyte, Schwann, astrocyte, microglia, ependyma
- **Organs** — one tissue arrangement plus a supporting stroma
- **Tubes** — concentric layers (gut, vessels, ducts, ureter)
- **Solid organs** — capsule, cortex, medulla, hilum (kidney, ovary, lymph node)
- **Glands** — ducts or no ducts (exocrine versus endocrine)
- **Systems** — organs that share a transport medium
- **Circulatory** — heart, vessels, blood, lymph (Chapters 11–13)
- **Immune** — thymus, lymph node, spleen, MALT (Chapter 14)
- **Digestive** — gut tube plus liver, pancreas, salivary glands (Chapters 15–16)
- **Respiratory** — conducting airway plus alveoli (Chapter 17)
- **Integumentary** — skin and appendages (Chapter 18)
- **Urinary** — filtration, reabsorption, storage (Chapter 19)
- **Endocrine** — pituitary, thyroid, adrenal, islets, parathyroid, pineal (Chapter 20)
- **Reproductive** — testis and ovary, then ducts and glands (Chapters 21–22)
- **Special senses** — eye and ear (Chapter 23)

**HIGH-YIELD PEARL**
Read the map downward when you revise a chapter, and upward when you face an unknown slide. Downward gives you the function; upward gives you the name.

## The 60-second organ identification algorithm

### Step 1 — Is there a lumen?

- **No lumen, solid cords or follicles** → thyroid (follicles), adrenal (cords), liver (cords + central veins), parathyroid (cords), pituitary (cell sheets), lymphoid organ (nodules).
- **Yes, a lumen** → go to Step 2.

### Step 2 — What is the epithelium?

- **Stratified squamous** → esophagus, oral cavity, vagina, anal canal, epidermis (no lumen).
- **Simple columnar with crypts** → stomach, intestine, gallbladder, uterus.
- **Pseudostratified ciliated** → respiratory tract. **Pseudostratified with stereocilia** → epididymis, vas deferens.
- **Transitional** → ureter, bladder, renal pelvis.
- **Simple cuboidal/columnar tubules** → kidney, salivary ducts, epididymis.

### Step 3 — What surrounds it?

- **Cartilage rings** → trachea. **Cartilage plates** → bronchus. **No cartilage** → bronchiole.
- **Three muscle layers** → stomach. **Taeniae coli** → colon. **Thick muscle wall** → vas deferens, ureter.
- **Serosa** → intraperitoneal; **adventitia** → retroperitoneal.

## The ten master tables

### 1. Epithelium by location

| Epithelium | Where |
|---|---|
| Simple squamous | Alveoli, endothelium, mesothelium, Bowman capsule |
| Simple cuboidal | Kidney tubules, thyroid follicles, small ducts, ovary surface |
| Simple columnar | Stomach, intestine, gallbladder, uterine tube |
| Pseudostratified ciliated | Trachea, bronchi, nasal cavity |
| Pseudostratified with stereocilia | Epididymis, vas deferens |
| Stratified squamous keratinized | Epidermis |
| Stratified squamous non-keratinized | Oral cavity, esophagus, vagina, cornea, anal canal |
| Transitional | Renal pelvis, ureter, bladder, proximal urethra |
| Epithelioid | Adrenal cortex, islets, liver |

### 2. Collagen types

| Type | Where | Disease |
|---|---|---|
| I | Bone, skin, tendon, cornea | Osteogenesis imperfecta |
| II | Cartilage, vitreous | Chondrodysplasia |
| III | Reticulin: lymphoid organs, marrow, liver | Ehlers–Danlos vascular |
| IV | Basement membrane | Alport syndrome |
| V | Cornea, bone, placenta | Ehlers–Danlos |

### 3. Muscle

| Feature | Skeletal | Cardiac | Smooth |
|---|---|---|---|
| Striations | Yes | Yes | **No** |
| Nuclei | Many, **peripheral** | 1–2, **central** | 1, **central** |

| Feature | Skeletal | Cardiac | Smooth |
|---|---|---|---|
| Junctions | — | **Intercalated discs** | Gap junctions |
| T-tubules | Triads at A–I | Diads at Z | Caveolae |
| $Ca^{2+}$ sensor | Troponin C | Troponin C | **Calmodulin** |
| Regeneration | Limited (satellite) | **None** | Hypertrophy/hyperplasia |

## 4. Cartilage

| Feature | Hyaline | Elastic | Fibrocartilage |
|---|---|---|---|
| Collagen | II | II + elastic | **I + II** |
| Perichondrium | Yes | Yes | **No** |
| Where | Trachea, ribs, joints, nose | Epiglottis, pinna | Disc, symphysis, menisci |
| Cells | Isogenous groups | Numerous | **Rows** |

## 5. Bone cells

| Cell | Origin | Job |
|---|---|---|
| Osteoprogenitor | Mesenchyme | Precursor |
| Osteoblast | Mesenchyme | **Make osteoid**, RANKL |
| Osteocyte | Osteoblast | Maintain, sense load |
| Osteoclast | **Monocyte** | **Resorb**, ruffled border |

## 6. Blood cells

| Cell | % | Nucleus | Signature |
|---|---|---|---|
| Neutrophil | 60–70 | 3–5 lobes | Pale granules |
| Eosinophil | 2–4 | **Bilobed** | **Orange-red** granules |
| Basophil | 0.5–1 | S-shaped, hidden | Dark blue granules |
| Lymphocyte | 20–30 | Round, dense | Scant cytoplasm |
| Monocyte | 3–8 | **Kidney-shaped** | Largest cell |

## 7. Lymphoid organs

| Organ | Follicles | Afferent lymph | Unique |
|---|---|---|---|
| Thymus | **No** | No | **Hassall corpuscles** |
| Lymph node | Yes | **Yes** | Cortex/paracortex/medulla |
| Spleen | Yes | **No** | White + red pulp |
| Tonsil | Yes | No | Crypts, partial capsule |

## 8. Gut segments

| Segment | Diagnostic feature |
|---|---|
| Esophagus | Stratified squamous, skeletal muscle above |
| Stomach | **Three muscle layers**, parietal + chief cells, no goblet |
| Duodenum | **Brunner glands (submucosa)** |

| Segment | Diagnostic feature |
|---|---|
| Jejunum | Tallest villi; no Brunner glands; Peyer's patches belong to the ileum |
| Ileum | **Peyer's patches** |
| Colon | **No villi**, crypts, many goblet, taeniae |
| Appendix | Lymphoid tissue all round, blind tube |

### 9. Endocrine

| Gland | Pattern | Hormone |
|---|---|---|
| Thyroid | **Follicles with colloid** | T3, T4; C cells → calcitonin |
| Parathyroid | Cords of chief cells | PTH |
| Adrenal cortex G/F/R | Cords | Aldosterone / cortisol / androgens |
| Adrenal medulla | Chromaffin | Epinephrine |
| Pituitary anterior | Acidophils + basophils | GH, PRL / ACTH, TSH, FSH, LH |
| Pituitary posterior | Fibrillary, Herring bodies | Oxytocin, ADH |
| Islets | Clusters | Insulin, glucagon, somatostatin |

### 10. Airway

| Level | Cartilage | Glands | Muscle |
|---|---|---|---|
| Trachea | **Rings** | Yes | Trachealis |
| Bronchus | **Plates** | Yes | Complete ring |
| Bronchiole | **None** | **None** | Prominent |
| Respiratory bronchiole | None | None | + alveoli in wall |

## One hundred facts in one hundred lines

1. Basophilic = binds hematoxylin = nucleic acid.
2. Eosinophilic = binds eosin = protein.
3. PAS stains glycogen and basement membrane.
4. Silver stains reticulin (type III collagen).
5. Light microscope resolution = 0.2 µm.
6. Frozen section = rapid diagnosis + lipid preservation.
7. Plasma membrane thickness = 7–10 nm.
8. $Na^+/K^+$-ATPase pumps 3 $Na^+$ out, 2 $K^+$ in.
9. Clathrin mediates receptor-mediated endocytosis.
10. Mitochondria are maternally inherited.
11. Steroid-secreting cells have **tubular** cristae.
12. Nissl bodies = rough ER + polysomes.
13. Golgi **modifies and sorts**; RER **synthesizes**.
14. Lysosomes are acidic (pH ≈ 5).
15. Lipofuscin = residual body pigment.
16. Peroxisomes contain catalase.
17. Proteasomes degrade ubiquitin-tagged protein.
18. Kinesin = anterograde; dynein = retrograde.

19. Cilia = 9 + 2; basal body = 9 + 0.
20. Keratin = epithelium; vimentin = mesenchyme; desmin = muscle; GFAP = astrocyte.
21. Lamin A mutation = progeria.
22. Nucleolus makes rRNA; NORs on 13, 14, 15, 21, 22.
23. Barr body = inactivated X.
24. p53 = guardian of the genome.
25. Meiosis = four haploid cells; crossing over in pachytene.
26. Apoptosis = no inflammation.
27. Labile tissue = epidermis, gut, marrow; permanent = neurons, cardiac muscle.
28. Epithelium is avascular.
29. Basal lamina = lamina lucida + lamina densa; the basement membrane adds the reticular lamina (type III collagen).
30. Type IV collagen is in the basal lamina.
31. Desmosomes link to keratin; hemidesmosomes to laminin via integrin.
32. E-cadherin is calcium-dependent.
33. Gap junctions = connexins.
34. Microvilli core = actin.
35. Stereocilia are long microvilli (epididymis).
36. Endothelium and mesothelium are simple squamous.
37. Pseudostratified epithelium is truly one layer.
38. Transitional epithelium has umbrella cells and uroplakins.
39. Goblet cell = unicellular mucous gland.
40. Sebaceous gland = holocrine.
41. Mammary gland = modified apocrine gland; lipid by apocrine secretion, protein by merocrine secretion.
42. Pancreas = merocrine.
43. Myoepithelial cells compress acini.
44. Collagen synthesis needs **vitamin C** and **copper**.
45. Scurvy = defective hydroxylation.
46. Marfan = fibrillin.
47. Ehlers–Danlos = collagen processing.
48. Plasma cell nucleus = clock-face, eccentric.
49. Mast cell = histamine + heparin.
50. Brown fat = multilocular + UCP1 + central nucleus.
51. Leptin reduces appetite; adiponectin increases insulin sensitivity.
52. Hyaline cartilage = type II collagen.
53. Fibrocartilage = **no perichondrium**.
54. Interstitial growth = from within lacunae.
55. Osteoclast = monocyte lineage, TRAP-positive.
56. Osteoblast = RANKL; OPG is the decoy.
57. Osteon = concentric lamellae + Haversian canal.
58. Epiphyseal zones: reserve → proliferate → hypertrophy → calcify → ossify.
59. Intramembranous ossification = flat bones.
60. PTH raises calcium; calcitonin lowers it.
61. Oligodendrocyte = CNS, many axons; Schwann = PNS, one segment.

62. Microglia are mesodermal.
63. Wallerian degeneration occurs distal to injury.
64. Regeneration ≈ 1 mm/day.
65. A band never shortens.
66. Skeletal T-tubules form triads at the A–I junction.
67. Cardiac T-tubules form diads at the Z line.
68. Smooth muscle uses calmodulin, not troponin.
69. Intercalated discs = fascia adherens + desmosome + gap junction.
70. Purkinje fibers conduct; they are not neurons.
71. Elastic artery = elastic lamellae in media.
72. Muscular artery = prominent internal elastic lamina.
73. Sinusoidal capillaries = liver, spleen, marrow.
74. Blood–brain barrier = continuous capillaries + tight junctions.
75. Lymphatic capillaries are blind-ended.
76. Erythrocyte lifespan = 120 days.
77. Neutrophil = 3–5 lobes, acute inflammation.
78. Eosinophil = parasites and allergy.
79. Basophil = histamine, IgE.
80. Monocyte = largest leukocyte.
81. Platelets = 2–4 µm, from megakaryocytes.
82. EPO from kidney; TPO from liver.
83. M:E ratio = 2:1 to 4:1.
84. CD4 ↔ MHC II; CD8 ↔ MHC I.
85. Thymus has no follicles; spleen has no afferent lymphatics.
86. Hassall corpuscles = thymus.
87. PALS = T-cell zone of spleen.
88. Germinal center = B-cell proliferation.
89. Enteric plexuses: Meissner (submucosa), Auerbach (myenteric).
90. Parietal cell = HCl + intrinsic factor.
91. Chief cell = pepsinogen.
92. Paneth cell = lysozyme and defensins.
93. Brunner glands = duodenum.
94. Peyer's patches = ileum.
95. Portal triad = artery, portal vein, bile duct.
96. Bile flows toward the triad; blood flows away.
97. Type I pneumocyte = gas exchange; type II = surfactant.
98. Club (formerly Clara) cells = bronchioles; they secrete surfactant-like material and detoxify inhaled compounds.
99. Podocytes form the filtration slit diaphragm (nephrin).
100. Macula densa = distal tubule; JG cells = afferent arteriole.

## Twenty-five distinctions worth knowing

| Do not confuse | With | The difference |
|---|---|---|
| Basophilic | Eosinophilic | DNA/RNA versus protein |
| Mucous | Mucosa | Secretion versus lining layer |

| Do not confuse | With | The difference |
|---|---|---|
| Endothelium | Mesothelium | Vessels versus body cavities |
| Pseudostratified | Stratified | One layer versus many |
| Hyaline cartilage | Fibrocartilage | Type II + perichondrium versus I + II + none |
| Osteoblast | Osteoclast | Builds versus resorbs; mesenchymal versus monocyte |
| Astrocyte | Oligodendrocyte | Support/BBB versus myelination |
| Schwann cell | Oligodendrocyte | One segment versus many axons |
| Neutrophil | Eosinophil | 3–5 lobes versus bilobed |
| Monocyte | Lymphocyte | Kidney-shaped versus round |
| Proximal tubule | Distal tubule | Brush border, eosinophilic versus clean, pale |
| Type I pneumocyte | Type II | Thin for gas versus cuboidal for surfactant |
| Parietal cell | Chief cell | HCl versus pepsinogen |
| Brunner gland | Peyer's patch | Duodenum submucosa versus ileum lymphoid |
| Zona glomerulosa | Zona fasciculata | Aldosterone versus cortisol |
| Adrenal cortex | Adrenal medulla | Steroid versus catecholamine |
| Anterior pituitary | Posterior pituitary | Glandular versus neural |
| Sertoli cell | Leydig cell | Barrier/nurture versus testosterone |
| Granulosa cell | Theca cell | Estrogen (aromatase) versus androgen |
| Proliferative | Secretory endometrium | Straight glands versus tortuous secreting glands |
| Follicular | Luteal phase | Estrogen-dominant versus progesterone-dominant |
| Rod | Cone | Dim light versus color and acuity |
| Endolymph | Perilymph | High $K^+$ versus high $Na^+$ |
| Apoptosis | Necrosis | No inflammation versus inflammation |
| Labile | Permanent tissue | Continuous turnover versus no division |

## Eight common errors

1. Saying the osteoclast is mesenchymal. It is **hematopoietic**.
2. Saying the A band shortens. It **does not**.
3. Saying the thymus has follicles. It **does not**.
4. Saying the spleen has afferent lymphatics. It **does not**.
5. Saying the adrenal medulla makes steroids. It makes **catecholamines**.
6. Saying FSH acts on Leydig cells. It acts on **Sertoli** cells; LH acts on Leydig.
7. Saying rods are at the fovea. The fovea has **cones only**.
8. Saying the stomach has goblet cells. It has **surface mucous cells**, not goblet cells.

# Appendix A · Stains and What They Prove

| Stain | Target | Color | Classic exam use |
|---|---|---|---|
| **Hematoxylin** | Nucleic acids | Blue-purple | Nuclei, rough ER |
| **Eosin** | Proteins | Pink-red | Cytoplasm, collagen, RBC |
| **PAS** | Glycogen, mucin, glycoproteins, basement membrane | Magenta | Goblet cells, liver glycogen, fungal walls |
| **Masson trichrome** | Collagen versus muscle | Blue/green versus red | Liver fibrosis, kidney biopsy |
| **Silver (reticulin)** | Type III collagen, nerve fibers | Black | Lymph node, spleen, marrow |
| **Orcein / resorcin–fuchsin** | Elastin | Dark brown-black | Aorta, lung, skin |
| **Sudan black / oil red O** | Lipid | Black/red | Steatosis, adipocytes (frozen) |
| **Osmium tetroxide** | Lipid, myelin | Black | Fat, myelin |
| **Cresyl violet (Nissl)** | Rough ER | Violet | Neurons |
| **Feulgen** | DNA | Magenta | DNA quantification |
| **Prussian blue** | Iron (hemosiderin) | Blue | Iron overload, old hemorrhage |
| **Congo red** | Amyloid | Apple-green birefringence | Amyloidosis |
| **Gram** | Bacterial wall | Purple (positive) / pink (negative) | Infection |
| **Ziehl–Neelsen** | Acid-fast bacilli | Red | Tuberculosis |
| **Immunohistochemistry** | Specific antigen | Brown (DAB) or fluorescent | Tumor typing, cell identification |

## The stains the atlas uses

The textbook's own appendix is short but examinable, because every plate caption in the atlas names its stain. The table below follows its list and adds the chemistry behind each stain; the table above is the quick-recall version.

| Stain | Chemistry | What it stains | Why it is used |
|---|---|---|---|
| **Hematoxylin & eosin (H&E)** | Hematoxylin is a **basic** dye; eosin is an **acidic** counterstain | Nuclei and rough ER blue-purple; collagen and most cytoplasmic proteins, especially mitochondria, pink-red | The default survey stain; the source of the basophilic/eosinophilic vocabulary |
| **Pararosaniline–toluidine blue (PT)** | Basic dye combination | Chromatin purple; cytoplasm and collagen lighter violet | Penetrates **plastic (resin) sections**, giving finer detail than H&E |
| **Mallory trichrome** | Sequential stain combination | Nuclei purple; cytoplasm, keratin, and erythrocytes red-orange; **collagen blue** | Demonstrates cells and small vessels of connective tissue. Masson and Gomori trichromes are equivalent, with collagen blue-green or green |
| **Picrosirius–hematoxylin (PSH)** | Sirius red in picric acid | Collagen red; cytoplasm pink-violet; nuclei purple | Collagen becomes **birefringent** under polarized light — a specific collagen stain |

| Stain | Chemistry | What it stains | Why it is used |
|---|---|---|---|
| **Periodic acid–Schiff (PAS)** | Periodic acid oxidizes vicinal diols; Schiff reagent detects them | Carbohydrate-rich structures **magenta** | Mucin granules, glycogen, glycocalyx, basement membranes |
| **Wright–Giemsa** | Two related Romanowsky combinations | Nuclei purple; erythrocytes pink-orange; leukocyte granules differentially | Blood and bone-marrow smears |
| **Silver and gold impregnation** | Metal salts reduce on filaments | Nerve filaments and **reticulin (type III collagen)** dark brown or black | Largely replaced by immunohistochemistry |
| **Elastin stains** — Weigert resorcin fuchsin, aldehyde fuchsin, orcein | Dye binds elastin | Elastic fibers brown or purple | Distinguishing elastic from collagen (aorta, lung, skin) |
| **Lipid stains** — oil red O, Sudan black, osmium tetroxide | Lipophilic dye partitions into lipid | Lipid droplets and myelin | Requires frozen sections or special processing, because solvents used in paraffin embedding extract lipid |
| **Other aniline dyes** — azures, cresyl violet, brilliant cresyl blue, luxol fast blue, light green | Basic dyes | Various cellular and extracellular components | Permanence and color contrast in paraffin sections; many began as textile dyes |

**UNDER THE MICROSCOPE**

Read the caption before you read the picture. "PT" tells you the section is plastic-embedded and shows fine detail; "Mallory" tells you collagen will be blue and cytoplasm red; "PSH" tells you collagen is red and can be examined under polarized light. The stain is part of the evidence.

# Appendix B · Vocabulary You Should Know

| Term | Meaning |
|---|---|
| **Parenchyma** | The functional cells of an organ |
| **Stroma** | The supportive connective tissue framework |
| **Capsule** | Dense connective tissue enclosing an organ |
| **Hilum** | The entry/exit point for vessels, ducts, and nerves |
| **Cortex / medulla** | Outer versus inner region |
| **Trabecula** | A partition or supporting beam of connective tissue |
| **Sinusoid** | A wide, discontinuous capillary |
| **Acinus** | A berry-like cluster of secretory cells |
| **Tubuloacinar** | A gland with both tubular and acinar secretory units |
| **Metaplasia** | Change from one mature cell type to another |
| **Dysplasia** | Disordered growth with atypical cells |
| **Neoplasia** | New, autonomous growth |
| **Hypertrophy** | Increase in cell size |
| **Hyperplasia** | Increase in cell number |
| **Atrophy** | Decrease in size or number |
| **Apoptosis** | Programmed cell death |
| **Necrosis** | Pathological cell death with inflammation |
| **Autolysis** | Self-digestion after death |
| **Fixation** | Chemical preservation of tissue |
| **Artefact** | A structure produced by processing, not present in life |

# PART VII — ASSESS YOUR KNOWLEDGE

One hundred examination questions with explained answers

# 100 Examination Questions

**EXAM TIP**
Answer all 100 before turning to the key. Mark every guess with a dot; guesses, not errors, are what you must revise. A score of 80 or more suggests that the material is secure; below 65, return to the tables in Part VI. Neither figure predicts an examination result.

## Cells and techniques (1–12)

**1.** A tissue section is treated with a basic dye. Which structure will stain most intensely?
A. Nuclei and rough endoplasmic reticulum
B. Erythrocytes
C. Elastic fibers
D. Collagen bundles

**2.** Which stain would you choose to demonstrate a basement membrane?
A. Eosin
B. Orcein
C. PAS
D. Sudan black

**3.** A surgeon requires an immediate diagnosis during an operation. The correct specimen preparation is:
A. Resin embedding
B. Whole-mount preparation
C. Routine paraffin section
D. Frozen section

**4.** The resolving power of the light microscope is approximately:
A. 2 µm
B. 0.2 µm
C. 20 µm
D. 0.2 nm

**5.** Which technique localizes a specific protein in a tissue section using an antibody?
A. Autoradiography
B. In situ hybridization
C. Immunohistochemistry
D. Polarizing microscopy

**6.** A cell specialized for steroid hormone synthesis would show abundant:
A. Rough ER and Golgi
B. Smooth ER and lipid droplets
C. Lysosomes and residual bodies
D. Free ribosomes and proteasomes

**7.** Which organelle contains catalase and performs β-oxidation of very long-chain fatty acids?
A. Mitochondrion
B. Lysosome
C. Peroxisome
D. Proteasome

**8.** Retrograde axonal transport of viruses toward the neuronal soma is mediated by:
A. Myosin
B. Dynein and kinesin equally
C. Kinesin

D. Dynein

**9.** Which intermediate filament is characteristic of astrocytes?

A. Vimentin
B. GFAP
C. Neurofilament
D. Desmin

**10.** A cell with a 9 + 0 arrangement of microtubules is:

A. A basal body
B. A sperm flagellum
C. A microvillus
D. A motile cilium

**11.** A mutation in lamin A produces:

A. Alport syndrome
B. Progeria
C. Duchenne dystrophy
D. Marfan syndrome

**12.** The nucleolar organizer regions are located on which chromosomes?

A. 16, 17, 18, 19, 20
B. X and Y
C. 1, 2, 3, 4, 5
D. 13, 14, 15, 21, 22

## Epithelium, connective tissue, cartilage and bone (13–32)

**13.** Which junctional complex is principally responsible for sealing the intercellular space?

A. Hemidesmosome
B. Desmosome
C. Gap junction
D. Tight junction

**14.** Pemphigus vulgaris results from autoantibodies against:

A. Connexin
B. Laminin
C. Desmoglein
D. Integrin

**15.** Which epithelium lines the normal ureter?

A. Simple columnar
B. Pseudostratified ciliated
C. Stratified squamous
D. Transitional

**16.** Stereocilia are found in the:

A. Epididymis
B. Fallopian tube
C. Intestinal villi
D. Trachea

**17.** The basement membrane's lamina reticularis is composed mainly of:

A. Type I collagen
B. Type II collagen
C. Type III collagen
D. Type IV collagen

**18.** Which collagen type is defective in osteogenesis imperfecta?

A. Type II
B. Type III
C. Type IV
D. Type I

**19.** Hydroxylation of proline and lysine during collagen synthesis requires:

A. Vitamin C
B. Vitamin D
C. Copper
D. Vitamin A

**20.** A cell with an eccentric nucleus showing a clock-face chromatin pattern and a pale perinuclear halo is a:

A. Macrophage
B. Eosinophil
C. Mast cell
D. Plasma cell

**21.** The main function of UCP1 in brown adipose tissue is:

A. ATP synthesis
B. Uncoupling of oxidative phosphorylation to produce heat
C. Fatty acid storage
D. Leptin secretion

**22.** Adiponectin, secreted by adipocytes, acts to:

A. Increase insulin sensitivity
B. Increase blood pressure
C. Stimulate lipolysis in brown fat
D. Reduce appetite

**23.** Which cartilage type lacks a perichondrium?

A. Elastic
B. Fibrocartilage
C. Tracheal hyaline
D. Hyaline

**24.** The intervertebral disc's annulus fibrosus is composed of:

A. Dense regular connective tissue
B. Hyaline cartilage
C. Elastic cartilage
D. Fibrocartilage

**25.** Which cell is responsible for bone resorption?

A. Osteocyte
B. Osteoclast
C. Osteoprogenitor cell
D. Osteoblast

**26.** Osteoclasts are derived from:

A. Monocyte–macrophage lineage
B. Osteoblasts
C. Neural crest
D. Mesenchymal cells

**27.** In endochondral ossification, the zone in which chondrocytes enlarge and the matrix calcifies is the:

A. Ossification zone
B. Reserve zone
C. Proliferative zone
D. Hypertrophic zone

**28.** Which hormone lowers serum calcium?
A. Calcitriol
B. Calcitonin
C. Growth hormone
D. PTH

**29.** The primary ossification center of a long bone appears in the:
A. Diaphysis
B. Metaphysis
C. Periosteum
D. Epiphysis

**30.** A multinucleated cell lying in a scalloped resorption pit is:
A. An osteocyte
B. An osteoclast
C. A macrophage
D. An osteoblast

**31.** Dense regular connective tissue is characterized by:
A. Parallel collagen bundles
B. Abundant elastic fibers
C. A reticular network
D. Randomly oriented collagen bundles

**32.** Which tissue is the site of hematopoiesis in the adult?
A. Liver
B. Spleen
C. Red bone marrow
D. Yellow bone marrow

## Nerve and muscle (33–44)

**33.** Nissl bodies are composed of:
A. Rough ER and polysomes
B. Neurofilaments
C. Synaptic vesicles
D. Smooth ER and Golgi

**34.** Which glial cell myelinates several axons in the central nervous system?
A. Schwann cell
B. Microglia
C. Astrocyte
D. Oligodendrocyte

**35.** Microglia are derived from:
A. Neural tube
B. Mesoderm (monocyte lineage)
C. Endoderm
D. Neural crest

**36.** The blood–brain barrier is formed principally by:
A. Pericytes alone
B. Astrocyte end-feet alone

C. Tight junctions between capillary endothelial cells
D. Ependymal cells

**37.** In a peripheral nerve, the connective tissue sheath surrounding a single axon is the:

A. Perineurium
B. Endoneurium
C. Mesoneurium
D. Epineurium

**38.** During muscle contraction, which band does not change in length?

A. I band
B. H band
C. A band
D. None of these

**39.** Calcium binds which molecule in skeletal muscle to initiate contraction?

A. Myosin light chain
B. Actinin
C. Calmodulin
D. Troponin C

**40.** Intercalated discs contain all of the following except:

A. Fascia adherens
B. Desmosomes
C. Gap junctions
D. Tight junctions

**41.** Which feature distinguishes cardiac from skeletal muscle?

A. Mitochondria
B. Striations
C. Central nucleus
D. Myofibrils

**42.** Smooth muscle contraction differs from striated muscle because it uses:

A. Tropomyosin only
B. No calcium
C. Troponin C
D. Calmodulin and myosin light-chain kinase

**43.** T-tubules and terminal cisternae form triads at the:

A. H band
B. Z line
C. A–I junction
D. M line

**44.** A muscle fiber that is rich in mitochondria and myoglobin and resistant to fatigue is:

A. All equally
B. Type I
C. Type IIa
D. Type IIx

## Blood, hemopoiesis and immunity (45–60)

**45.** The most abundant leukocyte in normal peripheral blood is the:

A. Neutrophil
B. Monocyte
C. Eosinophil

D. Lymphocyte

**46.** A bilobed nucleus with bright orange-red granules identifies the:

A. Eosinophil
B. Monocyte
C. Basophil
D. Neutrophil

**47.** The largest leukocyte, with a kidney-shaped nucleus, is the:

A. Neutrophil
B. Basophil
C. Lymphocyte
D. Monocyte

**48.** Platelets are derived from:

A. Erythroblasts
B. Megakaryocytes
C. Monocytes
D. Lymphocytes

**49.** Erythropoietin is produced mainly by the:

A. Kidney
B. Spleen
C. Bone marrow
D. Liver

**50.** The normal myeloid-to-erythroid ratio in bone marrow is:

A. 10:1
B. 1:1
C. 2:1 to 4:1
D. 1:2

**51.** Which immunoglobulin is the main secretory antibody at mucosal surfaces?

A. IgA
B. IgE
C. IgG
D. IgM

**52.** CD4-positive T cells recognize antigen presented by:

A. Antibody alone
B. MHC class I
C. MHC class II
D. MHC class III

**53.** Hassall corpuscles are found in the:

A. Thymus
B. Tonsil
C. Spleen
D. Lymph node

**54.** Which organs lack afferent lymphatic vessels?

A. Lymph node
B. Thymus and spleen
C. Tonsil
D. Appendix

**55.** The periarteriolar lymphoid sheath of the spleen contains mainly:

A. Plasma cells
B. B cells
C. T cells
D. Macrophages

**56.** A germinal center is a site of:
A. Erythropoiesis
B. Platelet production
C. T-cell maturation
D. B-cell proliferation

**57.** The red pulp of the spleen functions mainly to:
A. Secrete antibodies
B. Produce T cells
C. Filter blood and remove aged erythrocytes
D. Store lymphocytes

**58.** The lifespan of a normal erythrocyte is approximately:
A. 120 days
B. 1 year
C. 7 days
D. 30 days

**59.** During the second trimester, hemopoiesis occurs mainly in the:
A. Liver and spleen
B. Bone marrow
C. Thymus
D. Yolk sac

**60.** Which cell is the principal antigen-presenting cell of the epidermis?
A. Keratinocyte
B. Melanocyte
C. Langerhans cell
D. Merkel cell

## Digestive system (61–72)

**61.** Which layer of the gut wall contains the myenteric (Auerbach) plexus?
A. Between the circular and longitudinal muscle layers
B. Serosa
C. Mucosa
D. Submucosa

**62.** Parietal cells of the gastric glands secrete:
A. HCl and intrinsic factor
B. Mucus
C. Gastrin
D. Pepsinogen

**63.** Brunner glands are found in the:
A. Duodenum
B. Jejunum
C. Ileum
D. Colon

**64.** Peyer's patches are characteristic of the:
A. Jejunum
B. Ileum

C. Stomach
D. Duodenum

**65.** Paneth cells are located at the base of intestinal crypts and secrete:

A. Lysozyme and defensins
B. Gastrin
C. Intrinsic factor
D. Mucus

**66.** Which feature is absent in the colon but present in the small intestine?

A. Villi
B. Enteroendocrine cells
C. Goblet cells
D. Crypts

**67.** The portal triad contains all of the following except:

A. Hepatic artery
B. Portal vein
C. Bile duct
D. Central vein

**68.** Kupffer cells are:

A. Fat-storing cells
B. Endothelial cells
C. Hepatocytes
D. Sinusoidal macrophages

**69.** Bile flows in which direction relative to the portal triad?

A. Away from the triad
B. Toward the triad
C. Parallel to blood flow
D. Into the central vein

**70.** The exocrine pancreas is identified by:

A. Basophilic acini and zymogen granules
B. Mucous acini only
C. Follicles of colloid
D. Islets and striated ducts

**71.** Which cells secrete insulin?

A. Delta cells
B. PP cells
C. Alpha cells
D. Beta cells

**72.** The esophagus is lined by:

A. Transitional epithelium
B. Pseudostratified ciliated epithelium
C. Simple columnar epithelium
D. Stratified squamous non-keratinized epithelium

## Respiratory system and skin (73–82)

**73.** Type II pneumocytes are characterized by:

A. Cilia and goblet granules
B. Phagocytic vacuoles
C. A squamous shape and no organelles

D. Lamellar bodies and surfactant production

**74.** Clara (club) cells are found in the:

A. Bronchioles
B. Pleura
C. Alveoli
D. Trachea

**75.** Cartilage in the bronchial tree changes in which sequence?

A. Rings → none → plates
B. None → plates → rings
C. Rings → plates → none
D. Plates → rings → none

**76.** The blood–air barrier does not include:

A. Basement membrane
B. Type II pneumocyte
C. Surfactant
D. Type I pneumocyte

**77.** The layer of the epidermis present only in thick skin is the:

A. Lucidum
B. Basale
C. Spinosum
D. Granulosum

**78.** Melanocytes are derived from:

A. Endoderm
B. Ectoderm
C. Neural crest
D. Mesoderm

**79.** Sebaceous glands secrete by which mechanism?

A. Apocrine
B. Holocrine
C. Endocrine
D. Merocrine

**80.** Pacinian corpuscles are located in the:

A. Basal epidermis
B. Dermal papillae
C. Deep dermis and hypodermis
D. Stratum corneum

**81.** Which cell type is most numerous in the epidermis?

A. Langerhans cell
B. Keratinocyte
C. Merkel cell
D. Melanocyte

**82.** Which receptor is responsible for fine tactile discrimination in the fingertips?

A. Pacinian corpuscle
B. Meissner corpuscle
C. Ruffini ending
D. Free nerve ending

## Urinary and endocrine (83–92)

**83.** The filtration slit diaphragm between podocyte foot processes contains:
A. Desmoglein
B. Connexin
C. Laminin only
D. Nephrin

**84.** The macula densa is located in the:
A. Proximal tubule
B. Thin limb
C. Distal tubule
D. Collecting duct

**85.** Renin is secreted by:
A. Mesangial cells
B. Macula densa cells
C. Juxtaglomerular cells
D. Podocytes

**86.** Which tubule segment is most eosinophilic because of abundant mitochondria?
A. Collecting duct
B. Proximal convoluted tubule
C. Thin descending limb
D. Distal convoluted tubule

**87.** The epithelium of the ureter is:
A. Pseudostratified
B. Simple columnar
C. Transitional
D. Stratified squamous

**88.** Thyroid follicles are lined by:
A. Stratified cuboidal epithelium
B. Pseudostratified epithelium
C. Transitional epithelium
D. Simple cuboidal epithelium

**89.** Calcitonin is secreted by:
A. Follicular cells
B. Parafollicular C cells
C. Chief cells
D. Oxyphil cells

**90.** The zona fasciculata of the adrenal cortex secretes:
A. Epinephrine
B. Aldosterone
C. Cortisol
D. Androgens

**91.** Which is derived from neural crest?
A. Adrenal cortex
B. Adrenal medulla
C. Thyroid follicle
D. Parathyroid chief cells

**92.** Herring bodies are found in the:
A. Pineal gland
B. Thyroid
C. Anterior pituitary

D. Posterior pituitary

## Reproductive system, eye and ear (93–100)

**93.** The blood–testis barrier is formed by tight junctions between:

A. Peritubular myoid cells
B. Spermatogonia
C. Sertoli cells
D. Leydig cells

**94.** LH acts primarily on:

A. Spermatogonia
B. Epididymal cells
C. Sertoli cells
D. Leydig cells

**95.** The usual site of fertilization is the:

A. Uterine cavity
B. Ampulla of the uterine tube
C. Isthmus
D. Cervix

**96.** Straight, narrow endometrial glands with numerous mitoses indicate which phase?

A. Menstrual
B. Proliferative
C. Secretory
D. Gestational

**97.** The corpus luteum secretes mainly:

A. FSH
B. Estrogen
C. Progesterone
D. hCG

**98.** The fovea centralis contains:

A. Both equally
B. Neither
C. Rods only
D. Cones only

**99.** Aqueous humor drains into the:

A. Canal of Schlemm
B. Ciliary body
C. Choroid
D. Vitreous body

**100.** The sensory cells of the organ of Corti are:

A. Spiral ganglion cells
B. Tectorial cells
C. Hair cells
D. Supporting cells

# Answer Key with Explanations

**1. A** — Basic dyes such as hematoxylin bind acidic structures, and nucleic acids (nucleus, rough ER) are the most acidic components of the cell. **Trap:** D — collagen is basic and binds eosin, not hematoxylin. B — hemoglobin is basic. C — elastin is not particularly acidic.

**2. C** — PAS stains the glycoprotein-rich basement membrane magenta; eosin stains it only weakly and orcein stains elastin. **Trap:** A — eosin stains protein, not carbohydrate. B — orcein stains elastin. D — Sudan black stains lipid.

**3. D** — A frozen section gives a diagnosis within minutes, at the cost of some morphological distortion. **Trap:** C takes a day or more. A and B are research or teaching preparations, not urgent diagnostics.

**4. B** — The light microscope resolves about 0.2 µm; electron microscopy is required for anything smaller. **Trap:** D is electron-microscope resolution. A and C are far too coarse.

**5. C** — Immunohistochemistry uses a specific antibody, visualized by an enzyme reaction or a fluorophore. **Trap:** A detects newly synthesized molecules. B detects nucleic acid sequence. D detects birefringence.

**6. B** — Steroid synthesis requires smooth ER and lipid (cholesterol) stores; such cells appear pale and foamy. **Trap:** A is the profile of a protein-secreting cell. C is a phagocyte. D is a stem or rapidly dividing cell.

**7. C** — Peroxisomes contain catalase and carry out β-oxidation of very long-chain fatty acids. **Trap:** B is the lysosome (acid hydrolases). D degrades ubiquitinated proteins. A is the main ATP source.

**8. D** — Dynein moves cargo toward the minus end, i.e. toward the cell body (retrograde). **Trap:** C moves cargo the other way (anterograde). A is a muscle motor. B is wrong because transport is directional.

**9. B** — GFAP is the characteristic intermediate filament of astrocytes. **Trap:** D is muscle. A is connective tissue and some neural cells. C is the neuronal cytoskeleton.

**10. A** — The basal body is 9 + 0; the motile cilium and flagellum are 9 + 2. **Trap:** D and B are 9 + 2. C has no microtubules at all.

**11. B** — Lamin A mutations destabilize the nuclear envelope and cause Hutchinson–Gilford progeria. **Trap:** C is dystrophin. D is fibrillin. A is type IV collagen.

**12. D** — The acrocentric chromosomes 13, 14, 15, 21, and 22 carry the nucleolar organizer regions. **Trap:** C and A are metacentric or submetacentric autosomes. B carries few rRNA genes.

**13. D** — The tight junction (zonula occludens) seals the intercellular space and creates polarity. **Trap:** B anchors cells to each other but does not seal. C communicates. A anchors epithelium to basal lamina.

**14. C** — Anti-desmoglein antibodies disrupt desmosomes and cause intraepithelial blistering. **Trap:** D links integrins to matrix (epidermolysis bullosa). A is a gap junction protein. B is a basal lamina protein.

**15. D** — The ureter is lined by transitional epithelium, which allows distension. **Trap:** C is the vagina and esophagus. A is the gallbladder. B is the trachea.

**16. A** — Long, stiff microvilli (stereocilia) are found in the epididymis and inner ear. **Trap:** D has motile cilia. B has motile cilia. C has microvilli of normal length.

**17. C** — The lamina reticularis contains type III (reticular) collagen, produced by fibroblasts. **Trap:** A is dense connective tissue. B is cartilage. D is the lamina densa.

**18. D** — Osteogenesis imperfecta results from defective type I collagen. **Trap:** A causes chondrodysplasias. B causes Ehlers–Danlos type IV. C causes Alport syndrome.

**19. A** — Vitamin C is required for prolyl and lysyl hydroxylase; deficiency causes scurvy. **Trap:** D is needed for vision. B is calcium homeostasis. C is a cofactor for lysyl oxidase.

**20. D** — The plasma cell has an eccentric clock-face nucleus and a pale perinuclear Golgi zone. **Trap:** C has granules and a central nucleus. A has a bean-shaped nucleus and vacuoles. B has a bilobed nucleus.

**21. B** — UCP1 uncouples the proton gradient so that energy is released as heat. **Trap:** A is the normal function that UCP1 bypasses. C is unilocular white fat. D is a white-adipocyte hormone.

**22. A** — Adiponectin increases insulin sensitivity; it is reduced in obesity. **Trap:** D is leptin. B and C are not adiponectin actions.

**23. B** — Fibrocartilage has no perichondrium, which contributes to its poor healing. **Trap:** D and A have perichondrium. C is hyaline and is covered by perichondrium.

**24. D** — The annulus fibrosus is fibrocartilage, with type I and type II collagen. **Trap:** B is the articular surface. C is the epiglottis. A is tendon.

**25. B** — The osteoclast resorbs bone using a ruffled border and acid secretion. **Trap:** D forms bone. A maintains it. C is the precursor.

**26. A** — Osteoclasts arise from the monocyte–macrophage lineage, not from mesenchyme. **Trap:** D gives osteoblasts. B are the bone-forming cells themselves. C gives melanocytes and Schwann cells.

**27. D** — In the hypertrophic zone chondrocytes enlarge and the matrix begins to calcify. **Trap:** B is small, sparse cells. C is active division. A is where osteoid is deposited.

**28. B** — Calcitonin from thyroid C cells inhibits osteoclasts and lowers serum calcium. **Trap:** D raises calcium. A raises calcium by increasing absorption. C has no direct effect.

**29. A** — The primary center appears in the diaphysis; secondary centers appear in the epiphyses. **Trap:** D is a secondary center. B is the growth plate region. C is not an ossification center.

**30. B** — A multinucleated cell in a Howship lacuna is an osteoclast. **Trap:** D is cuboidal and lines the surface. A sits in a lacuna without a pit. C does not make Howship lacunae.

**31. A** — Dense regular connective tissue has parallel collagen bundles, as in tendon. **Trap:** D is dense irregular. B describes elastic tissue. C is reticular tissue.

**32. C** — Red bone marrow is the adult hematopoietic site. **Trap:** A is the fetal site. B is a lymphoid filter. D is a fat store, not hemopoietic.

**33. A** — Nissl bodies are stacked rough ER and polysomes. **Trap:** D describes steroid-producing cells. B is the axonal cytoskeleton. C are synaptic, not somatic.

**34. D** — One oligodendrocyte myelinates segments of several CNS axons. **Trap:** C is a supporting astrocyte. A myelinates in the PNS. B is phagocytic.

**35. B** — Microglia are mesodermal, derived from monocytes — the exception among glia. **Trap:** D gives melanocytes and Schwann cells. A gives neurons and macroglia. C gives epithelia.

**36. C** — Tight junctions between endothelial cells are the anatomical basis; astrocyte feet and pericytes reinforce it. **Trap:** B contributes a barrier but is not the seal. D lines ventricles. A support the capillary.

**37. B** — Endoneurium surrounds individual nerve fibers; perineurium surrounds fascicles. **Trap:** D surrounds the whole nerve. A surrounds a fascicle. C does not exist as a standard layer.

**38. C** — The A band is the length of the thick filaments and does not change during contraction. **Trap:** A and B shorten. D is wrong because the A band is fixed.

**39. D** — Calcium binds troponin C, which moves tropomyosin and exposes the myosin-binding site. **Trap:** C is the smooth-muscle calcium sensor. A is the smooth-muscle target. B is a structural protein.

**40. D** — Intercalated discs contain fascia adherens, desmosomes, and gap junctions. **Trap:** A anchors actin, B resists shear, and C couples the cells electrically; a tight junction is not part of the disc.

**41. C** — Cardiac muscle has one or two central nuclei; skeletal muscle nuclei are peripheral and multiple. Striations, myofibrils, and mitochondria are shared, so they cannot distinguish the two. **Trap:** B is shared. D and A are shared.

**42. D** — Smooth muscle uses calmodulin and myosin light-chain kinase; it has no troponin. **Trap:** C refers to striated muscle. A is part of the thin filament but not the calcium sensor. B is wrong.

**43. C** — In mammalian skeletal muscle, triads lie at the A–I junction; in cardiac muscle, diads lie at the Z line. **Trap:** B is the site of cardiac diads. D is the middle of the sarcomere. A is the H zone, which lies inside the A band.

**44. B** — Type I fibers are slow, oxidative, and fatigue-resistant. **Trap:** C is fast, oxidative-glycolytic and fatigable. D is fast glycolytic. A is wrong.

**45. A** — Neutrophils constitute 60–70% of circulating leukocytes. **Trap:** D is second in adults. B and C are minorities.

**46. A** — The eosinophil has a bilobed nucleus and bright orange-red granules. **Trap:** C has basophilic granules. D has a multilobed nucleus and pale granules. B is agranular and larger.

**47. D** — The monocyte is the largest leukocyte and has a kidney-shaped nucleus. **Trap:** C has a round dark nucleus and scant cytoplasm. A has a segmented nucleus. B is smaller with coarse granules.

**48. B** — Platelets are fragments of megakaryocyte cytoplasm. **Trap:** A is an erythroid precursor. C and D do not make platelets.

**49. A** — Peritubular interstitial cells of the kidney produce erythropoietin. **Trap:** D produces it in fetal life. B and C are not the main source.

**50. C** — The normal M:E ratio is 2:1 to 4:1, as Chapter 13 states. **Trap:** A (10:1) suggests myeloid hyperplasia; B and D are not normal adult values.

**51. A** — IgA is the main secretory immunoglobulin; it exists as a dimer with a secretory component. **Trap:** C crosses the placenta and dominates serum. D is the first antibody response. B mediates allergy.

**52. C** — $CD4^+$ helper T cells recognize antigen on MHC class II. **Trap:** B is read by CD8 cells. D is not a classical antigen-presenting molecule. A is wrong.

**53. A** — Hassall corpuscles are concentric epithelial whorls unique to the thymic medulla. **Trap:** C has no corpuscles. D has follicles and sinuses. B has crypts and lymphoid tissue.

**54. B** — Neither the thymus nor the spleen has afferent lymphatics. **Trap:** A has afferent lymphatics. C and D are MALT, not encapsulated organs.

**55. C** — The PALS is the T-cell zone of the spleen; follicles are the B-cell zone. **Trap:** B occupies follicles. D and A are present but not the defining population.

**56. D** — Germinal centers are sites of B-cell proliferation and affinity maturation. **Trap:** C occurs in the thymus. A occurs in marrow. B occurs in marrow.

**57. C** — The red pulp filters blood and removes aged or damaged erythrocytes. **Trap:** B is thymic. D and A describe white pulp.

**58. A** — Erythrocytes circulate for about 120 days. **Trap:** C is a platelet. D and B are wrong.

**59. A** — In the second trimester, the liver and spleen are the main hemopoietic organs. **Trap:** D is the earliest site. B and C are later or lymphoid.

**60. C** — Langerhans cells are the epidermal antigen-presenting cells; they originate in bone marrow. **Trap:** A is structural. B is pigment-producing. D is a mechanoreceptor.

**61. A** — Auerbach's plexus lies between the circular and longitudinal layers of the muscularis externa. **Trap:** D contains the submucosal (Meissner) plexus. C is the mucosa; B is the outer covering.

**62. A** — Parietal cells secrete hydrochloric acid and intrinsic factor. **Trap:** D is secreted by chief cells. B by mucous cells. C by G cells.

**63. A** — Brunner glands lie in the submucosa of the duodenum. **Trap:** B, C, and D contain no Brunner glands.

**64. B** — Peyer's patches are characteristic of the ileum. **Trap:** Peyer's patches are not a feature of the duodenum or jejunum, and the stomach has no organized lymphoid nodules of this kind.

**65. A** — Paneth cells secrete lysozyme and defensins, contributing to innate immunity. **Trap:** D is goblet cells. B is G cells. C is parietal cells.

**66. A** — Villi are present in the small intestine and absent in the colon. **Trap:** C, D, and B are present in the colon.

**67. D** — The central vein is at the center of the lobule, not in the portal triad. **Trap:** A, B, and C form the triad. The central vein is at the lobule center.

**68. D** — Kupffer cells are the resident macrophages of the hepatic sinusoids. **Trap:** C are the parenchymal cells. A are Ito cells. B line the sinusoid.

**69. B** — Bile flows toward the portal triad while blood flows away from it toward the central vein. **Trap:** A is the direction of blood. C is wrong for bile. D is the direction of blood.

**70. A** — The exocrine pancreas has basophilic acini with eosinophilic zymogen granules. **Trap:** D describes salivary gland. B describes mucous glands. C describes thyroid.

**71. D** — Beta cells secrete insulin and are about twice as abundant as alpha cells. **Trap:** C is glucagon. A is somatostatin. B is pancreatic polypeptide.

**72. D** — The esophagus is lined by stratified squamous non-keratinized epithelium. **Trap:** C is the stomach and gut. A is the urinary tract. B is the respiratory tract.

**73. D** — Type II pneumocytes contain lamellar bodies and secrete surfactant. **Trap:** C is the type I cell. A describes airway epithelium. B describes a macrophage.

**74. A** — Clara (club) cells line bronchioles. **Trap:** The alveoli are lined by type I and type II pneumocytes, the trachea by respiratory epithelium with goblet cells, and the pleura by mesothelium.

**75. C** — Cartilage forms rings in the trachea, plates in bronchi, and is absent in bronchioles. **Trap:** B and D reverse the sequence. A is not the pattern.

**76. B** — Type II pneumocytes do not form part of the thin blood–air barrier. **Trap:** C, D, and A are all part of the barrier.

**77. A** — The stratum lucidum is present only in thick skin of palms and soles. **Trap:** B, C, and D are present in thin skin too.

**78. C** — Melanocytes derive from the neural crest. **Trap:** B is the origin of the neural tube. D gives the dermis. A gives gut epithelium.

**79. B** — Sebaceous glands are holocrine; the whole cell disintegrates into the secretion. **Trap:** D is eccrine sweat. A is mammary lipid. C is hormonal release.

**80. C** — Pacinian corpuscles lie deep in the dermis and hypodermis and sense vibration. **Trap:** B contains Meissner corpuscles; A is supplied by free nerve endings. D has no corpuscles.

**81. B** — Keratinocytes make up about 90% of epidermal cells. **Trap:** D, A, and C together are a small minority.

**82. B** — Meissner corpuscles in the dermal papillae mediate fine tactile discrimination. **Trap:** A senses deep pressure and vibration. C senses stretch. D senses pain and temperature.

**83. D** — Nephrin is the key protein of the slit diaphragm. **Trap:** A is a desmosomal protein. B is a gap junction protein. C is a basal lamina protein.

**84. C** — The macula densa is the thickened plaque of the distal straight tubule (thick ascending limb) where it touches the vascular pole. **Trap:** A, B, and D do not sense tubular fluid.

**85. C** — Juxtaglomerular cells of the afferent arteriole secrete renin. **Trap:** B senses NaCl. D are podocytes. A are mesangial cells.

**86. B** — The proximal tubule is intensely eosinophilic because of abundant mitochondria and brush border. **Trap:** C is thin. D is pale. A is pale.

**87. C** — The ureter is lined by transitional epithelium. **Trap:** B and A are respiratory or gut. D is the distal urethra and vagina.

**88. D** — Thyroid follicles are lined by a simple epithelium whose cells range from squamous (inactive) to low columnar (active), classically described as cuboidal. The key point is that the epithelium is always simple; squamous and columnar are activity states, not different cell types. **Trap:** C lines the urinary tract. A and B are not found in thyroid follicles.

**89. B** — Parafollicular C cells secrete calcitonin. **Trap:** A make thyroid hormone. C are parathyroid. D are parathyroid.

**90. C** — The zona fasciculata secretes cortisol under ACTH control. **Trap:** B is glomerulosa. D is reticularis. A is medulla.

**91. B** — The adrenal medulla is neural crest derived; the cortex is mesodermal. **Trap:** A is mesoderm. C is endoderm. D is pharyngeal pouch endoderm.

**92. D** — Herring bodies are neurosecretory dilatations in the posterior pituitary. **Trap:** C contains no Herring bodies. A and B are not neural.

**93. C** — Tight junctions between adjacent Sertoli cells form the blood–testis barrier. **Trap:** B are germ cells. D are interstitial. A surround the tubule.

**94. D** — LH stimulates Leydig cells; FSH acts on Sertoli cells. **Trap:** C responds to FSH. A and B are not the primary LH targets.

**95. B** — Fertilization normally occurs in the ampulla of the uterine tube. **Trap:** A is where the embryo implants. C is a narrow transport segment. D is the birth canal.

**96. B** — Straight, narrow glands with mitoses are typical of the proliferative phase. **Trap:** A shows shedding. C shows tortuous secreting glands. D shows decidual change.

**97. C** — The corpus luteum secretes mainly progesterone. **Trap:** B is a minor product. D comes from trophoblast. A comes from pituitary.

**98. D** — The fovea contains cones only, giving maximal acuity and color vision. **Trap:** C is the periphery. A is false. B is the optic disc, not the fovea.

**99. A** — Aqueous humor drains through the trabecular meshwork into the canal of Schlemm. **Trap:** D is behind the lens. B produces it. C is not a drainage route.

**100. C** — The inner and outer hair cells of the organ of Corti are the transducing receptors. **Trap:** D support the hair cells. A are afferent neurons. B is a membrane, not cells.

# Scoring and What to Do Next

| Score | Interpretation | Action |
|---|---|---|
| **90–100** | Excellent | Skim the rapid review once daily; do slide identification only |
| **80–89** | Exam-ready | Reread the twenty-five distinctions table; revisit missed chapters |
| **65–79** | Adequate | Redo the master tables in Part VI, then retake the questions |
| **50–64** | At risk | Read each chapter's "core idea" and tables, not the prose |
| **Below 50** | Start again from Part II | Build the six core tables from memory before anything else |

**HIGH-YIELD PEARL**

The final revision loop is not reading — it is **recall**. Close the book and write the ten master tables from memory. Whatever you cannot reproduce is exactly what you must study. Students who can reproduce the tables usually answer the recall questions far more quickly.